Phase-Matching Second-Order Optical Nonlinear Interactions using Bragg Reflection Waveguides: A Platform for Integrated Parametric Devices

by

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Abstract

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Bragg reflection waveguides (BRW) or one-dimensional photonic bandgap structures have been demonstrated for phase-matching $\boldsymbol{\chi}^{(2)}$ nonlinearities in $\mathrm{Al}_x\mathrm{Ga}_{1-x}\mathrm{As}$. The method exploits strong modal dispersion of a Bragg mode and total internal reflection modes co-propagating inside the waveguide. It is shown that phase-matching is attained among the lowest order modes of interacting harmonics, which allows maximizing the utilization of harmonics powers for nonlinear interactions.

As our first demonstration, we report second-harmonic generation (SHG) of a 2-ps telecommunication pump in a 2.4 mm long slab BRW. The conversion efficiency is estimated as 2.0 $\%\mathrm{W}^{-1}\mathrm{cm}^{-2}$ with a generated SH power of 729 nW. This efficiency has been considerably improved by introducing lateral confinement of optical modes in ridge structures. Characterizations denote that efficiency of SHG in ridge BRWs can increase by over an order of magnitude in comparison to that of the slab device. Also, we report continuous-wave SHG in BRWs. Using a telecommunication pump with a power of 98 mW, the continuous-wave SH power of 23 nW is measured in a 2.0 mm long device.

Significant enhancements of $\boldsymbol{\chi}^{(2)}$ interactions is obtained in the modified design of matching-layer enhanced BRW (ML-BRW). For the first time, we report type-II SHG in

ML-BRW, where the second-harmonic power of 60 μW is measured for a pump power of 3.3 mW in a 2.2 mm long sample. Also, we demonstrate the existence of type-0 SHG, where both pump and SH signal have an identical TM polarization state. It is shown that the efficiency of the type-0 process is comparable to type-I and type-II processes with the phase-matching wavelengths of all three interactions lying within a spectral window as small as 17 nm. ML-BRW is further reported for sum-frequency and difference-frequency generations. For applications requiring high pump power, a generalized ML-BRW design is proposed and demonstrated. The proposed structure offsets the destructive effects of third-order nonlinearities on $\boldsymbol{\chi}^{(2)}$ processes when high power harmonics are involved. This is carried out through incorporation of larger bandgap materials by using high aluminum content $Al_xGa_{1-x}As$ layers without undermining the nonlinear conversion efficiency.

Theoretical investigations of BRWs as integrated sources of photon-pairs with frequency correlation properties are discussed. It is shown that the versatile dispersion properties in BRWs enables generation of telecommunication anti-correlated photon-pairs with bandwidth tunablity between 1 nm and 450 nm.

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Chapter 1

Introduction

1.1 Background

Since their advent over five decades ago, lasers [14, 15] have influenced nearly every aspect of human societies, from fiber-optic communications to medical diagnosis/treatment to DVDs, CDs and display technologies. Without the tremendous bandwidth offered by laser enabled optical communications, the explosive growth of Internet access would not be possible. In eye surgeries, lasers have offered low-cost treatment of diabetic retinopathy, one of the major causes of blindness in Europe and North America. In display technologies, emerging laser TVs have superior colour gamut and are expected to dominate the display market. In addition to the lasing mechanism, coherent radiation of light can be generated through optical parametric processes using nonlinear light-mater interaction [16]. For example, optical parametric oscillators (OPOs) have been widely used for generating tunable optical radiation at wavelengths where no laser material can access. However, today, OPOs have limited practicalities due to their large foot-prints, lack of portability, alignment complexity, large power consumption and high initial/maintenance costs.

Integrated parametric devices harnessing second-order optical nonlinearities in III-V compound semiconductors have attracted increasing attentions in recent years as versatile, compact, alignment free and cost efficient photonics elements for optical frequency mixing. Examples of all-semiconductor parametric devices include electrically pumped optical parametric oscillators, frequency conversion elements and on-chip sources of entangled photon-pairs. Such parametric devices are key photonics elements for a vast number of established and emerging applications, including infrared spectroscopy, environmental monitoring, chemical sensing, medical diagnostics/treatments, THz generation and quantum communications.

$Al_xGa_{1-x}As$ is an excellent ternary semiconductor for on-chip optical frequency mixing. It exhibits large second order effective nonlinearity ($d_{\mathrm{eff}}^{\mathrm{GaAs}} \approx 100$ pm/V at 1550 nm) [4], broad transparency window ($0.9 - 17$ μm), good thermal conductivity [17] and well-established fabrication technology. However, the cubic structure ($\bar{4}3m$ symmetry) of $Al_xGa_{1-x}As$ renders these crystals optically isotropic and lacking in intrinsic birefringence. As a result, phase-matching (PM) $\boldsymbol{\chi}^{(2)}$ nonlinearities involve significant challenges in these crystals. In a parametric device, efficiency of nonlinear interaction strongly depends on the employed phase-matching scheme.

The physical meaning of phase-matching has been well described by Butcher and Cotter [18], where the concept of nonlinear electric polarization wave was introduced. According to their interpretation, the effect of a coherent optical field on the medium establishes an assembly of coherently oscillating electric dipoles. From the microscopic point of view, such an assembly of oscillating dipoles comprises an electric polarization vector with a phase-velocity equals to the phase-velocity of the generated harmonic. The phase-matching condition then simply indicates that the polarization wave should be phase synchronized with the sum of the phases of the individuals harmonics which initiated the nonlinear interaction, a condition which is essential for constructive superposition of oscillating dipoles hence complete power transfer to the generated harmonic.

For a given parametric process, the phase-matching condition is generally expressed

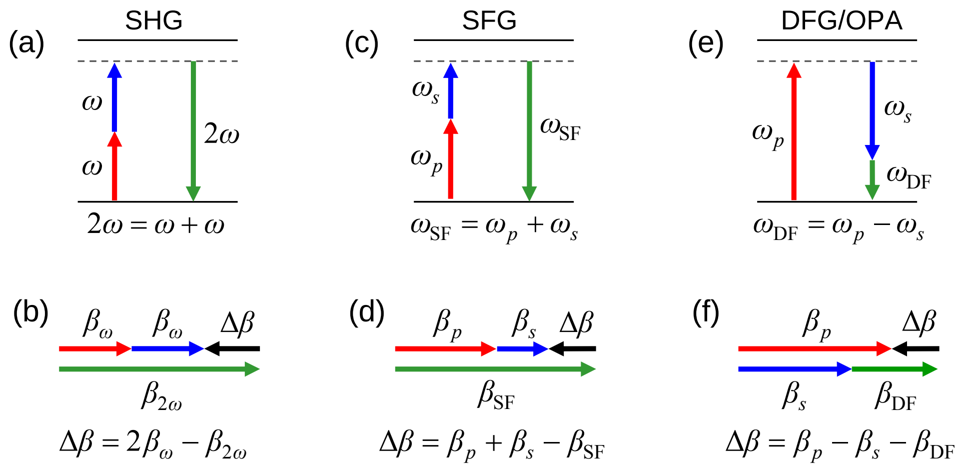

Figure 1.1: Diagrams of the conservation of energy and the wavevector mismatch for (a)-(b) colinear SHG, (c)-(d) colinear SFG and (e)-(f) colinear DFG/OPA.

by the wavenumber mismatch between the interacting harmonics. Figure 1.1 illustrates the conservation of energy along with the wavenumber mismatch for second-order nonlinear processes, including second-harmonic generation (SHG), sum-frequency generation (SFG), difference-frequency generation (DFG) and optical parametric amplification (OPA).

Phase-matching second-order nonlinear interactions in $Al_xGa_{1-x}As$ waveguides have been extensively investigated by different research centers. Among these, the most successful techniques are the form birefringence phase-matching (BPM) [19], developed at Laboratoire Matriaux et Phnomnes Quantiques (CNRS) in France [20] and the quasi-phase-matching (QPM) [21] investigated at Ginzton Lab in Stanford University [22]. Despite the greatest successes of form birefringence waveguides in phase-matching various nonlinear interactions, the necessity of incorporating Al_2O_3 layers with high absorption above 7.5 μm along with GaAs with large absorption below 870 nm limits the operating window of BPM devices particularly when generation of long wavelength radiations is desired [23]. The existence of oxide layers in BPM waveguides also has serious drawbacks when integration of these devices with laser diode pump is required. For example, in an electically pumped OPO system where diode pump and frequency conversion element are monolithically integrated, the Al_2O_3 layers prohibit the current injection into the active region.

In 2006, our research group at the University of Toronto proposed a novel phase-matching technique using Bragg reflection waveguides or one-dimensional photonic bandgap structures [24, 25]. BRW phase-matching is an exact modal phase-matching technique which exploits modal dispersions of a photonic bandgap mode (Bragg mode) and conventional total internal reflection (TIR) modes. Using strong modal discrimination of Bragg mode and TIR modes, we have shown that phase-matching can be attained among the lowest order modes of involved harmonics. Modal phase-matching (MPM) has been previously shown in $Al_xGa_{1-x}As$ waveguides where, for example for second-harmonic generation, PM was attained between the lowest order TIR mode of the pump and high order modes of second-harmonic signal [26, 27]. However, the involvement of higher order modes in MPM devices significantly reduces the efficiency of nonlinear interaction due to a poor modal overlap factor as well as increased propagation losses adhering to the propagation of high order modes. In contrast, phase-matching in BRWs is achieved between the lowest order optical modes which benefit from maximum utilization of associated powers in interacting harmonics. Phase-matched BRWs additionally benefit from fabrication simplicity as well as being more indulgent for monolithic integration with laser diode pumps. These unique characteristics make Bragg reflection waveguides a promising platform for the development of all-semiconductor parametric devices.

1.2 Dissertation Overview

This dissertation describes the design and characterization of $Al_xGa_{1-x}As$ Bragg reflection waveguides for phase-matching second-order nonlinear interactions. The demonstrated phase-matching technique can be applied to compound semiconductors other than $Al_xGa_{1-x}As$ as well as to non-semiconductor material systems. The organization of the dissertation is as follows.

Chapter 2 develops the theoretical basis required for the analysis of nonlinear interactions in guided-wave structures. Using coupled-mode equations in waveguides, detailed analyses of second-order processes including second-harmonic generation, sum-frequency generation and difference-frequency generation are presented. A summary of available phase-matching techniques in $Al_xGa_{1-x}As$ is also provided.

Chapter 3 sets forth the analysis of multi-layer dielectric structures in general and Bragg reflection waveguides in particular. Modal dispersion properties of slab BRWs as well as ridge structures are theoretically investigated through numerical simulations.

In Chapter 4, the design of phase-matched quarter-wave BRWs is examined in detail. Second-harmonic generation in slab and ridge structures is also demonstrated and characterization results of continuous-wave frequency doubling of telecommunication signal are discussed.

Chapter 5 proposes the modified design of matching-layer enhanced Bragg reflection waveguides (ML-BRWs). Significant enhancement offered by ML-BRW designs in improving the efficiency of second-harmonic generation is demonstrated. Furthermore, the ML-BRWs are employed for degenerate continuous-wave sum-frequency generation and pulsed difference-frequency generation.

Chapter 6 provides an overview of extended ML-BRW designs suitable for applications involving high pump power. Through fabrication and characterizations, the potential of generalized ML-BRWs for optical frequency mixing is highlighted.

In Chapter 7, we examine the theoretical application of phase-matched BRWs as integrated sources of photon-pairs for quantum optical communications. We focus in particular on the generation of frequency anti-correlated biphotons and on tailoring their dispersion properties for spectral bandwidth control.

Finally, in Chapter 8, the contributions of this dissertation are summarized and potential avenues to the further advancement of this research are delineated.

Chapter 2

Guided-Wave Nonlinear Optics

This chapter provides the mathematical formulation used throughout this dissertation for the analysis of second-order nonlinear interactions in guided-wave structures. The formulation is based on coupled-mode theory in nonlinear waveguides [28]. Using coupled-mode equations, nonlinear interactions including second-harmonic generation (SHG), sum-frequency generation (SFG), difference frequency generation (DFG) are discussed.

2.1 Nonlinear Electric Polarization and Maxwell's Equations

When an optical signal is incident on a dielectric medium, the orientation and oscillation of the electric dipoles of the subject medium specifies the medium response to the applied external field. The medium response is characterized by the electric polarization vector

$$\mathbf{P} = \epsilon_0 \left(\boldsymbol{\chi}^{(1)}\mathbf{E} + \boldsymbol{\chi}^{(2)}\mathbf{E}\mathbf{E} + \boldsymbol{\chi}^{(3)}\mathbf{E}\mathbf{E}\mathbf{E} + \ldots \right) = \mathbf{P}^{(1)} + \mathbf{P}^{(\mathrm{NL})} \tag{2.1}$$

where $\mathbf{E}$ is the applied electric field, $\boldsymbol{\chi}^{(1)}$ is the linear susceptibility tensor, $\boldsymbol{\chi}^{(2)}$ is the second-order susceptibility tensor, ... In Eq. (2.1), $\mathbf{P}^{(1)} = \epsilon_0 \boldsymbol{\chi}^{(1)}\mathbf{E}$ is the linear electric polarization vector which accounts for linear optical phenomenon such as refraction, diffraction and dispersion and $\mathbf{P}^{(\mathrm{NL})} = \epsilon_0 \left(\boldsymbol{\chi}^{(2)}\mathbf{EE} + \boldsymbol{\chi}^{(3)}\mathbf{EEE} + ...\right)$ is the nonlinear electric polarization vector which initiates the generation of new optical harmonics. To properly catalogue possible nonlinear processes for a given order of optical nonlinearity, we adopt the notation of Butcher and Cotter [18] in expanding the electric polarization vector in terms of its component as

$$\left[\mathbf{P}^{(n)}_{\omega_\sigma}\right]_\mu = \epsilon_0 \sum_{\omega} K(-\omega_\sigma;\omega_1,...,\omega_n)\chi^{(n)}_{\mu\alpha_1...\alpha_n}(-\omega_\sigma;\omega_1,...,\omega_n)(E_{\omega_1})_{\alpha_1}...(E_{\omega_n})_{\alpha_n} \tag{2.2}$$

where, by convention, the summation over repeated indices hold and the summation over σ denotes that the equation should be summed over all distinct sets of $\omega_1, \omega_2, \ldots, \omega_n$. In Eq. (2.2), K is the degeneracy factor defined as

$$K(-\omega_\sigma;\omega_1,...,\omega_n) = 2^{l+m-n} \tag{2.3}$$

where

$$p = \text{the number of distinct permutations of } \omega_1,...,\omega_n, \tag{2.4}$$

$$n = \text{the order of nonlinear interaction,} \tag{2.5}$$

$$m = \text{the number of frequencies } \omega_k \text{ that are zero, and} \tag{2.6}$$

$$l = \begin{cases} 1, & \text{if } \omega_\sigma = 0 \\ 0, & \text{otherwise} \end{cases} \tag{2.7}$$

In Eq. (2.4), m is regarded as the number of existing DC electric fields and $l = 0$ denotes optical rectification.

In the presence of optical nonlinearities, propagation of an optical signal is completely governed by the Maxwell's equations

$$\nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t \tag{2.8a}$$

$$\nabla \times \mathbf{H} = \partial \mathbf{D}/\partial t + \mathbf{J}_i + \mathbf{J}_c \tag{2.8b}$$

$$\nabla . \mathbf{D} = \rho_e \tag{2.8c}$$

$$\nabla . \mathbf{B} = 0 \tag{2.8d}$$

where $\mathbf{H}$ is the magnetic field, $\mathbf{B}$ is the magnetic flux density, $\mathbf{D}$ is the electric displacement, ρ_e is the free electric charge density, $\mathbf{J}_i$ is the impressed current density and $\mathbf{J}_c = \sigma\mathbf{E}$ is the conduction current density where σ denotes the medium conductivity. We assume optical signals propagate in a source free medium where $\rho_e = 0$ and $\mathbf{J}_i = 0$. Using the constitutive relations, the medium properties and electromagnetic fields are related to each other as

$$\mathbf{B} = \boldsymbol{\mu}\mathbf{H} \tag{2.9a}$$

$$\mathbf{D} = \boldsymbol{\epsilon}\mathbf{E} = \epsilon_0(\boldsymbol{I} + \boldsymbol{\chi}^{(1)})\mathbf{E} + \mathbf{P}^{(\mathrm{NL})} \tag{2.9b}$$

where $\boldsymbol{\epsilon}$ and $\boldsymbol{\mu}$ are the permittivity and permeability tensors, respectively, and $\boldsymbol{I}$ is the identity tensor. Only non-magnetic materials are considered here for which $\boldsymbol{\mu} = \mu_0$ where μ_0 is the permeability of free space. Using Maxwell's curl equations in frequency domain, the wave equation, with the existence of nonlinear medium, is expressed as

$$\nabla \times \nabla \times \mathbf{E}(\mathbf{r}, \omega) - \frac{\omega^2}{c^2}(\boldsymbol{I} + \boldsymbol{\chi}^{(1)})\mathbf{E}(\mathbf{r}, \omega) = \mu_0\omega^2\mathbf{P}^{(\mathrm{NL})}(\mathbf{r}, \omega) \tag{2.10}$$

The right hand side of Eq. (2.10) accounts for the modification of propagation in free space due to light-matter nonlinear interaction. In this regard, $\mathbf{P}^{(\mathrm{NL})}$ serves as a source of electromagnetic radiation which converts the homogenous differential equation of wave propagation in free space into a non-homogenous equation.

2.2 Coupled-Mode Equations

Coupled-mode theory (CMT) is a powerful tool for the analysis of nonlinear interactions in guided-wave structures [28]. CMT not only establishes the required mathematical approach for analysis of nonlinear interactions but also it offers physical insight for better understanding the interconnections among the interacting harmonics in nonlinear waveguides.

Consider a nonlinear waveguide which supports the propagation of many optical modes along the $+z$-axis. In the absence of optical nonlinearity, the spatial-temporal dependence of electromagnetic fields of the m-th mode can be expressed as

$$\mathbf{E}_m(x, y, z, t) = \mathbf{E}_m(x, y) \exp[i(\beta_m z - \omega_m t)] = (\mathbf{E}_{tm} + \mathbf{E}_{zm}) \exp[i(\beta_m z - \omega_m t)] \quad (2.11a)$$

$$\mathbf{H}_m(x, y, z, t) = \mathbf{H}_m(x, y) \exp[i(\beta_m z - \omega_m t)] = (\mathbf{H}_{tm} + \mathbf{H}_{zm}) \exp[i(\beta_m z - \omega_m t)] \quad (2.11b)$$

where β_m is the propagation constant, ω_m is the angular frequency, $\mathbf{E}_{tm}$ and $\mathbf{H}_{tm}$ are the electric and magnetic field vectors in the transverse plane, respectively, and $\mathbf{E}_{zm}$ and $\mathbf{H}_{zm}$ are those along the propagation direction. The orthogonality of the m-th and m'-th modes requires that the spatial transverse components of the two modes satisfy the condition that

$$\frac{1}{4} \iint_{-\infty}^{+\infty} \left(\mathbf{E}_{tm} \times \mathbf{H}^*_{tm'} + \mathbf{E}^*_{tm'} \times \mathbf{H}_{tm} \right) .\hat{z}\, ds = \delta_{mm'} \quad (2.12)$$

where $\delta_{mm'}$ is the Kronecker delta function.

Equations (2.11)–(2.12) are valid only for the case in which the optical nonlinearities of the waveguide are disregarded. In the presence of nonlinear effects, the envelope of mode profiles along the propagation direction will not remain constant and varies as a result of nonlinear mode coupling. In order to include the envelope variation, we express the spatial dependency of electromagnetic fields as

$$\mathbf{E}_m(x,y,z) = \mathbf{E}_m(x,y)A_m(z)\exp(i\beta_m z) = (\mathbf{E}_{tm} + \mathbf{E}_{zm})A_m(z)\exp(i\beta_m z) \tag{2.13a}$$

$$\mathbf{H}_m(x,y,z) = \mathbf{H}_m(x,y)A_m(z)\exp(i\beta_m z) = (\mathbf{H}_{tm} + \mathbf{H}_{zm})A_m(z)\exp(i\beta_m z) \tag{2.13b}$$

where $A_m(z)$ denotes the slowly varying envelope (SVE) function. In Eq. (2.13), we considered only the spatial variation of the envelope function, $A_m(z)$, in the presence of optical nonlinearities with any given order. A more comprehensive description of the nonlinear mode propagation requires the inclusion of the temporal dependence of the envelope function which is beyond the focus of this work. Using the orthogonality relation of Eq. (2.12), variation of the envelope function can be expressed according to

$$\frac{\partial A_m(z)}{\partial z} = \frac{i\omega}{4}\iint_{-\infty}^{+\infty} \mathbf{E}_m^*(x,y).\mathbf{P}^{(\mathrm{NL})}\exp(-i\beta_m z)dxdy \tag{2.14}$$

Equation (2.14) serves as the generic coupled-mode equation for a nonlinear guided-wave structure in vectorial form. By using the above equation, one needs to be careful when expanding the term $\mathbf{E}_m^*(x,y)$ within the integration. $\mathbf{E}_m^*(x,y)$ enters the formulation as the transverse field profile in the linear regime, and its spatial dependency does not include the slowly varying envelope function. However, in the equation of $\mathbf{P}^{(\mathrm{NL})}$, which includes the electric fields of the coupled modes, variation of the field amplitudes along the propagation direction should be considered in the calculations.

Optical modes in 2-D guided-wave structures such as ridge waveguides are hybrid modes with vectorial mode profiles. However, it is usually possible to use approximate scalar expressions for the major field components. We assume that the field components are normalized to have unity power in the propagation direction. For the interacting harmonic at frequency ω_σ, the scalar field profile is taken in form of $C_{\omega_\sigma}E_{\omega_\sigma}(x,y)$ where

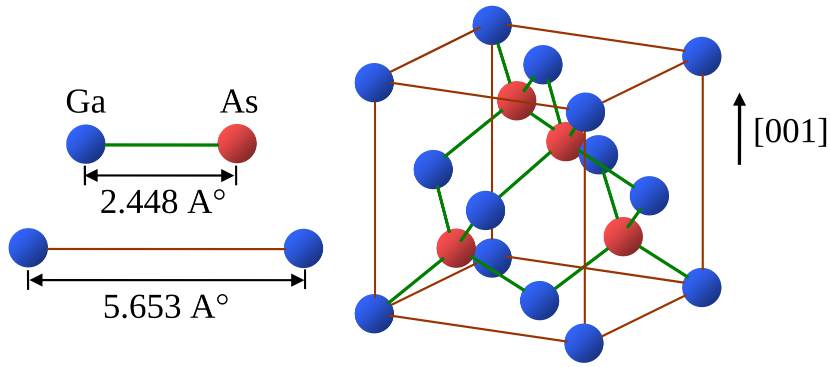

Figure 2.1: Unit cell of GaAs with [001] growth direction. The crystal has zinc blende structure where the As sublatttice has a diagonal shift by a quarter unit cell.

C_{ω_σ} is the normalization factor expressed as

$$C_{\omega_\sigma} = \left(\frac{k_{\omega_\sigma}}{2\omega_\sigma \mu_0} \iint_{-\infty}^{+\infty} |E_{\omega_\sigma}(x,y)|^2 \, dxdy \right)^{-1/2} \tag{2.15}$$

For an optical harmonic with angular frequency ω_σ, the scalar coupled-mode equation then becomes

$$\frac{\partial A_{\omega_\sigma(z)}}{\partial z} = \frac{i\omega_\sigma}{4} \iint_{-\infty}^{+\infty} E^*_{\omega_\sigma}(x,y) P_{\mathrm{NL}} \exp(-i\beta_{\omega_\sigma} z) dxdy \tag{2.16}$$

Equation (2.16) is the core coupled-mode equation which has been extensively used for expressing various nonlinear processes in waveguides. In the following, we employ this formulation for the analysis of $\chi^{(2)}$ interactions in $\mathrm{Al}_x\mathrm{Ga}_{1-x}\mathrm{As}$ guided-wave structures.

2.3 $\chi^{(2)}$ Interactions in $\mathrm{Al}_x\mathrm{Ga}_{1-x}\mathrm{As}$ Waveguides

GaAs has a non-centrosymmetric cubic structure with $\bar{4}3m$ point symmetry group. This renders the crystal optically isotropic with lack of natural birefringence. A unit cell of GaAs is schematically illustrated in Fig. 2.1. The crystal is composed of two face-centered cubic sublattices of Ga and As where one sublattice has a diagonal shift by a quarter-unit-cell.

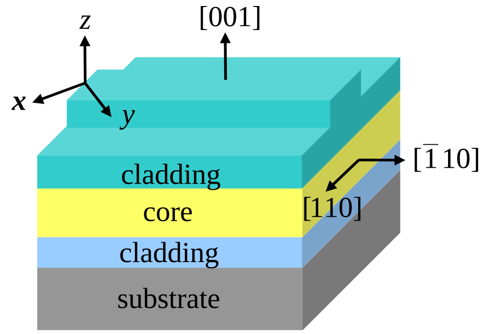

Figure 2.2: Schematic of a typical planar $Al_xGa_{1-x}As$ waveguide with [001] growth direction and propagation direction along [$\bar{1}$10]. The xyz coordinate system denotes the crystal frame.

The point symmetry group of a crystal structure determines the accessible elements of the $\boldsymbol{\chi}^{(2)}$ tensor for nonlinear interactions. In GaAs material system, the only non-zero second-order susceptibility tensor element is $\chi^{(2)}_{xyz}$, which implies that nonlinear interaction can only take place when interacting harmonics have non-vanishing field components along all three crystal axes. Due to the zinc blende crystal structure of GaAs, there exist two distinct growth directions which allows one to access $\chi^{(2)}_{xyz}$. These include the [001] and [111] directions. The growth direction of [111] is usually not desired for integrated devices where waveguides with parallel facets are required. In this case, cleaving the wafer results in waveguides where the facets make an angle of 60° with respect to each other. Despite this, the [111] growth direction favours surface emitting nonlinear devices since the effective second-order nonlinearity is the highest [29]. Devices with [001] growth direction and [$\bar{1}$10] cleavage planes are the most widely used configurations for nonlinear interaction in $Al_xGa_{1-x}As$ waveguides. A representative of such a device is schematically shown in Fig. 2.2, where the coordinate system of xyz denotes the crystal reference frame.

Consider the general nonlinear process of three-wave mixing where the two harmonics with angular frequencies ω_1, and ω_2 generate a new harmonic at ω_3. Using Eq. (2.2), the components of the second-order nonlinear polarization vector at frequency ω_3 are

$$[P^{(2)}_{\omega_3}]_x = \epsilon_0 K(-\omega_3;\omega_1,\omega_2)\chi^{(2)}_{xyz}\left(E^y_{\omega_1}E^z_{\omega_2} + E^z_{\omega_1}E^y_{\omega_2}\right) \tag{2.17a}$$

$$[P^{(2)}_{\omega_3}]_y = \epsilon_0 K(-\omega_3;\omega_1,\omega_2)\chi^{(2)}_{xyz}\left(E^x_{\omega_1}E^z_{\omega_2} + E^z_{\omega_1}E^x_{\omega_2}\right) \tag{2.17b}$$

$$[P^{(2)}_{\omega_3}]_z = \epsilon_0 K(-\omega_3;\omega_1,\omega_2)\chi^{(2)}_{xyz}\left(E^x_{\omega_1}E^y_{\omega_2} + E^y_{\omega_1}E^x_{\omega_2}\right) \tag{2.17c}$$

where K is the degeneracy factor defined in Eq. (2.3). Equations (2.17) are used for determining possible second-order nonlinear interactions in a $Al_xGa_{1-x}As$ waveguide.

2.3.1 Second-harmonic generation

A second-harmonic generation waveguide is the most fundamental nonlinear device where an input pump wave (FH) at ω is frequency doubled to generate a second-harmonic signal (SH) at 2ω. The most widely investigated SHG interactions in $Al_xGa_{1-x}As$ waveguides include the type-I process where a TE-polarized pump generates TM-polarized SH and the type-II process where a hybrid pump with TE+TM polarization state generates a TE-polarized SH.

For type-I process, a TE_ω polarized pump with an effective mode index of $n^{(TE)}_\omega$ interacts with $TM_{2\omega}$ polarized SH with an index of $n^{(TM)}_{2\omega}$. Using Eq. (2.17), the components of the electric polarization vector at 2ω frequency become

$$[P^{(2)}_{2\omega}]_x = 0 \tag{2.18a}$$

$$[P^{(2)}_{2\omega}]_y = 0 \tag{2.18b}$$

$$[P^{(2)}_{2\omega}]_z = \epsilon_0\chi^{(2)}_{xyz}E^x_\omega E^y_\omega \tag{2.18c}$$

The existence of only $[P^{(2)}_{2\omega}]_z$ indicates that the generated SH has only $TM_{2\omega}$ polarization state. Using Eq. (2.16), the coupled-mode equations for type-I SHG are obtained as

$$\frac{\partial A_{\omega}^{(\mathrm{TE})}}{\partial z} = i\kappa_{\omega}\xi^{*}A_{\omega}^{*(\mathrm{TE})}A_{2\omega}^{(\mathrm{TM})}\exp(-i\Delta\beta z) - \frac{\alpha_{\omega}^{(\mathrm{TE})}}{2}A_{\omega}^{(\mathrm{TE})} \tag{2.19a}$$

$$\frac{\partial A_{2\omega}^{(\mathrm{TM})}}{\partial z} = i\kappa_{2\omega}\xi A_{\omega}^{(\mathrm{TE})}A_{\omega}^{(\mathrm{TE})}\exp(i\Delta\beta z) - \frac{\alpha_{2\omega}^{(\mathrm{TM})}}{2}A_{2\omega}^{(\mathrm{TM})} \tag{2.19b}$$

where $\Delta\beta$ is the wavenumber mismatch defined as

$$\begin{aligned}\Delta\beta &= 2\beta_{\omega}^{(\mathrm{TE})} - \beta_{2\omega}^{(\mathrm{TM})} \\ &= \frac{4\pi}{\lambda_{\omega}}\left[n_{\omega}^{(\mathrm{TE})} - n_{2\omega}^{(\mathrm{TM})}\right]\end{aligned} \tag{2.20}$$

In Eq. (2.19), κ_{ω_σ} is the nonlinear coupling factor, $\alpha_{\omega}^{(\mathrm{TE})}$ and $\alpha_{2\omega}^{(\mathrm{TM})}$ are the propagation losses of pump and SH, respectively, and ξ is the nonlinear spatial overlap factor, which are defined as

$$\kappa_{\omega_\sigma} = \left(\frac{8\pi^2 d_{\mathrm{eff}}^2}{(n_{\omega}^{(\mathrm{TE})})^2 n_{2\omega}^{(\mathrm{TM})}\epsilon_0 c\lambda_{\omega_\sigma}^2}\right)^{1/2}, \quad \omega_\sigma \in \{\omega, 2\omega\} \tag{2.21}$$

$$d_{\mathrm{eff}} = \frac{\iint_{-\infty}^{+\infty} E_{2\omega}^{*(\mathrm{TM})} d(x,y)[E_{\omega}^{(\mathrm{TE})}]^2 dxdy}{\iint_{-\infty}^{+\infty} E_{2\omega}^{*(\mathrm{TM})}[E_{\omega}^{(\mathrm{TE})}]^2 dxdy} \tag{2.22}$$

$$\xi = \frac{\iint_{-\infty}^{+\infty} E_{2\omega}^{*(\mathrm{TM})}[E_{\omega}^{(\mathrm{TE})}]^2 dxdy}{\left(\iint_{-\infty}^{+\infty} |E_{\omega}^{(\mathrm{TE})}|^2 dxdy\right)\left(\iint_{-\infty}^{+\infty} |E_{2\omega}^{(\mathrm{TM})}|^2 dxdy\right)^{1/2}} \tag{2.23}$$

where d_{eff} is the effective second-order nonlinear coefficient of the waveguide and $d(x,y)$ denotes the spatial dependence of effective $\boldsymbol{\chi}^{(2)}$ coefficient in the transverse plane. The spatial modal overlap factor, ξ, is widely expressed as $A_{\mathrm{eff}}^{(2)} = \xi^{-2}$ where $A_{\mathrm{eff}}^{(2)}$ is defined as the effective area of the second-harmonic generation. A small $A_{\mathrm{eff}}^{(2)}$ is usually desired for increased intensity of optical modes hence enhanced second-harmonic generation. Although, we defined the $A_{\mathrm{eff}}^{(2)}$ for second-harmonic generation, the relation between $A_{\mathrm{eff}}^{(2)}$ and ξ is universal and can be used for other $\boldsymbol{\chi}^{(2)}$ processes including sum-frequency generation and difference-frequency generation.

In the low conversion limit where pump depletion can be ignored, the variation of pump amplitude along the waveguide length, L, is negligible indicating that $A_{\omega}^{(\mathrm{TE})}(0) = A_{\omega}^{(\mathrm{TE})}(L)$. Using Eq. (2.19), one can calculate the generated SH power at the waveguide output as [30]

$$\mathcal{P}_{2\omega}^{(\mathrm{TM})}(L) = \kappa_{\omega}^{2}\xi^{2}L^{2}\mathcal{P}_{\omega}^{(\mathrm{TE})}(0)^{2}\exp[-(\alpha_{\omega}^{(\mathrm{TE})} + \alpha_{2\omega}^{(\mathrm{TM})}/2)L] \\ \times \frac{\sin^{2}(\Delta\beta L/2) + \sinh^{2}[(\alpha_{\omega}^{(\mathrm{TE})} - \alpha_{2\omega}^{(\mathrm{TM})}/2)L/2]}{(\Delta\beta L/2)^{2} + [(\alpha_{\omega}^{(\mathrm{TE})} - \alpha_{2\omega}^{(\mathrm{TM})}/2)L/2]} \tag{2.24}$$

In obtaining Eq. (2.24) we assumed that the envelope functions are normalized such that

$$\mathcal{P}_{2\omega}^{(\mathrm{TM})}(L) = \left|A_{2\omega}^{(\mathrm{TM})}(L)\right|^{2}, \quad \mathcal{P}_{\omega}^{(\mathrm{TE})}(0) = \left|A_{\omega}^{(\mathrm{TE})}(0)\right|^{2} \tag{2.25}$$

When propagation losses can be ignored, Eq. (2.24) transforms to the well known sinc function as

$$\mathcal{P}_{2\omega}^{(\mathrm{TM})}(L) = \kappa_{\omega}^{2}\xi^{2}L^{2}\mathcal{P}_{\omega}^{(\mathrm{TE})}(0)^{2}\mathrm{sinc}^{2}(\frac{\Delta\beta L}{2}) \tag{2.26}$$

From Eq. (2.24), $\mathcal{P}_{2\omega}^{(\mathrm{TM})}(L)$ has a quadratic relation with nonlinear coupling factor (κ), nonlinear overlap factor (ξ) as well as the device length (L). As such, one can expect to increase SH power by improving the structure d_{eff}, enhancing the nonlinear overlap factor and increasing the device length. However, in practice, there exists a trade off between increasing the device length and generated SH power due to increased propagation losses and the dispersion of the propagating frequencies.

The existence of propagation losses not only reduce the SH power but also increase the bandwidth of SHG process. To illustrate this, in Fig. 2.3 we have plotted the dependency of SH power on phase-mismatch for different values of SH absorption ($\alpha_{2\omega}L$). From the figure, for small values of $\alpha_{2\omega}L$, SH power follows the ideal sinc function as is the case of a lossless process. However for large values of $\alpha_{2\omega}L$, the sinc function transforms to a Lorentzian shape with broadened bandwidth. This transformation from an ideal sinc

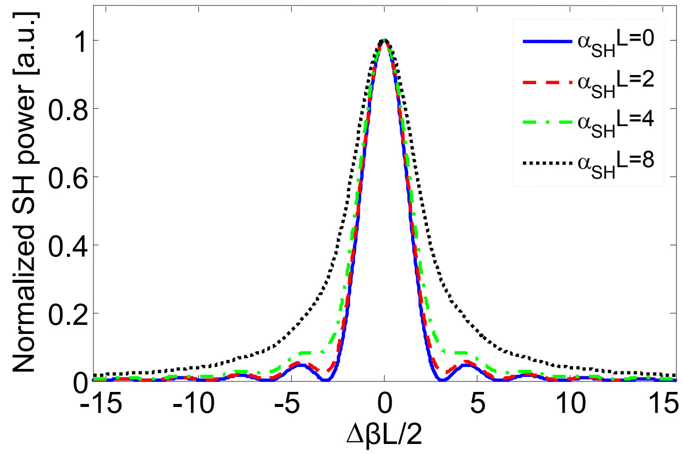

Figure 2.3: Normalized second-harmonic power as a function of phase-mismatch including the effect of absorption at second-harmonic frequency.

shape to a Lorentzian shape is widely used to obtain a rough estimation of SH losses in characterization.

Also, the dispersion of the interacting harmonics may degrade the generated second-harmonic as the device length is increased. For example, in SHG with ultra-short pulsed pump in femtosecond regime, the temporal pulse walk-off due to the group velocity mismatch (GVM) between the pump and SH signal and the pulse distortion due to group velocity dispersion (GVD) play significant roles in reducing the generated second-harmonic power. The degrading effects of GVM and GVD for SHG devices are further addressed in Chapter 5. The above discussion reveals the fact that an optimum design of a $\boldsymbol{\chi}^{(2)}$ device, such as a frequency doubling element, is an involved procedure which demands careful considerations of the trade-offs existing among key waveguide parameters including the harmonics losses, the device length and the modal dispersions.

A figure of merit which denotes the performance of a SHG device is the nonlinear conversion efficiency which is defined as

$$\eta = \frac{\mathcal{P}_{2\omega}^{(\mathrm{TM})}(L)}{\mathcal{P}_{\omega}^{(\mathrm{TE})}(0)^2} \quad \text{or} \quad \eta_{\mathrm{norm}} = \frac{\eta}{L^2} \tag{2.27}$$

Type-II SHG involves the interaction of a hybrid pump where both TE_ω and TM_ω

components exist to generate a $\text{TE}_{2\omega}$ second-harmonic signal. In this case, the pump electric field makes and angle of 45° with respect to [001] crystal axis. The elements of electric polarization vector in type-II SHG are

$$[P_{2\omega}^{(2)}]_x = \epsilon_0 \chi_{xyz}^{(2)} E_\omega^y E_\omega^z \tag{2.28a}$$

$$[P_{2\omega}^{(2)}]_y = \epsilon_0 \chi_{xyz}^{(2)} E_\omega^x E_\omega^z \tag{2.28b}$$

$$[P_{2\omega}^{(2)}]_z = \epsilon_0 \chi_{xyz}^{(2)} E_\omega^x E_\omega^y \tag{2.28c}$$

From (2.28), the non-zero values of all three components of electric polarization vector denote that in type-II interaction, the generated SH signal has mixed $\text{TE}_{2\omega}$ and $\text{TM}_{2\omega}$ polarizations. However, in the xy plane, the effective second-order nonlinearity is twice that along the z-axis. Although the SH signal is mixed polarized in this case, it is expected that the frequency doubled signal to be mainly $\text{TE}_{2\omega}$ polarized. Using Eq. (2.16), the coupled-mode equations for type-II process are expressed according to

$$\frac{\partial A_\omega^{(\text{TE})}}{\partial z} = i\kappa_\omega \xi^* A_\omega^{*(\text{TM})} A_{2\omega}^{(\text{TE})} \exp(-i\Delta\beta z) - \frac{\alpha_\omega^{(\text{TE})}}{2} A_\omega^{(\text{TE})} \tag{2.29a}$$

$$\frac{\partial A_\omega^{(\text{TM})}}{\partial z} = i\kappa_\omega \xi^* A_\omega^{*(\text{TE})} A_{2\omega}^{(\text{TE})} \exp(-i\Delta\beta z) - \frac{\alpha_\omega^{(\text{TM})}}{2} A_\omega^{(\text{TM})} \tag{2.29b}$$

$$\frac{\partial A_{2\omega}^{(\text{TE})}}{\partial z} = i\kappa_{2\omega} \xi A_\omega^{(\text{TE})} A_\omega^{(\text{TM})} \exp(i\Delta\beta z) - \frac{\alpha_{2\omega}^{(\text{TE})}}{2} A_{2\omega}^{(\text{TE})} \tag{2.29c}$$

where the wavenumber mismatch is given by

$$\begin{aligned} \Delta\beta &= \beta_\omega^{(\text{TE})} + \beta_\omega^{(\text{TM})} - \beta_{2\omega}^{(\text{TE})} \\ &= \frac{4\pi}{\lambda_\omega} \left[\frac{1}{2} \left(n_\omega^{(\text{TE})} + n_\omega^{(\text{TM})} \right) - n_{2\omega}^{(\text{TE})} \right] \end{aligned} \tag{2.30}$$

In Eqs. (2.29), the nonlinear parameters κ, d_{eff} and ξ are defined as

$$\kappa_{\omega_\sigma} = \left(\frac{8\pi^2 d_{\text{eff}}^2}{n_\omega^{(\text{TE})} n_\omega^{(\text{TM})} n_{2\omega}^{(\text{TM})} \epsilon_0 c \lambda_{\omega_\sigma}^2} \right)^{1/2}, \quad \omega_\sigma \in \{\omega, 2\omega\} \tag{2.31}$$

$$d_{\text{eff}} = \frac{\iint_{-\infty}^{+\infty} E_{2\omega}^{*(\text{TE})} d(x,y) E_{\omega}^{(\text{TE})} E_{\omega}^{(\text{TM})} dxdy}{\iint_{-\infty}^{+\infty} E_{2\omega}^{*(\text{TM})} E_{\omega}^{(\text{TE})} E_{\omega}^{(\text{TM})} dxdy} \tag{2.32}$$

$$\xi = \frac{\iint_{-\infty}^{+\infty} E_{2\omega}^{*(\text{TE})} E_{\omega}^{(\text{TE})} E_{\omega}^{(\text{TM})} dxdy}{\left(\iint_{-\infty}^{+\infty} |E_{\omega}^{(\text{TE})}|^2 dxdy\right)^{1/2} \left(\iint_{-\infty}^{+\infty} |E_{\omega}^{(\text{TM})}|^2 dxdy\right)^{1/2} \left(\iint_{-\infty}^{+\infty} |E_{2\omega}^{(\text{TE})}|^2 dxdy\right)^{1/2}} \tag{2.33}$$

For type-II SHG the generated SH power with the existence of propagation losses is expressed as [30]

$$\begin{aligned} \mathcal{P}_{2\omega}^{(\text{TE})}(L) = \kappa_{\omega}^2 \xi^2 L^2 \mathcal{P}_{\omega}^{(\text{TE})}(0) \mathcal{P}_{\omega}^{(\text{TM})}(0) \exp[-(\alpha_{\omega}^{(\text{TE})} + \alpha_{\omega}^{(\text{TM})} + \alpha_{2\omega}^{(\text{TE})})L/2] \\ \times \frac{\sin^2(\Delta\beta L/2) + \sinh^2[(\alpha_{\omega}^{(\text{TE})} + \alpha_{\omega}^{(\text{TM})} - \alpha_{2\omega}^{(\text{TE})})L/4]}{(\Delta\beta L/2)^2 + [(\alpha_{\omega}^{(\text{TE})} + \alpha_{\omega}^{(\text{TM})} - \alpha_{2\omega}^{(\text{TE})})L/4]} \end{aligned} \tag{2.34}$$

Also, the conversion efficiency of type-II SHG is defined as

$$\eta = \frac{\mathcal{P}_{2\omega}^{(\text{TE})}(L)}{\mathcal{P}_{\omega}^{(\text{TE})}(0)\mathcal{P}_{\omega}^{(\text{TM})}(0)} \quad \text{or} \quad \eta_{\text{norm}} = \frac{\eta}{L^2} \tag{2.35}$$

Type-I and type-II SHG are the mostly widely investigated second-harmonic generation processes in z-grown $\text{Al}_x\text{Ga}_{1-x}\text{As}$ waveguides. There exist another SHG scheme in $\text{Al}_x\text{Ga}_{1-x}\text{As}$ devices namely type-0 interaction where both pump and SH signal are TM-polarized [2]. This process has not received any attention in the literature partly owing to its weak nonlinear interaction in comparison to type-I and type-II interactions. Type-0 SHG in $\text{Al}_x\text{Ga}_{1-x}\text{As}$ waveguides is comprehensively studied in section 5.3.4.

2.3.2 Sum-frequency generation

The process of sum-frequency generation (SFG) involves the interaction of a pump (p) at ω_p, a signal (s) at ω_s and a generated sum-frequency wave (SF) at ω_{SF}. The conservation of energy requires that $\omega_{\text{SF}} = \omega_p + \omega_s$. Analysis of SFG is similar to second-harmonic generation. Two mostly examined SFG processes include type-I SFG where both pump and signal are TE-polarized while sum-frequency signal is TM-polarized and type-II

SFG where the polarization states of the pump and signal are orthogonal and the sum-frequency signal has mainly TE polarization state. Consider for example a type-II SFG with TE-polarized pump and TM-polarized signal. Using Eq. (2.16), the coupled-mode equations are expressed as

$$\frac{\partial A_p}{\partial z} = i\kappa_p \xi^* A_s^* A_{\mathrm{SF}} \exp(-i\Delta\beta z) - \frac{\alpha_p}{2} A_p \tag{2.36a}$$
$$\frac{\partial A_s}{\partial z} = i\kappa_s \xi^* A_p^* A_{\mathrm{SF}} \exp(-i\Delta\beta z) - \frac{\alpha_s}{2} A_s \tag{2.36b}$$
$$\frac{\partial A_{\mathrm{SF}}}{\partial z} = i\kappa_{\mathrm{SF}} \xi A_p A_s \exp(i\Delta\beta z) - \frac{\alpha_{\mathrm{SF}}}{2} A_{\mathrm{SF}} \tag{2.36c}$$

where the wavenumber mismatch is given by

$$\begin{aligned} \Delta\beta &= \beta_p + \beta_s - \beta_{\mathrm{SF}} \\ &= 2\pi \left(\frac{n_p}{\lambda_p} + \frac{n_s}{\lambda_s} - \frac{n_{\mathrm{SF}}}{\lambda_{\mathrm{SF}}} \right) \end{aligned} \tag{2.37}$$

Under the assumption that both the amplitudes of the signal and sum-frequency wave vary along the interaction length and assuming no pump depletion, the boundary conditions are expressed as

$$\mathcal{P}_p = |A_p(0)|^2, \quad \mathcal{P}_s(0) = |A_s(0)|^2, \quad \mathcal{P}_{\mathrm{SF}}(0) = 0 \tag{2.38}$$

Assuming propagation losses can be ignored, the solution of the coupled-mode equations become

$$\mathcal{P}_p(L) = \mathcal{P}_p \tag{2.39a}$$
$$\mathcal{P}_s(L) = \mathcal{P}_s(0) \left[\cos^2(g_{\mathrm{SF}} L) + (\frac{\Delta\beta}{2g_{\mathrm{SF}}})^2 \sin^2(g_{\mathrm{SF}} L) \right] \tag{2.39b}$$
$$\mathcal{P}_{\mathrm{SF}}(L) = \frac{\kappa_{\mathrm{SF}}^2 \xi^2 \mathcal{P}_p \mathcal{P}_s(0)}{g_{\mathrm{SF}}^2} \sin^2(g_{\mathrm{SF}} L) \tag{2.39c}$$

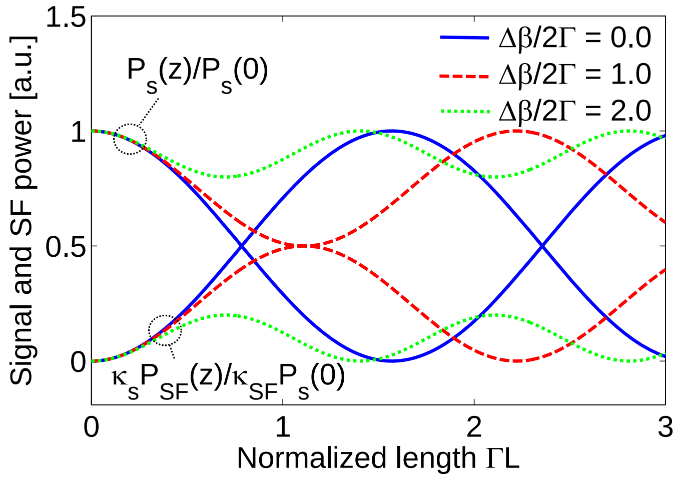

Figure 2.4: Variation of the normalized power of signal, $\mathcal{P}_s(L)/\mathcal{P}_s(0)$, and sum-frequency, $\kappa_s\mathcal{P}_{\mathrm{SF}}(L)/\kappa_{\mathrm{SF}}\mathcal{P}_s(0)$, on normalized interaction length.

where the parameter g_{SF} is defined as

$$g_{\mathrm{SF}} = \sqrt{\Gamma + \left(\frac{\Delta\beta}{2}\right)^2}, \quad \Gamma = \kappa_s\kappa_{\mathrm{SF}}|\xi|^2\mathcal{P}_p \tag{2.40}$$

Figure 2.4 illustrates the dependency of the signal and sum-frequency powers on the normalized interaction length, ΓL, for various phase-matching condition. From the figure, it can be seen that $\mathcal{P}_s(L)$ and $\mathcal{P}_{\mathrm{SF}}(L)$ are periodic functions of the interaction length. For the case that $\Delta\beta = 0$, the sum-frequency power increases with L to a maximum value of $\omega_{\mathrm{SF}}\mathcal{P}_s(0)/\omega_s$ and decreases afterward. The oscillating behaviour of signal and SF powers can be explained by noticing that a complete cycle of SFG involves two consecutive parametric processes [28, 31]. The first process is the sum-frequency between ω_p and ω_s which gives rise to the depletion of signal. At this point no ω_s photons exist for further sum-frequency generation. The second process involves the regeneration of signal photons through difference-frequency generation of $\omega_s = \omega_{\mathrm{SF}} - \omega_p$ which depletes the SF wave. These complementary parametric processes are then repeated along the interaction length giving rise to the oscillation of ω_p and ω_{SF} frequencies. Also from Fig. 2.4, it can be observed that the maximum intensities of the generated waves decrease as the $\Delta\beta$

increases. Using Eq. (2.39), the power of the signal and SF are inversely proportional to g^2. Hence, as $\Delta\beta$ increases the intensity of generated waves decrease by a factor of $\kappa_{\mathrm{SF}}^2/g_{\mathrm{SF}}^2$.

In the low conversion limit where the amplitudes of both pump and signal remain constant, the variation of sum-frequency power, with the presence of propagation losses, is given by [30]

$$\begin{aligned}\mathcal{P}_{\mathrm{SF}}(L) = \kappa_{\mathrm{SF}}^2 \xi^2 L^2 \mathcal{P}_p \mathcal{P}_s(0) \exp\left[-(\alpha_p + \alpha_s + \alpha_{\mathrm{SF}})L/2\right] \\ \times \frac{\sin^2(\Delta\beta L/2) + \sinh^2\left[(\alpha_p + \alpha_s - \alpha_{\mathrm{SF}})L/4\right]}{(\Delta\beta L/2)^2 + \left[(\alpha_p + \alpha_s - \alpha_{\mathrm{SF}})L/4\right]^2}\end{aligned} \tag{2.41}$$

Ignoring harmonic losses, Eq. (2.41) can be further simplified to the well-known sinc function for SF power

$$\mathcal{P}_{\mathrm{SF}}(L) = \kappa_{\mathrm{SF}}^2 \xi^2 L^2 \mathcal{P}_p \mathcal{P}_s(0) \mathrm{sinc}^2(\frac{\Delta\beta L}{2}) \tag{2.42}$$

The conversion efficiency of SFG process is defined as

$$\eta = \frac{\mathcal{P}_{\mathrm{SF}}(L)}{\mathcal{P}_p \mathcal{P}_s(0)} \quad \text{or} \quad \eta_{\mathrm{norm}} = \frac{\eta}{L^2} \tag{2.43}$$

For a SFG process with $\Delta\beta = 0$, for high conversion limit where the coupling between both signal and sum-frequency wave should be taken into account, the conversion efficiency is an oscillating function of normalized interaction length, ΓL, and reaches the maximum value of $\omega_{\mathrm{SF}}/(\omega_s \mathcal{P}_p)$. On the other hand, at low conversion limit, η has a quadratic relation with the device length, L, and can be increased quadratically.

2.3.3 Difference-frequency generation

Difference-frequency generation involves the interaction of a strong pump at ω_p, a signal at ω_s and a difference frequency wave (DF) at ω_{DF}. The conservation of energy requires that $\omega_{\mathrm{DF}} = \omega_p - \omega_s$. Similar to second-harmonic generation and sum-frequency generation in $\mathrm{Al}_x\mathrm{Ga}_{1-x}\mathrm{As}$, most widely investigated DFG processes are type-I and type-II interactions. Consider a type-II DFG where a TE-polarized pump and TM-polarized signal generate a difference-frequency wave with TE polarization state. The coupled-mode equations are expressed as

$$\frac{\partial A_p}{\partial z} = i\kappa_p \xi A_s A_{\mathrm{DF}} \exp(i\Delta\beta z) - \frac{\alpha_p}{2} A_p \tag{2.44a}$$

$$\frac{\partial A_s}{\partial z} = i\kappa_s \xi^* A_p A_{\mathrm{DF}}^* \exp(-i\Delta\beta z) - \frac{\alpha_s}{2} A_s \tag{2.44b}$$

$$\frac{\partial A_{\mathrm{DF}}}{\partial z} = i\kappa_{\mathrm{DF}} \xi^* A_p A_s^* \exp(-i\Delta\beta z) - \frac{\alpha_{\mathrm{DF}}}{2} A_{\mathrm{DF}} \tag{2.44c}$$

where $\Delta\beta$ is the DFG wavenumber mismatch defined as

$$\begin{aligned} \Delta\beta &= \beta_p - \beta_s - \beta_{\mathrm{DF}} \\ &= 2\pi \left(\frac{n_p}{\lambda_p} - \frac{n_s}{\lambda_s} - \frac{n_{\mathrm{DF}}}{\lambda_{\mathrm{DF}}} \right) \end{aligned} \tag{2.45}$$

Under the no pump-depletion approximation the boundary conditions of DFG process are given as

$$\mathcal{P}_p = |A_p(0)|^2, \quad \mathcal{P}_s(0) = |A_s(0)|^2, \quad \mathcal{P}_{\mathrm{DF}}(0) = 0 \tag{2.46}$$

Ignoring propagation losses, the power of harmonics involved in the DFG process are calculated according to

$$\mathcal{P}_p(L) = \mathcal{P}_p \tag{2.47a}$$

$$\mathcal{P}_s(L) = \mathcal{P}_s(0) \left[\cosh^2(g_{\mathrm{DF}} L) + (\frac{\Delta\beta}{2g_{\mathrm{DF}}})^2 \sinh^2(g_{\mathrm{DF}} L) \right] \tag{2.47b}$$

$$\mathcal{P}_{\mathrm{DF}}(L) = \frac{\kappa_{\mathrm{DF}}^2 \xi^2 \mathcal{P}_p \mathcal{P}_s(0)}{g_{\mathrm{DF}}^2} \sinh^2(g_{\mathrm{DF}} L) \tag{2.47c}$$

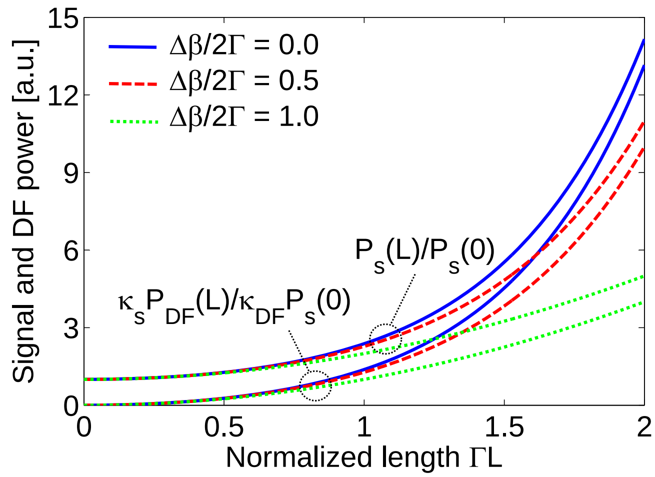

Figure 2.5: Dependency of normalized signal power, $\mathcal{P}_L/\mathcal{P}_0$, and DF power ,$\kappa_s\mathcal{P}_{\rm DF}(L)/\kappa_{\rm DF}\mathcal{P}_s(0)$, on normalized interaction length.

where the DFG gain parameter $g_{\rm DF}$ is defined as

$$g_{\rm DF} = \sqrt{\Gamma - \left(\frac{\Delta\beta}{2}\right)^2}, \quad \Gamma = \kappa_s\kappa_{\rm DF}|\xi|^2\mathcal{P}_p \tag{2.48}$$

The relative powers of signal and difference-frequency as functions of normalized device length, ΓL, are plotted in Fig. 2.5 for different values of normalized wavenumber mismatch, $\Delta\beta/2\Gamma$. From the figure, for real values of $g_{\rm DF}$ where $\Gamma > \Delta\beta/2$, both signal and difference-frequency experience monotonic growth as the interaction length is increased. For $\Gamma L >> 1$, the growth of the signal and DF become asymptotically exponential. The monotonic growth of the signal wave is regarded as the parametric amplification which simultaneously takes place with the DFG process. For $\Gamma < \Delta\beta/2$, the value of $g_{\rm DF}$ becomes imaginary and the hyperbolic functions in Eqs. (2.47) transform to sinusoidal functions. In this case, signal and DF wave oscillate along the interaction length and no parametric amplification takes place. For the DFG process the conversion efficiency is defined as

$$\eta = \frac{\mathcal{P}_{\rm DF}(L)}{\mathcal{P}_p\mathcal{P}_s(0)} \quad \text{or} \quad \eta_{\rm norm} = \frac{\eta}{L^2} \tag{2.49}$$

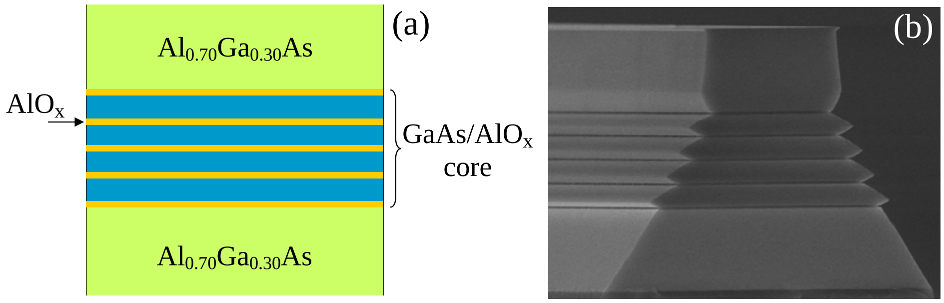

Figure 2.6: Schematic of a form birefringence phase-matched waveguide (a) and the scanning-electron-microscope image of the partially oxidized waveguide [5].

Using Eqs. (2.47), the conversion efficiency of DFG process can increase substantially with the increase of the device length. This in contrast to the SFG process where the conversion efficiency is limited to $\omega_{\mathrm{SF}}/(\omega_s \mathcal{P}_p)$ [28].

2.4 Phase-Matching Techniques

Compound III-V semiconductors have zinc blende crystal structure and are optically isotropic and lacking in natural birefringence. This renders the phase-matching $\chi^{(2)}$ process in semiconductors extremely challenging. To date, several engineering techniques have been incorporated to achieve phase-matching in $Al_xGa_{1-x}As$. Examples include quasi phase-matching (QPM) [32, 22, 29], artificial birefringence (BPM) [20], modal phase-matching (MPM) [26] and phase-matching using Bragg reflection waveguides. A short overview of each technique is discussed here.

2.4.1 Artificial birefringence phase-matching (BPM)

Form birefringence phase-matching has been the most successful technique to achieve PM in $Al_xGa_{1-x}As$ waveguides. A schematic representative of a BPM device along with a SEM of a recently characterized waveguide is shown in Fig. 2.6 [5]. BPM is obtained by using the artificial birefringence introduced in a multilayer core composed of GaAs and thin AlAs (AlO_x) layers. The technique was initially proposed by van der Ziel in

1975 [19], however, its first illustration could not be realized till 1997 when fabrication techniques of lateral selective oxidation for implanting the oxide layers with acceptable quality became feasible.

The principle of BPM relies on breaking the symmetry of GaAs with thin oxide layers. As mentioned earlier, GaAs belongs to $4\bar{3}m$ point symmetry group. For the [001] growth direction, implantation of AlO_x layers along with GaAs, results in breaking the 3-fold rotation axes of bulk GaAs, hence transforming the point symmetry group of the multilayer core to $4\bar{2}m$. The transformed symmetry of $4\bar{2}m$ is identical to that of the KDP crystal [5]. In this regard, BPM waveguide inherits optical nonlinear properties of GaAs while in linear regime it shows the birefringence properties of KDP.

In a form birefringence waveguide, the guided mode has an effective mode index which strongly depends on the polarization state of the involved harmonics. Due to high refractive index contrast between the AlO_x layers ($n \approx 1.6$) and GaAs ($n \approx 3.3$), the modal effective index between TE and TM polarizations are generally large enough to attain phase-matching for a particular nonlinear interaction. BPM waveguides have been successfully illustrated for PM of second-harmonic generation [33], difference-frequency generation [34], parametric fluorescence [35] and generation of photon-pairs through spontanous parametric down-conversion (SPDC) [36]. However, despite exceptional characteristics of BPM devices, these frequency conversion elements impose serious challenges once integration with active devices is of interest. For example, in an integrated OPO where both pump and nonlinear frequency mixing section should be monolithically integrated on the same wafer, the presence of oxide layers severely undermines the current injection of the laser diode pump. Also, the necessity of incorporating AlO_x layers with high absorption above 7.5 μm along with the absorption of GaAs above 870 nm limits the operating window of BPM devices for infrared generation [23].

2.4.2 Quasi phase-matching

Quasi phase-matching is an alternative to exact phase-matching which relies on periodically resetting the accumulated phase-mismatch between the harmonics [21]. QPM in $Al_xGa_{1-x}As$ waveguides is usually obtained by either periodically modulating the sign (domain-reversal QPM) [22, 29, 37, 38, 39, 40] or the magnitude (domain-suppressed QPM) [32, 41, 42, 43] of the nonlinear coefficient. The maximum growth rate of the generated harmonics can be achieved using domain-reversal QPM (DR-QPM) where the sign of $\boldsymbol{\chi}^{(2)}$ coefficient changes from positive to negative every coherence length, which guarantees constructive superposition of generated harmonics within each coherence length. In domain-suppressed QPM (DS-QPM), the magnitude of $\boldsymbol{\chi}^{(2)}$ coefficient periodically varies between a high and low value. While power coupling between pump and generated harmonic is carried out within one coherence length with high nonlinearity, no coupling takes place in the low nonlinear region. As a result in phase superposition of the generated harmonics only occurs in every other coherent length and the efficiency reduces in comparison to DR-QPM. A graphical comparison among exact phase-matching using BPM and quasi-phase matching techniques for second-harmonic generation is illustrated in Fig. 2.7. In addition to lower conversion efficiency of QPM devices in comparison to BPM counterparts, quasi-phase matched structures additionally suffer from fabrication complexities involved in writing QPM periods.

2.4.3 Modal phase-matching

Modal phase-matching (MPM) exploits the modal dispersion properties of guided modes to achieve exact phase-matching [26, 27]. Due to large modal dispersions in $Al_xGa_{1-x}As$ waveguides, MPM cannot be achieved between the lowest order modes of the interacting frequencies. As such, MPM is generally achieved between the lowest order mode of one harmonic and higher order modes of other harmonics. For example in [26], MPM has been demonstrated for type-I second-harmonic generation between TE_0 pump and TM_2

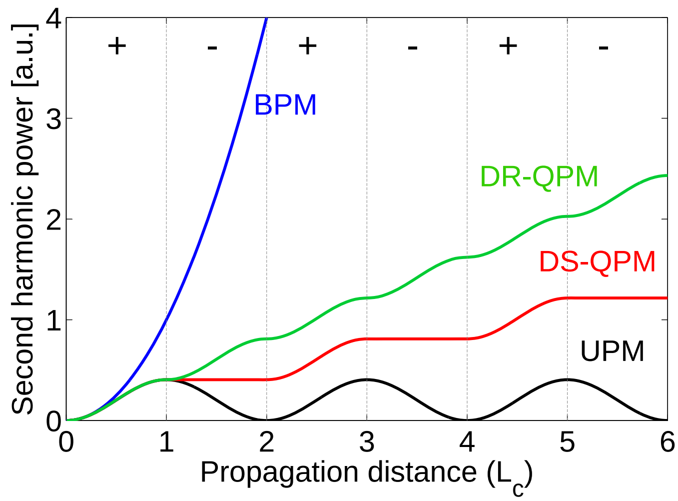

Figure 2.7: Second-harmonic power as a function of propagation distance for exact phase-matching, domain-reversal QPM, domain-suppressed QPM and unphase-matched process. The propagation distance is expressed in term of coherence length $L_c = \pi/\Delta\beta$. The + and − signs denote the sign of the $\boldsymbol{\chi}^{(2)}$ coefficient.

second-harmonic mode. Also, type-II SHG has been demonstrated where phase-matching is achieved between TE_0+TM_0 pump and TE_2 second-harmonic [26]. Unlike BPM devices, MPM waveguides benefit from fabrication simplicity and being more susceptible for monolithic integration with electrically pumped diode laser sources. However, the nonlinear conversion efficiency in these structures is generally low as a result of low nonlinear overlap factor as well as the large propagation losses adhered to high order optical modes.

2.4.4 Phase-matching using Bragg reflection waveguides

Phase-matching using Bragg reflection waveguides is an exact modal phase-matching technique which uses the strong modal dispersion properties in photonic bandgap structures [24, 25]. In this technique, the coupled frequencies propagate as different kinds of optical modes. For example in a second-harmonic generation, the pump mode is a bounded mode where its propagation relies on total internal reflection while the SH is a leaky mode where its propagation is carried out through Bragg reflections from periodic claddings. BRWs offer phase-matching between the lowest order modes hence the nonlinear interaction can benefit from maximum power utilization of the interacting harmonics. BRWs are attractive for applications where tailoring the dispersion of frequencies is desired [44]. As the operation of BRWs is based on cavity resonances, Bragg modes generally are more dispersive compare to their TIR counterparts. This provides additional flexibility in controlling the dispersion of different harmonics to the desired value.

BRWs have been successfully illustrated for phase-matching second-harmonic generation [6, 7, 9], sum-frequency generation [10] as well as difference-frequency generation [11]. Detailed examination of phase-matched BRWs will be discussed in the following chapters.

2.4.5 Other phase-matching techniques

There are several research groups that have investigated other phase-matching techniques in zinc blende semiconductors. For example, Huang *et al.* [45] have proposed directional couplers for obtaining quasi phase-matching. In this technique, unlike conventional QPM which requires spatial periodic modulation of the nonlinear coefficient, phase-matching is obtained using spatial periodic modulation of the light intensity along the propagation direction in a directional coupler. The method was experimentally illustrated by Dong [46] for frequency doubling of CW telecommunication wavelength. However, no further improvements have been reported since the initial illustrations. Another technique proposed by Di Falco *et al.* [47] uses slotted air-clad waveguide for phase-matching second-harmonic generation. The method lends itself to the artificial birefringence formed by the large index contrast between $Al_xGa_{1-x}As$ and a nanometre scale air gap in suspended rod waveguides. In [48], slot waveguides are further investigated for difference-frequency generation in mid-infrared, however no characterization results either for SHG or DFG have yet reported. In a recently proposed design [49], a device consisting of two coupled 2-D photonic crystal waveguides with a row of micro cavities tuned at the second-harmonic frequency to assist selective coupling is used. And finally, several phase-matching techniques in various resonator structures have been proposed [50, 51, 52] which rely on the resonance effects in artificially structured materials to enhance the intensity of light and light-matter interaction, hence the increased efficiency of the nonlinear interaction.

Chapter 3

Planar Bragg Reflection Waveguides

In this chapter, we discuss the guiding properties of planar Bragg reflection waveguides (BRWs) or one-dimensional photonic crystals [53, 54, 55]. Optical waveguiding in BRWs relies on Bragg reflections from periodic claddings. In this class of waveguides, propagation can take place in a core layer with a low material index. This is in contrast to the propagation of bounded modes, where waveguiding relies on total internal reflection (TIR) between a high index core and low index claddings. Optical modes in BRWs are generally leaky in nature, indicating that the propagation constant has both real and imaginary components. Our numerical analysis here for determining the propagation constant in BRWs is based on the transfer-matrix method (TMM) [56, 57, 58, 59] and on Block-Floquet formalism [60, 61, 62].

3.1 Transfer Matrix Method

A planar multi-layer dielectric waveguide is schematically illustrated in Fig. 3.1. The optical field is assumed to propagate along the z-axis, with no fields variations along the y-axis ($\partial/\partial y = 0$). General solutions of the electromagnetic field are taken as

$$\psi_\omega(x, y, z, t) = \psi_\omega(x) \exp[i(\beta z - \omega t)] \tag{3.1}$$

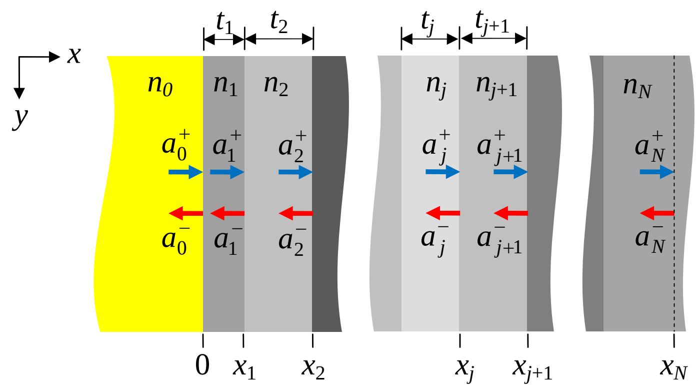

Figure 3.1: Schematic of a multilayer dielectric structure. The arrows in each layer indicate the field amplitudes of the counter propagating plane waves in the transverse direction. The EM fields variations along the y-axis are assumed to vanish. The propagation direction is taken along the z-axis.

where $\psi_\omega(x) = E^y_\omega(x)$ for TE propagation with non-zero field components $(H^x_\omega, E^y_\omega, H^z_\omega)$ while $\psi_\omega(x) = H^y_\omega$ for TM propagation with field components $(E^x_\omega, H^y_\omega, E^z_\omega)$. The function $\psi_\omega(x)$ denotes the transverse mode profile which satisfies the scalar Helmholtz equation

$$\frac{d^2\psi_\omega(x)}{dx^2} + k_0^2[n^2_\omega(x) - n^2_{\text{eff},\omega}]\psi_\omega(x) = 0. \tag{3.2}$$

In Eq. (3.2), $n^2_\omega(x)$ indicates the variation of the refractive index of the multilayer dielectric structure, which is a piecewise continuous function of x. The field distribution within each layer can then be expressed as the superposition of the right $(+x)$ and left $(-x)$ propagating plane waves. Using Maxwell's equations for TE propagation, the field distribution for tangential components, E^y_ω and H^z_ω, in the j-th layer is written as

$$E^y_{\omega,j}(x) = a^+_j \exp[ik^x_j(x - x_j)] + a^-_j \exp[-ik^x_j(x - x_j)] \tag{3.3a}$$

$$H^z_{\omega,j}(x) = \frac{k^x_j}{\omega\mu_0}\left[a^+_j \exp[ik^x_j(x - x_j)] - a^-_j \exp[-ik^x_j(x - x_j)]\right] \tag{3.3b}$$

where a_j^+ and a_j^- are the amplitudes of the right and left propagating fields in the j-th layer and k_j^x is the transverse wavevector defined as

$$k_j^x = k_0\sqrt{n_j^2 - n_{\text{eff}}^2} \tag{3.4}$$

For TM propagation, the tangential field components are expressed as

$$H_{\omega,j}^y(x) = a_j^+ \exp[ik_j^x(x - x_j)] + a_j^- \exp[-ik_j^x(x - x_j)] \tag{3.5a}$$

$$E_{\omega,j}^z(x) = \frac{k_j^x}{\omega n_j^2}\left[-a_j^+ \exp[ik_j^x(x - x_j)] + a_j^- \exp[-ik_j^x(x - x_j)]\right] \tag{3.5b}$$

For TE polarization, the field solutions in the j-th and $(j+1)$-st adjacent layers satisfy the boundary conditions of continuity of $E_{\omega,j}^y(x_j)$ and $H_{\omega,j}^z(x_j)$. For TM polarization, the continuity of $H_{\omega,j}^y(x_j)$ and $E_{\omega,j}^z(x_j)$ should be satisfied. Applying the appropriate boundary conditions, the forward transfer matrix which relates the amplitudes of the right and left propagating waves in the j-th layer to those of the $(j+1)$-st layer is expressed as

$$\begin{bmatrix} a_{j+1}^+ \\ a_{j+1}^- \end{bmatrix} = \mathbf{M}_{j+1,j}\begin{bmatrix} a_j^+ \\ a_j^- \end{bmatrix} \tag{3.6}$$

$$\mathbf{M}_{j+1,j} = \frac{1}{2}\begin{bmatrix} (1+f_{\mathrm{P}})\exp(+i\phi_{j+1}) & (1-f_{\mathrm{P}})\exp(+i\phi_{j+1}) \\ (1-f_{\mathrm{P}})\exp(-i\phi_{j+1}) & (1+f_{\mathrm{P}})\exp(-i\phi_{j+1}) \end{bmatrix} \tag{3.7}$$

where $\mathrm{P} = \{\mathrm{TE}, \mathrm{TM}\}$ is the subscript used for indicating the polarization state of the propagating field, $\phi_j = k_j^x d_j$ is the accumulated phase in the j-th layer and

$$f_{\mathrm{P}} = \begin{cases} \dfrac{k_j^x}{k_{j+1}^x} & \text{for TE polarization} \\ \dfrac{n_{j+1}^2 k_j^x}{n_j^2 k_{j+1}^x} & \text{for TM polarization} \end{cases} \tag{3.8}$$

Similarly, the backward transfer matrix which relates the field amplitudes in the $j+1$-th layer to those in the j-th layer is given by

$$\begin{bmatrix} a_j^+ \\ a_j^- \end{bmatrix} = \mathbf{M}'_{j,j+1} \begin{bmatrix} a_{j+1}^+ \\ a_{j+1}^- \end{bmatrix} \tag{3.9}$$

$$\mathbf{M}'_{j,j+1} = \frac{1}{2} \begin{bmatrix} (1+f'_{\mathrm{P}})\exp(-j\phi_j) & (1-f'_{\mathrm{P}})\exp(+j\phi_j) \\ (1-f'_{\mathrm{P}})\exp(-j\phi_j) & (1+f'_{\mathrm{P}})\exp(+j\phi_j) \end{bmatrix} \begin{bmatrix} a_j^+ \\ a_j^- \end{bmatrix} \tag{3.10}$$

where

$$f'_{\mathrm{P}} = \begin{cases} \dfrac{k_{j+1}^x}{k_j^x} & \text{for TE polarization} \\ \dfrac{n_j^2 k_{j+1}^x}{n_{j+1}^2 k_j^x} & \text{for TM polarization} \end{cases} \tag{3.11}$$

Assuming that the amplitudes of counter propagating fields in a given layer are known, Eqs. (3.7)–(3.10) can be used to calculate the field amplitudes in all layers within the structure. Usually, a_0^+ and a_0^- in the first half-space are assumed to be known a priori. The field amplitudes in the j-th layer are then related to a_0^+ and a_0^- according to

$$\begin{bmatrix} a_0^+ \\ a_0^- \end{bmatrix} = \mathbf{M} \begin{bmatrix} a_j^+ \\ a_j^- \end{bmatrix} \quad , \quad \mathbf{M} = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix} = \prod_{l=1}^{j} \mathbf{M}_{l-1,l} \tag{3.12a}$$

$$\begin{bmatrix} a_j^+ \\ a_j^- \end{bmatrix} = \mathbf{M}' \begin{bmatrix} a_0^+ \\ a_0^- \end{bmatrix} \quad , \quad \mathbf{M}' = \begin{bmatrix} m'_{11} & m'_{12} \\ m'_{21} & m'_{22} \end{bmatrix} = \prod_{l=j}^{1} \mathbf{M}'_{l,l-1} \tag{3.12b}$$

Equations (3.12) can be used for calculating the reflection (r) and transmission (t) coefficients of a multilayer dielectric structure. For a waveguide composed on N layers, this is carried out by exciting the stack from the first half-space at $x = 0$ in Fig. 3.1, with a given excitation a_0^+, while the left propagating field amplitude in the N layer is set to zero, $a_N^- = 0$. Using forward transfer matrix formulation, the reflection and transmission coefficients are derived as

$$r = \frac{a_0^-}{a_0^+} = -\frac{m_{21}}{m_{22}} \tag{3.13a}$$

$$t = \frac{a_N^+}{a_0^+} = -\frac{m_{11}m_{22} - m_{12}m_{21}}{m_{22}} \tag{3.13b}$$

Similarly, using backward transfer matrix formulation, the reflection and transmission coefficients are given as

$$r = \frac{a_0^-}{a_0^+} = \frac{m'_{21}}{m'_{11}} \tag{3.14a}$$

$$t = \frac{a_N^+}{a_0^+} = \frac{1}{m'_{11}} \tag{3.14b}$$

The above discussion provide the required analytical tools for the analysis of multi-layer dielectric structures in the most general form, where the dielectric layers do not establish any particular periodicity in the material indices and thicknesses. In applications where such periodicities exist, analysis of the problem can be simplified using Bloch-Floquet formalism.

3.2 Bloch-Floquet Formalism

Consider a simple periodic medium consisting of nonmagnetic bi-layers with refractive indices of n_1 and n_2, associated thicknesses of t_1 and t_2. The periodicity of the structure is denoted as $n(x) = n(x + \Lambda)$, where $\Lambda = t_1 + t_2$ is the lattice constant of the unit cell composed of bi-layers. A schematic of such a periodic structure is shown in Fig. 3.2. The electromagnetic fields in the transverse direction (x-axis) can be written as the

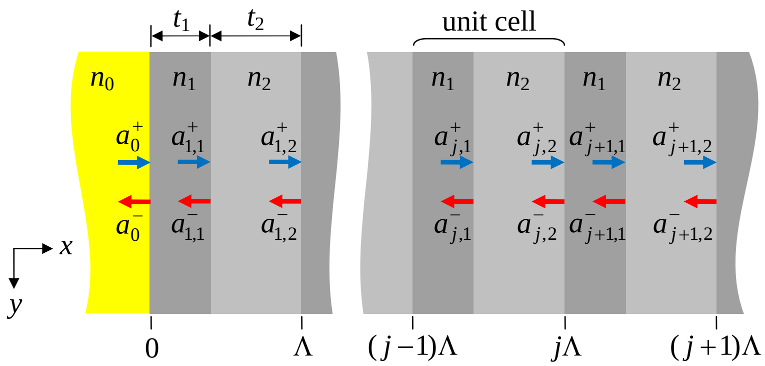

Figure 3.2: Schematic of a periodic dielectric structure with a unit cell composed of bi-layers with refractive indices of n_1 and n_2, associated thicknesses t_1 and t_2, and a lattice constant of $\Lambda = t_1 + t_2$.

superposition of the counter propagating plane waves. For the j-th unit cell

$$\psi_{\omega,j}(x) = \begin{cases} a_{j,2}^{+} \exp[i(k_2^x(x - j\Lambda))] + a_{j,2}^{-} \exp[-i(k_2^x(x - j\Lambda))] & j\Lambda - t_2 < x < j\Lambda \\ a_{j,1}^{+} \exp[i(k_1^x(x - \Delta_j))] + a_{j,1}^{-} \exp[-i(k_1^x(x - \Delta_j))] & (j-1)\Lambda < x < \Delta_j \end{cases} \tag{3.15}$$

where $\Delta_j = j\Lambda - t_2$ and $\psi_{\omega,j}(x) = E_{\omega,j}^{y}(x)$ for TE polarization and $\psi_{\omega,j}(x) = H_{\omega,j}^{y}(x)$ for TM polarization. The field amplitudes between adjacent unit cells, for example the j-th and $(j+1)$-st cells, can be related to each other by applying the boundary conditions of continuity of tangential electromagnetic fields at the interface $x = \Lambda_j$

$$\begin{bmatrix} a_{j,2}^{+} \\ a_{j,2}^{-} \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} a_{j+1,2}^{+} \\ a_{j+1,2}^{-} \end{bmatrix} \tag{3.16}$$

where the $ABCD$ matrix is the unit-cell translation matrix. For TE polarization, the elements of the $ABCD$ matrix are given as [60]

$$A = \exp(-ik_1^x t_1)\left[\cos(k_2^x t_2) + \frac{i}{2}\left(\frac{k_2^x}{k_1^x} + \frac{k_2^x}{k_1^x}\right)\sin(k_2^x t_2)\right] \tag{3.17a}$$

$$B = \exp(+ik_1^x t_1)\left[\frac{i}{2}\left(\frac{k_2^x}{k_1^x} - \frac{k_2^x}{k_1^x}\right)\sin(k_2^x t_2)\right] \tag{3.17b}$$

$$C = B^* \tag{3.17c}$$

$$D = A^* \tag{3.17d}$$

and for TM polarization they are

$$A = \exp(-ik_1^x t_1)\left[\cos(k_2^x t_2) + \frac{i}{2}\left(\frac{n_1^2 k_2^x}{n_2^2 k_1^x} + \frac{n_2^2 k_1^x}{n_1^2 k_2^x}\right)\sin(k_2^x t_2)\right] \tag{3.18a}$$

$$B = \exp(+ik_1^x t_1)\left[\frac{i}{2}\left(\frac{n_2^2 k_1^x}{n_1^2 k_2^x} - \frac{n_1^2 k_2^x}{n_2^2 k_1^x}\right)\sin(k_2^x t_2)\right] \tag{3.18b}$$

$$C = B^* \tag{3.18c}$$

$$D = A^* \tag{3.18d}$$

In obtaining the fields in each layer, the field amplitudes in the zeroth layer, a_0^+ and a_0^-, are usually assumed to be given. The field amplitudes within the remaining layers are then related to those of the zeroth layer according to

$$\begin{bmatrix} a_0^+ \\ a_0^- \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}^j \begin{bmatrix} a_{j,2}^+ \\ a_{j,2}^- \end{bmatrix} \tag{3.19}$$

Equation (3.19) can be rewritten as

$$\begin{bmatrix} a_{j,2}^+ \\ a_{j,2}^- \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}^{-j} \begin{bmatrix} a_0^+ \\ a_0^- \end{bmatrix} \tag{3.20}$$

Using backward or forward transfer matrices for the j-th unit cell, the field amplitudes $a_{j,1}^{+}$ and $a_{j,1}^{-}$ in n_1 layer can be derived from $a_{j,2}^{+}$ and $a_{j,2}^{-}$. The field profile of the structure can be considered complete once the transverse wavevectors in each layer of a unit cell are known.

The Block-Floquet theorem implies that, in a periodic structure, the solution of the electromagnetic fields can be expressed as a plane wave modulated by a periodic function which has a periodicity equal to the lattice constant. In mathematical terms

$$\psi_\omega(x, y, z, t) = \psi_{K_\omega}(x)\exp(iK_\omega x)\exp[i(\beta z - \omega t)] \tag{3.21}$$

where K_ω is regarded as the Bloch wavenumber, $\exp(iK_\omega x)$ is a plane wave and $\psi_{K_\omega}(x)$ is a periodic function with periodicity Λ [60]. In terms of the field amplitudes, the Block-Floquet theorem denotes that

$$\begin{bmatrix} a_{j+1,2}^{+} \\ a_{j+1,2}^{-} \end{bmatrix} = \exp(iK_\omega\Lambda)\begin{bmatrix} a_{j,2}^{+} \\ a_{j,2}^{-} \end{bmatrix} \tag{3.22}$$

From Eqs. (3.19) and (3.22), the phase factor $\exp(-iK_\omega\Lambda)$ is the eigenvalue of the unit-cell translation matrix which is given by

$$\exp(-iK_\omega\Lambda) = \mathrm{Re}(A) \pm \sqrt{[\mathrm{Re}(A)]^2 - 1} \tag{3.23}$$

with the corresponding eigen vectors are given by

$$\begin{bmatrix} a_{j,2}^{+} \\ a_{j,2}^{-} \end{bmatrix} = \exp(ijK_\omega\Lambda)\begin{bmatrix} a_0^{+} \\ a_0^{-} \end{bmatrix} = \exp(ijK_\omega\Lambda)\begin{bmatrix} B \\ \exp(-iK_{\omega\Lambda}A) \end{bmatrix} \tag{3.24}$$

Equation (3.24) provides the dispersion relation in terms of the angular frequency ω, the longitudinal wavevector and the Bloch wavevector

$$K_\omega = \frac{1}{\Lambda}\cos^{-1}\left[\mathrm{Re}(A)\right] \tag{3.25}$$

The magnitude of Re(A) determines three distinct regimes for the propagation of the Bloch wave [60]. The regimes where $\mathrm{Re}(A) < 1$ corresponds to a propagating Bloch wave where K_ω is real. On the other hand, for regimes where $\mathrm{Re}(A) > 1$, the Bloch wavenumber should be imaginary: $K_\omega = \mathrm{Re}(K_\omega) + \mathrm{Im}(K_\omega)$. In this case, $\mathrm{Re}(K_\omega)$ corresponds to a propagating plane wave while $\mathrm{Im}\{K_\omega\}$ denotes the existence of an evanescent field along the x-axis. The regimes where $\mathrm{Re}(A) > 1$ determine the photonic bandgaps or stopbands of the periodic structure. Finally, the regime for which $\mathrm{Re}(A) = 1$ define the band edges of the one-dimensional photonic crystal.

3.3 Slab Bragg Reflection Waveguide

The design of periodic dielectric structures offers unprecedented opportunities for the implementation of novel photonic devices [61]. An example of such devices is the Bragg reflection waveguide (BRW) which is composed of a core layer surrounded by two semi-infinite periodic claddings or transverse Bragg reflectors (TBRs) [53]. Some of the explored applications of BRWs include polarization discremination [63, 64], dispersion compensation [65], mode filtering [66], electron accelerators [67, 68] and soliton propagation [69].

A schematic of a planar BRW is shown in Fig. (3.3). Waveguiding in BRWs relies on Bragg reflections at the interfaces between the core and the periodic claddings. As a result, it is possible to confine and propagate the optical radiation in a core with a low refractive index such as air. BRWs additionally exhibit unique characteristics, including low-loss propagation and versatile dispersion properties [44, 70]. In the next section, we discuss the modal properties of an important class of BRWs, referred to as quarter-wave

BRWs, where the bi-layers of periodic claddings are quarter-wave-thick at the operating wavelength [71].

3.3.1 Quarter-wave Bragg reflection waveguide

The design of the Bragg mirrors in a BRW significantly influences the performance of these devices for a particular application. Usually, it is desired to design a BRW with highest optical confinement and lowest leakage loss. In this case, one can define the design specification for maximizing the reflectivity of the Bragg reflectors at the operating wavelength. The simplest approach for enhancing the reflectivity of TBRs is employing quarter-wave Bragg reflectors. These are periodic claddings where the thickness of each layer is one quarter of the operating wavelength with respect to the transverse wavevector. Equivalently, the quarter-wave condition is expressed by the condition that the accumulated phase in each layer of the Bragg reflectors is equal to an odd integer of $\pi/2$. In mathematical form,

$$k_1^x t_1 = (2m_1+1)\frac{\pi}{2} \Rightarrow t_1 = \frac{(2m_1+1)\pi}{k_0\sqrt{n_1^2 - n_{\text{eff}}^2}} \tag{3.26a}$$

$$k_2^x t_2 = (2m_2+1)\frac{\pi}{2} \Rightarrow t_2 = \frac{(2m_2+1)\pi}{k_0\sqrt{n_2^2 - n_{\text{eff}}^2}} \tag{3.26b}$$

where m_1 and m_2 are positive integers. It is well known that the quarter-wave condition sets the imaginary part of Bloch wavenumber, $\text{Im}(K_\omega)$, at the center of a photonic forbidden band, where it has the largest possible value for the specific bandgap [61]. This translates to maximum evanescent field decay within the Bragg reflectors. A comprehensive analysis of QtW-BRWs, using Bloch-Floquet formalism can be found in [71], where analytical expressions for the modal dispersion of Bragg modes were derived. Here, we adopt a different approach which relies on the unique field transitions at the interface between the core and periodic claddings. These field transitions simply denote that, in BRWs with quarter-wave reflectors, the electromagnetic field either vanishes (node) or

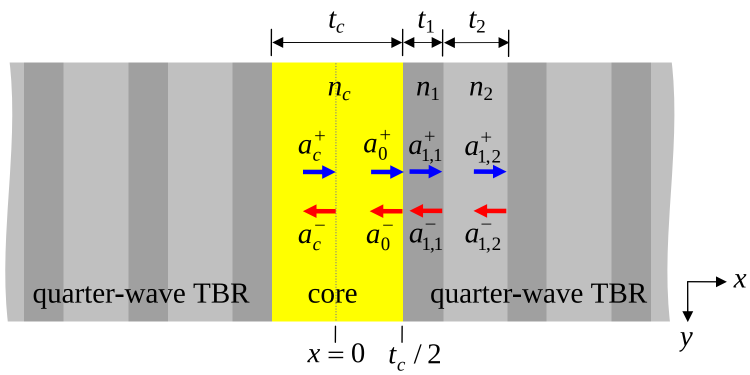

Figure 3.3: Schematic of a slab Bragg reflection waveguide. The core with thickness t_c and refractive index n_c is surrounded by transverse Bragg reflectors on each side. The bi-layers of reflectors consist of layers with refractive indices n_1 and n_2 with associated thicknesses t_1 and t_2. The optical wave propagates along the z-axis.

peaks (anti-node) at the interfaces between two adjacent layers [72]. In the former case, the quarter-wave stack acts as a perfect electric wall, where the tangential component of the electric field vanishes at the interface between core and claddings. In contrast, the latter case resembles a perfect magnetic wall where the first order derivative of the electric field vanishes at core-cladding interface.

Due to the symmetry in physical properties of the structure, one can assume that the propagating modes in the QtW-BRW shown in Fig. 3.3 are either even or odd modes. For optical frequency mixing, we are particularly interested in the lowest order even Bragg mode, where the electric field has its maximum at the core center and the confinement factor is the largest. As such, we disregard the analysis of higher order Bragg modes and focus only on the fundamental even mode.

From Fig. 3.3, for even modes, the amplitudes of the right and left propagating plane waves within the core must be equal. This indicates that $a_c^+ = a_c^-$. Also, without loss of generality, the overall field amplitude at the core center is taken to be unity; hence, $a_c^+ = a_c^- = 1/2$. Using the transfer matrix method, the field amplitudes at the interface of core-cladding become

$$\begin{bmatrix} a_0^+ \\ a_0^- \end{bmatrix} = \begin{bmatrix} \exp(+ik_c^x t_c/2) & 0 \\ 0 & \exp(-ik_c^x t_c/2) \end{bmatrix} \begin{bmatrix} a_c^+ \\ a_c^- \end{bmatrix} = \frac{1}{2} \begin{bmatrix} \exp(+ik_c^x t_c/2) \\ \exp(-ik_c^x t_c/2) \end{bmatrix} \tag{3.27}$$

For the electric field to have a node at $x = t_c/2$, the right and left propagating fields should cancel out each other at the core-cladding interface. As such, $a_0^+ + a_0^- = 0$. Using Eq. (3.27), we obtain the modal dispersion equation of even Bragg modes in a QtW-BRW as

$$\frac{k_c^x t_c}{2} = (2p+1)\frac{\pi}{2} \tag{3.28}$$

In Eq. (3.28), $p = 0, 1, 2, \ldots$ is the mode order, where for the lowest order even mode $p = 0$. Solving Eq. (3.28) for the effective mode index, we obtain

$$n_{\text{eff}} = \sqrt{n_c^2 - \left(\frac{\lambda}{2t_c}\right)^2} \tag{3.29}$$

The above equation determines the range of core thicknesses for which the propagation of the lowest order even Bragg mode is allowed. In BRW with infinite number of periods, n_{eff} should be real and smaller than any material indices within the structure. Using Eq. (3.29), the core thickness should satisfy the condition that

$$t_c > \frac{\lambda}{2n_c} \quad (k_c^x < k_1^x) \tag{3.30}$$

Equations (3.28) and (3.29) are valid regardless of the polarization state of the Bragg mode. A simple consequence is the fact, that in a QtW-BRW, the effective mode indices of TE and TM polarizations are identical [71]. It should be stressed that perfect polarization degeneracy is obtainable for structures with semi-infinite periodic claddings. In practice where the TBRs have a limited number of unit-cells, one can expect a modal birefringence

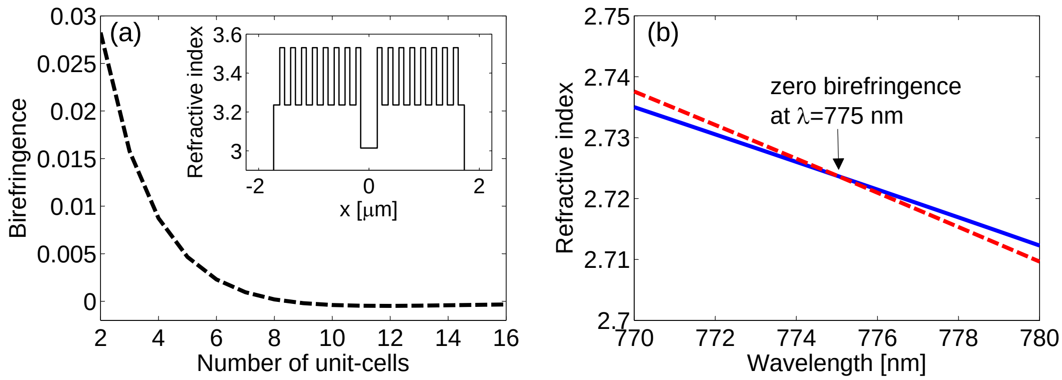

Figure 3.4: (a) Variation of modal birefringence of fundamental modes, $\Delta n = n_{\text{eff}}^{(\text{TE})} - n_{\text{eff}}^{(\text{TM})}$, as a function of the number of unit-cells of quarter-wave Bragg reflectors. (Inset) Index profile of the structure with eight unit-cells. (b) Dependency of effective indices of TE (solid-line) and TM (dashed-line) modes on operating wavelength.

between the TE and TM modes. To illustrate this, in Fig. 3.4, we have plotted the modal birefringence for the fundamental TE_0 and TM_0 modes of a typical QtW-BRW, composed of $\text{Al}_x\text{Ga}_{1-x}\text{As}$ elements as a function of the number of unit-cells. The refractive indices of the bi-layers were taken as $(n_1, n_2) = (3.5306, 3.2360)$, with associated thicknesses of $(t_1, t_2) = (86.2, 110.9)$ nm. The core thickness was $t_c = 300$ nm with an index of $n_c = 3.0145$. The operating wavelength was 775 nm. From Fig. 3.4(a), it can be seen that, for a sufficiently large number of unit-cells, the modal birefringence converges to zero, indicating the degeneracy of the orthogonal polarizations. Also in Fig. 3.4(b), we have shown the dependency of the effective mode indices of TE and TM polarizations as a function of the wavelength. From the figure, the crossing point of the dispersion curves clearly indicates the polarization degeneracy of the QtW-BRW structure at the wavelength of 775 nm.

The effective index and modal dispersion of Bragg modes are functions of waveguide materials and the geometry of the structure. Among different parameters, core characteristics – particularly thickness – have the strongest influence on the mode index. In Fig. 3.5(a), we have plotted the variation of n_{eff} as a function of core thickness in the range of $0.3 - 2.0$ μm for a set of QtW-BRWs, with a core index of $n_c = 3.0145$ and

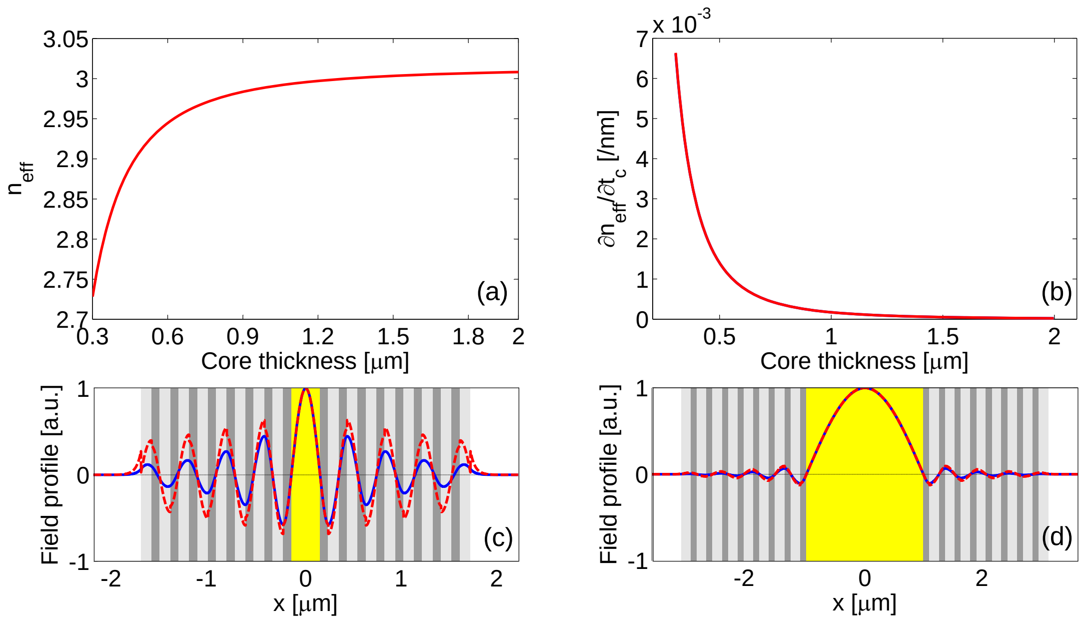

Figure 3.5: (a) Variation of effective index as a function of core thickness for quarter-wave BRW with bi-layers $(n_1, n_2) = (3.5306, 3.2360)$ and a core index of $n_c = 3.0145$. (b) Modal dispersion as a function of core thickness. (c) Field profiles of E^y_ω (solid-line) and E^x_ω (dashed-line) for the QtW-BRW with $t_c = 300$ nm, and (d) those for $t_c = 2.0$ μm.

bi-layer indices of $(n_1, n_2) = (3.5306, 3.2360)$. From the figure, it can be seen that the dependency of n_{eff} on core size is stronger for small core dimensions and converges to the core refractive index as the core thickness is increased. To further highlight the dependency of n_{eff} on t_c, in Fig. 3.5(b), we have plotted the sensitivities of n_{eff} with respect to t_c as a function of t_c for TE and TM polarizations. From the figure, although the mode indices for TE and TM modes are degenerate, waveguide dispersions with respect to core dimension are different for each polarization. The simulation denotes that TE polarization exhibits stronger modal dispersion in this case. Table 3.1 presents the geometry and effective indices of the structures with $t_c = 300$ nm and 2.0 μm. The Bragg reflectors are assumed to have eight periods each. In Figs. 3.5(c)-(d), the field profiles of E^x_ω and E^y_ω for these two designs are plotted. From the figure, the comparison between the mode profiles reveals the fact that, for BRWs with large core size, the oscillating nature of the Bragg mode is significantly suppressed and the near-field of the mode resembles a single-lobe optical mode. In practice, single-lobe field distribution is important in integrated optical circuits, where optical power should be transferred within a cascade of various devices. An example is waveguide coupling into a fiber, where spatial modal overlap between the waveguide mode and the fiber mode should be as close as possible for maximizing the spatial coupling efficiency.

Table 3.1: Quarter-wave BRW design examples

Parameter	QtW-BRW1	QtW-BRW2
λ	775 nm	775 nm
(x_1, n_1, t_1)	(0.20,3.5306,86 nm)	(0.20,3.5306,105 nm)
(x_2, n_2, t_2)	(0.60,3.2360,111 nm)	(0.60,3.2360,162 nm)
(x_c, n_c, t_c)	(1.00,3.0145,300 nm)	(1.00,3.0145,2000 nm)
$n_{\text{eff}}^{(\text{TE})}$	2.7254	3.0083
$n_{\text{eff}}^{(\text{TM})}$	2.7252	3.0083

The quarter-wave BRWs discussed here have symmetric Bragg mirrors, which simplifies their analysis while providing a clear insight into the functionality of this class of waveguides. In practice, it might not be possible to satisfy all design specifications using only symmetric structures. For example, in ultrafast applications, a good versatility in tailoring dispersion properties of Bragg reflectors over a large spectral range is desired, which cannot be easily achieved using QtW-BRW designs. In such cases, other techniques, such as introducing a chirp function in the refractive indices or a chirp in the layers thicknesses, can be considered as potential routes. Chirping the reflectors can also be used for reducing the leakage loss in BRWs [73]. In the next section, we discuss an alternative to symmetric QtW-BRWs, where one of the Bragg stacks is replaced by a bulk cladding while the structure is designed such that the remaining periodic cladding operates at the quarter-wave condition. Such a design idea can be useful, for example, in

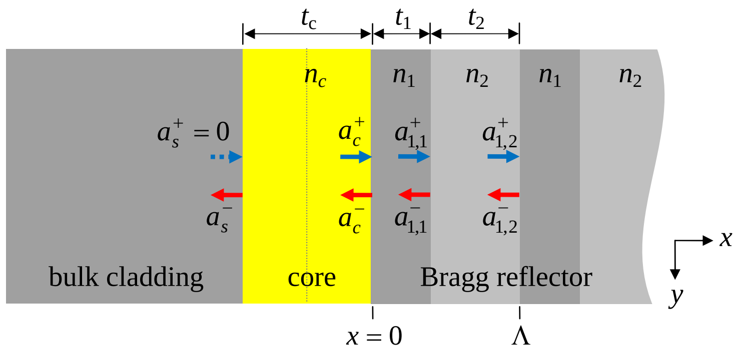

Figure 3.6: Schematic of an asymmetric Bragg reflection waveguide where the core is surrounded by bulk and periodic claddings. The field in the bulk cladding is evanescent with $a_s^+ = 0$.

reducing the overall thickness of BRWs, which facilitates the fabrication of these devices.

3.3.2 Mono-stack Bragg reflection waveguide

A planar Bragg reflection waveguide with a single Bragg mirror along with its index profile is illustrated in Fig. 3.6. We refer to this structure as mono-stack BRW (MS-BRW). Here, the core is surrounded on one side by a periodic cladding and on the other by a single-layer cladding. Waveguiding mechanism in MS-BRW relies on a mixture of Bragg reflections at the interface between the core and the single TBR and total internal reflection at the core single-cladding interface. To simplify the analysis, we assume that the periodic cladding is composed of quarter-wave-thick bi-layers.

Referring to Fig. 3.6, the field within the single-layer cladding is an evanescent field. As such, the amplitudes of the right and left propagating waves within this layer are taken as $a_s^+ = 0, a_s^- = 1$. Modal dispersion equations can then be obtained by applying the field transition at $x = 0$, where the field either vanishes or peaks at the interface of the core and the quarter-wave TBR. These conditions are satisfied by properly designing the thickness of the core.

Using the transfer matrix method, the field amplitudes of the single-layer cladding and those of the core are related according to

$$\begin{bmatrix} a_c^+ \\ a_c^- \end{bmatrix} = \frac{1}{2}\begin{bmatrix} (1+f_{\mathrm{P}})\exp(+j\phi_c) & (1-f_{\mathrm{P}})\exp(+j\phi_c) \\ (1-f_{\mathrm{P}})\exp(-j\phi_c) & (1+f_{\mathrm{P}})\exp(-j\phi_c) \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$
$$= \frac{1}{2}\begin{bmatrix} (1-f_{\mathrm{P}})\exp(+j\phi_c) \\ (1+f_{\mathrm{P}})\exp(-j\phi_c) \end{bmatrix} \tag{3.31}$$

where $\phi_c = k_c^x t_c$ and

$$f_{\mathrm{P}} = \begin{cases} \frac{k_s^x}{k_c^x} & \text{for TE polarization} \\ \frac{n_c^2 k_s^x}{n_s^2 k_c^x} & \text{for TM polarization} \end{cases} \tag{3.32}$$

Using Eqs. (3.31) and (3.32), modal dispersion of the TE polarization for $k_c^x < k_1^x$, when the electric field vanishes at $x = 0$, and for $k_c^x > k_1^x$, when the field peaks at $x = 0$, are expressed as

$$\begin{aligned} \cot(k_c^x t_c) &= -i\frac{k_s^x}{k_c^x} \quad (k_c^x < k_1^x) \\ \tan(k_c^x t_c) &= +i\frac{k_s^x}{k_c^x} \quad (k_c^x > k_1^x) \end{aligned} \tag{3.33}$$

Equations (3.33) can be solved for t_c to calculate the required core thickness, which makes the bi-layers of the TBR quarter-wave-thick. As such,

$$\begin{aligned} t_c &= \tfrac{1}{k_c^x}\left[\cot^{-1}\left(-i\frac{k_s^x}{k_c^x}\right) + p\pi\right] \quad (k_c^x < k_1^x) \\ t_c &= \tfrac{1}{k_c^x}\left[\tan^{-1}\left(+i\frac{k_s^x}{k_c^x}\right) + p\pi\right] \quad (k_c^x > k_1^x) \end{aligned} \tag{3.34}$$

where $p = 0, 1, 2, \ldots$ is the core thickness scaling factor which can be used for scaling the core dimension while maintaining quarter-wave operation of the periodic cladding.

For TM polarization, the required condition to have a node at $x = 0$ is $n_1^2 k_c^x > n_c^2 k_1^x$ and for an anti-node is $n_1^2 k_c^x < n_c^2 k_1^x$. The dispersion equations are obtained as

$$\begin{aligned} \cot(k_c^x t_c) &= -i\frac{n_c^2 k_s^x}{n_s^2 k_c^x} \quad (n_1^2 k_c^x > n_c^2 k_1^x) \\ \tan(k_c^x t_c) &= +i\frac{n_c^2 k_s^x}{n_s^2 k_c^x} \quad (n_1^2 k_c^x < n_c^2 k_1^x) \end{aligned} \tag{3.35}$$

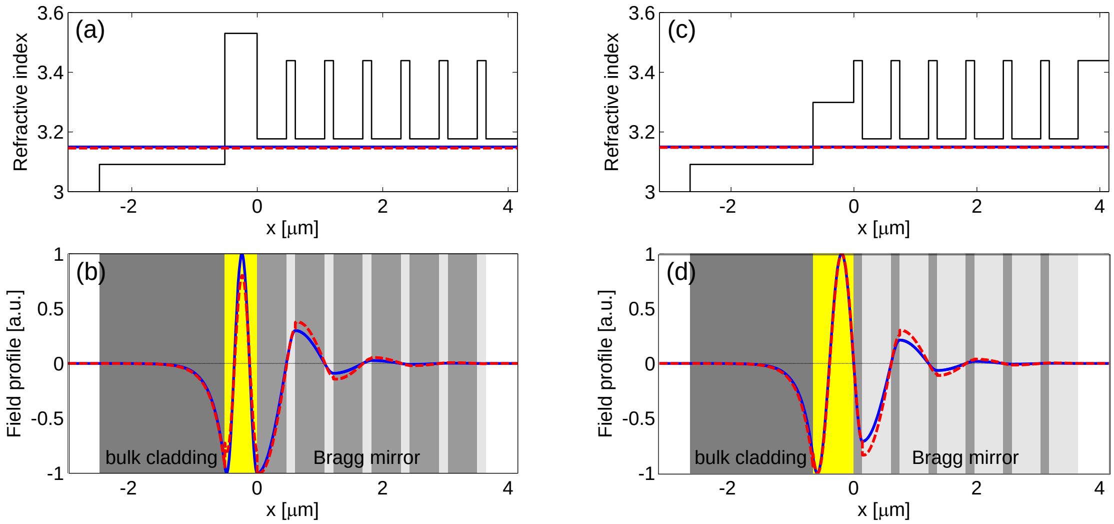

Figure 3.7: (a) Index profile and (b) field profiles of E^y_ω (solid-line) and E^x_ω (dashed-line) of SM-QtW-BRW1. The periodic cladding is designed to operate as a perfect magnetic wall at core-TBR interface. (c) Index profile and (d) field profiles of E^y_ω (solid-line) and E^x_ω (dashed-line) of SM-QtW-BRW2. The periodic cladding forms a perfect electric wall at core-TBR interface.

Solving Eq. (3.35) for core thickness, we obtain

$$\begin{aligned} t_c &= \tfrac{1}{k^x_c}\left[\cot^{-1}\left(-i\frac{n_c^2 k^x_s}{n_s^2 k^x_c}\right)+p\pi\right] \quad (n_1^2 k^x_c > n_c^2 k^x_1) \\ t_c &= \tfrac{1}{k^x_c}\left[\tan^{-1}\left(+i\frac{n_c^2 k^x_s}{n_s^2 k^x_c}\right)+p\pi\right] \quad (n_1^2 k^x_c < n_c^2 k^x_1) \end{aligned} \tag{3.36}$$

Unlike symmetric QtW-BRW, where the core thickness is a continuous function in the range specified by Eq. (3.30), for mono-stack BRW, the core thickness forms a discrete set of values. In Table-3.2, two single mirror BRW designs, SM-QtW-BRW1 and SM-QtW-BRW2, are presented, with their field profiles plotted in Fig. 3.7. Both designs employ identical materials for the bi-layers of the Bragg reflectors. The structure of SM-QtW-BRW1 was designed such that the quarter-wave stack operate as a perfect magnetic boundary at the core-TBR interface. This can be seen in Fig. 3.7(b), where the derivative of the field vanishes at the interface. On the other hand, the structure of SM-QtW-BRW2 was designed such that the Bragg mirror served as a perfect electric wall for which the electric field vanishes at the core-TBR interface (see Fig. 3.7(d)).

Table 3.2: Single mirror BRW design examples

Parameter	SM-QtW-BRW1	SM-QtW-BRW2
λ	770 nm	770 nm
(x_1, n_1, t_1)	(0.70,3.1801,441 nm)	(0.30,3.4444, 138 nm)
(x_2, n_2, t_2)	(0.30,3.4444,138 nm)	(0.70,3.1801, 441 nm)
(x_c, n_c, t_c)	(0.20,3.5381,138 nm)	(0.50,3.3029, 647nm)
(x_s, n_s, t_s)	(0.85,3.0942,2000 nm)	(0.85,3.0942,2000 nm)
$n_{\text{eff}}^{(\text{TE})}$	3.1500	3.1500
$n_{\text{eff}}^{(\text{TM})}$	3.1499	3.1482

3.4 Ridge Bragg Reflection Waveguide

The slab BRW structures discussed in the previous section offer the confinement of optical radiation in only the transfer direction. Many practical photonic devices such as semiconductor lasers, modulators and directional couplers require optical confinement in the lateral direction. In the simplest form, the lateral confinement is obtained using ridge structures.

A typical ridge BRW is shown schematically, in Fig. 3.8, where part of the top Bragg mirror is removed through the etch of the stab. In a ridge BRW, the confinement of light in the growth direction is obtained through Bragg reflections from the periodic claddings. In the region outside the ridge in the transverse direction, optical confinement is obtained by a combination of total internal reflection and Bragg reflection. In the lateral direction parallel to substrate interface, waveguiding completely relies on the total internal reflection. Analysis of ridge structures is much more involved in comparison to the slab waveguides. There is no analytical solution for studying the modal properties

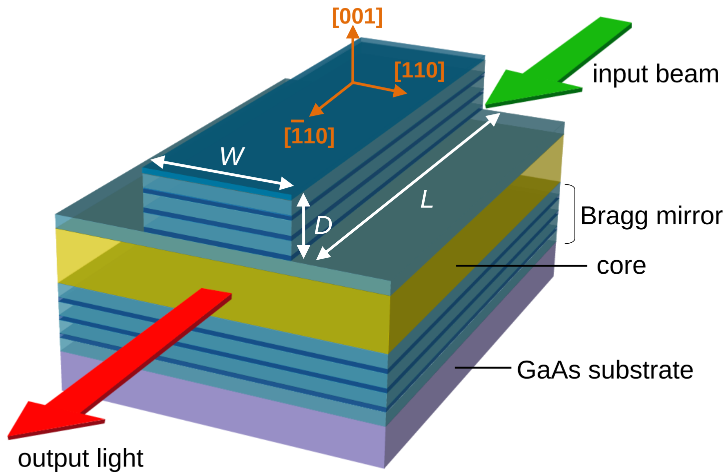

Figure 3.8: Schematic of a $Al_xGa_{1-x}As$ ridge Bragg reflection waveguide. The structure is grown on [001] GaAs substrate. The propagation direction is along the [$\bar{1}$10] crystal axis. Optical confinement in the transverse direction (normal to substrate) relies on Bragg reflections while in the lateral plane (parallel to substrate) it relies on total internal reflection.

of ridge devices. For waveguides with a rectangular cross section, several numerical techniques have been proposed as an approximation. For example, Marcatili has obtained approximate solutions for strip waveguides for optical modes far way from cut-off [74]. Another approximate method that provides relatively good predictions for the modal properties is the effective index method, which transforms the two-dimensional problem of solving the mode index into two one-dimensional planar problems [75, 76, 77]. Owing to the hybrid nature of the waveguiding in BRWs, the approximate numerical techniques fail to provide a fair approximation of the mode indices. As a result, we obtain the mode indices and the dispersion values throughout this work using the semi-vectorial approach proposed in [78] and the commercial full-vectorial mode solvers of Lumerical [12].

In order to address the effect of lateral confinement of Bragg modes through total internal reflection, we simulated the effective indices of a Bragg waveguide, operating at 775 nm, with core and bi-layers refractive indices given in Table 3.1. The core thickness was taken to 800 nm. The variation of the effective indices of orthogonal polarizations as functions of the ridge width, W, are illustrated in Fig. 3.9(a). The ridge depth was

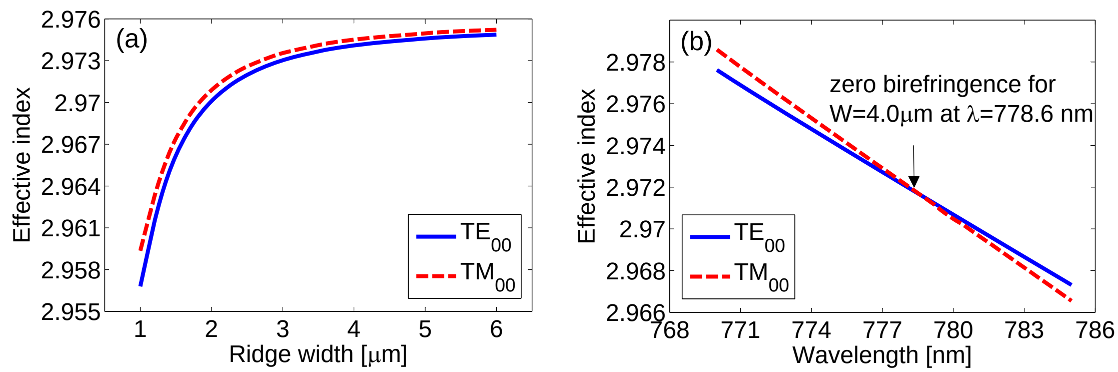

Figure 3.9: (a) Variation of effective index vs. ridge width and (b) variation of effective index vs. wavelength.

set to $D = 2.4$ nm. From the figure, it can be seen that, for small values of W, the effective mode index has a strong dependency on the ridge width. This suggests that the waveguide dispersion is a prominent factor in tailoring the modal dispersion for small ridge widths. Also from the figure, the effect of lateral confinement is violating the polarization degeneracy of orthogonal mode indices, a unique feature which was mentioned in the case of slab QtW-BRWs. As a result, although the bi-layers of the reflectors were initially designed to be quarter-wave thick for the slab case at 775 nm, the TBRs do not operate at the quarter-wave condition in the ridge structure. In Fig. 3.9(b), dispersion of TE_{00} and TM_{00} modes around 775 nm for $W = 4.0$ μm are plotted. The crossing point between the two curves at 778.6 nm denotes the polarization degenerate wavelength for the ridge structure, which is shifted toward longer wavelength compared to that of the slab structure. The intensity profiles of TE_{00} and TM_{00} mode for the design with $(W, D) = (4.0, 2.4)$ μm are shown in Fig. 3.10.

The analysis of BRW structures discussed in this chapter has been used as a tool for designing nonlinear waveguides for optical frequency mixing. In the next chapter, we demonstrate the potentials of BRWs for phase-matching second-order nonlinear interactions as a promising platform for the development of on-chip parametric devices.

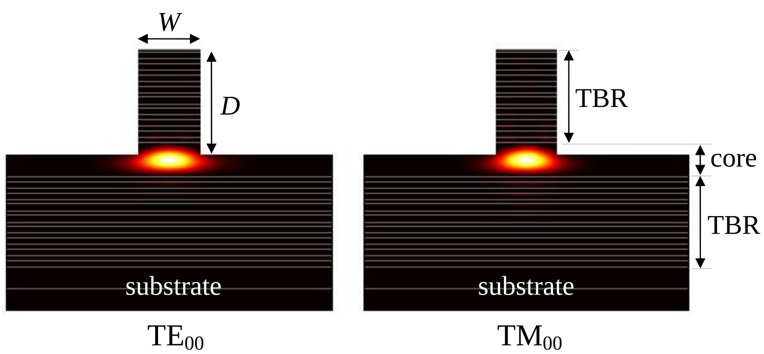

Figure 3.10: Intensity profiles of TE_{00} and TM_{00} Bragg modes of the ridge BRW design given in Table 3.1. The ridge width and its depth are $W = 4.0$ μm and $D = 2.6$ μm, respectively. The core thickness is 800 nm.

Chapter 4

Phase-Matching $\chi^{(2)}$ Nonlinear Processes using BRWs

For efficient optical frequency mixing and harmonic generation, phase-matching is a vital process. Owing to a lack of natural birefringence in bulk zinc blende semiconductors, birefringence phase-matching cannot be exploited in these crystals and other techniques such as quasi phase-matching [21] or artificial birefringence [19] should be employed. This chapter provides a detailed discussion of an alternative phase-matching technique using Bragg reflection waveguides [24, 25]. The method is an exact phase-matching technique which utilizes strong modal dispersion properties in photonic bandgap devices. The material of choice is $Al_xGa_{1-x}As$; however, the method can be generalized to other material systems. The technique is demonstrated by phase-matching second-harmonic generation in both slab and ridge Bragg reflection waveguides.

4.1 Phase-Matching SHG using Slab BRW

Waveguiding in BRWs is carried out through subsequent Bragg reflections at the interface between core and periodic claddings. However, the devices can be designed such that they additionally support the propagation of bounded modes formed by total internal reflection (TIR) between a high index core and periodic claddings with a lower effective index. Referring to Fig. 3.3, this requirement can be easily satisfied by a choice of the core index such that $n_c > \min\{n_1, n_2\}$. The existence of both Bragg and TIR modes in BRWs allows one to design these structures for exact phase-matching of $\chi^{(2)}$ processes. For example, consider type-II second-harmonic generation where a pump with a hybrid polarization of TE+TM at 1550 nm is used for the generation of a TM-polarized second-harmonic signal at 775 nm. In this process the pump is assumed to propagate as a TIR mode while the second-harmonic propagates as a Bragg mode. Figure 4.1 shows the index profiles along with effective mode indices of the involved frequencies for a typical unphase-matched BRW. Using the strong modal dispersion in BRWs one can align the mode indices of the two harmonics such that the phase-matching condition is satisfied. The recipe of designing phase-matched quarter-wave BRWs can be summarized as:

1. For a given nonlinear interaction, determine the type of the optical modes and their polarizations involved in the frequency mixing process. For example in a type-II second-harmonic generation, the pump has both TE and TM components and it propagates as a total internal reflection mode while the second-harmonic is TE-polarized and propagation as a Bragg mode

2. Choose the materials of core, n_c and the bi-layers of the Bragg reflectors, n_1 and n_2, considering material constraints such as absorption, oxidation, ...

3. For a given range of core thickness, $t_c \in [t_{c_1}, t_{c_2}]$, satisfying the condition of Eq. (3.30), calculate the effective index of Bragg mode using Eq. (3.29).

4. Using Eq. 3.28, calculate t_1 and t_2, the thicknesses of the bi-layers of Bragg reflectors.

5. Given the number of periods of bi-layers within each Bragg reflector, construct the structure using (n_c, n_1, n_2) and (t_c, t_1, t_2).

6. For $t_c \in [t_{c_1}, t_{c_2}]$, solve for TE- and TM-polarized total internal reflection modes at the relevant wavelength.

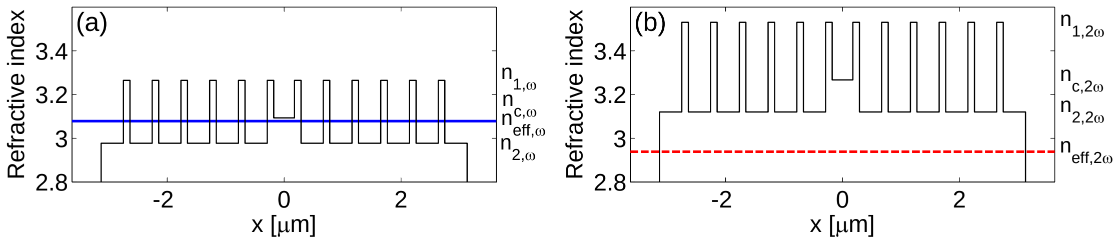

Figure 4.1: (a) Index profile of a TIR pump at frequency ω and its effective mode index (blue solid-line) and (b) those of a Bragg second-harmonic mode at frequency 2ω and its effective mode index (red dashed-line). The structure is unphase-matched resulting in misaligned effective mode indices of pump and second-harmonic signal.

7. Given the dispersion of Bragg mode and total internal reflection modes as a function of t_c, apply the appropriate phase-matching condition to determine the required core thickness for exact phase-matching.

8. Determine the nonlinear properties of the design including nonlinear overlap factor, nonlinear conversion efficiency and structure effective second-order nonlinearity.

9. Define an optimization problem by considering an objective function, such as the nonlinear conversion efficiency, and design parameters such as the material indices. Use a multi-variable optimization method to determine an optimum design for the slab Bragg reflection waveguide.

10. Using a 2-D mode solver, investigate the effect of waveguide ridge width and depth on nonlinear properties such as the detuning of the phase-matching wavelength from the nominal slab case.

To illustrate the method, consider quarter-wave Bragg waveguides with bi-layers composed of $Al_{0.20}Ga_{0.80}As/Al_{0.80}Ga_{0.20}As$ and a core of $Al_{0.55}Ga_{0.45}As$. The Bragg reflectors are assumed to have six periods each. The thicknesses of the layers are unknown as the phase-matched structure is to be found. In Fig. 4.2(a), we have plotted the effective indices of TIR and Bragg modes as a function of core thickness for the range of $320 - 350$ nm. For a given core thickness, the Bragg mirrors are then designed to satisfy the quarter-wave condition. From the figure, the indicated crossing point at $t_c = 351$ nm

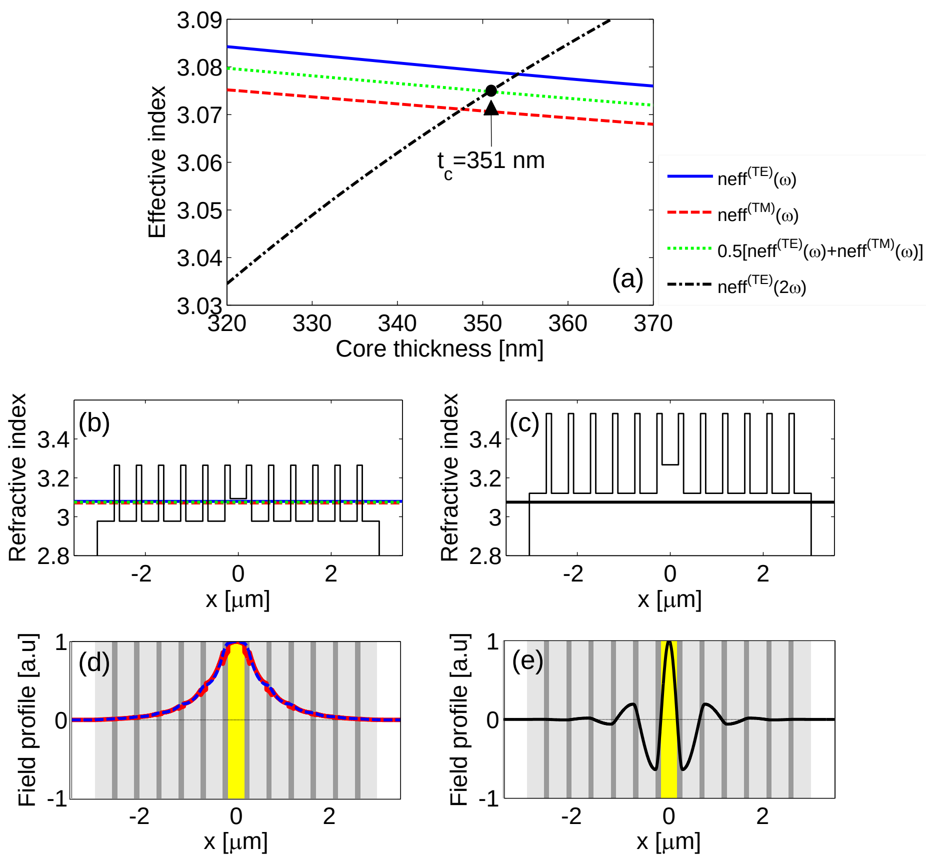

Figure 4.2: (a) Dependency of effective mode indices of pump (ω) and second-harmonic (2ω) on core thickness for the BRW design of Table-4.1. The crossing point at $t_c = 351$ nm denotes the type-II phase-matched structure at $\lambda_p = 1550$ nm. (b) Refractive index profile at pump wavelength and (c) at second-harmonic wavelength. The horizontal lines in each figure indicate the effective indices. (d) Field profiles of TE pump (blue dashed-line), TM pump (red solid-line) and (e) and TE second-harmonic.

denotes the thickness which offers exact phase-matching for the type-II interaction. The refractive index profiles at the pump and second-harmonic frequencies and the associated field profiles for the phase-matched structure are plotted in Figs. 4.2(b)-(e), respectively. Table 4.1 summarizes the pertinent parameters for this design.

Table 4.1: Sample type-II phase-matched Quarter-wave BRW

λ_p [nm]	(x_c, x_1, x_2)	(t_c, t_1, t_2) [nm]	$n_{\text{eff},\omega}^{(\text{TE})}$	$n_{\text{eff},\omega}^{(\text{TM})}$	$n_{\text{eff},2\omega}^{(\text{TE})}$
1550 nm	(0.55,0.20,0.80)	(351,112,365)	3.0790	3.0707	3.0748

The above example highlights the strong effect of the core dimension in aligning the effective mode indices, particularly in aligning the Bragg mode. It might appear that other waveguide parameters, including the material refractive indices, can also be considered as design parameters to obtain phase-matched BRWs. However, modal indices away from material resonances are not a strong function of the refractive index and it is not possible to rely entirely on the material dispersion to achieve PM in BRWs. In practice, one can utilize waveguide dispersion to design an initial phase-matched BRW while using material dispersion as a fine tuning mechanism for additional design optimization.

4.1.1 Pulsed characterization

Table-4.2 summarizes the geometry and material parameters of the first designed and fabricated phase-matched BRW for type-I SHG [25]. We refer to this design as QtW-BRW-SH1. In designing the structure the model of Adachi [79] was used for modeling the material indices. As stated earlier, refractive indices obtained from the Adachi model are less accurate for shorter wavelengths and for low aluminum concentrations in comparison to the more recent model proposed by Gehrsitz [80]. As a result, prior to linear/nonlinear characterizations a considerable mismatch between theory and experiment was anticipated. Also in Table-4.2, material parameters for the structure of QtW-BRW-SH1 redesigned using the Gehrsitz model are listed to provide a comparison between the two refractive index models.

Table 4.2: Type-I phase-matched QtW-BRW-SH1 design. The top and bottom Bragg mirrors had 8 and 10 periods, respectively.

Parameter	Value	Value
Refractive index model	Adachi [79]	Gehrsitz [80]
λ_p	1550 nm	1550 nm
(x_c, x_1, x_2)	(0.40,0.20,0.60)	(0.40,0.20,0.60)
(t_c, t_1, t_2)	(328,118,278) nm	(325,121,259) nm
$n^{(\mathrm{TE})}_{\mathrm{eff},\omega}$	3.2084	3.1483
$n^{(\mathrm{TM})}_{\mathrm{eff},2\omega}$	3.2084	3.1483

As the first illustration of phase-matched BRWs, slab QtW-BRW-SH1 samples were examined for both linear and nonlinear properties [6]. Using a slab arrangement rather than a ridge waveguide arrangement was chosen as the initial step to examine these structures. This helped in the study of the properties of the phase-matching technique while eliminating the tolerances inherent in the fabrication of ridge devices. In ridge waveguides, tolerances such as patterning the waveguides through etching would be convoluted with the introduced tolerances during molecular beam epitaxial growth, which will impede the ability to analyze, precisely, the properties of the phase-matching technique.

The linear propagation loss of the pump was extracted in a free-space optics setup with a tunable telecommunication laser source around 1550 nm at low powers. As the slab structures did not have lateral confinement, it was justifiable that loss estimation could be more effectively performed using the cut-back method instead of the Fabry-Pèrot method [81, 82, 83, 84]. In the cut-back technique, transmission of the device is measured as a function of the sample length, L. The propagation loss (α) and the linear coupling coefficient (C), defined as the spatial overlap factor between the incident beam and the excited mode, can be extracted. To show this, consider an incident pump power of $\mathcal{P}_p(0)$ measured before the front facet of the device and a transmitted power of $\mathcal{P}_p(L)$

measured after the exit facet. The transmission of the device, $\mathcal{P}_p(L)/\mathcal{P}_p(0)$, the facets reflectivity (R), the propagation loss and the coupling factor are related as

$$\mathcal{P}_p(L)/\mathcal{P}_p(0) = C(1-R)^2 \exp(-\alpha_p L) \tag{4.1a}$$

$$\ln\left(\mathcal{P}_p(L)/\mathcal{P}_p(0)\right) = \ln(C) + 2\ln(1-R) - \alpha_p L \tag{4.1b}$$

The above equations imply that the propagation loss and the coupling factor can be determined by plotting the device transmission, in ln scale, as a function of sample length and fitting the data to a linear function. The slope of the fit provides an estimation of the loss value. The facet reflectivity R is, usually, assumed to be given numerically. The interception of the linear fit in Eq. (4.1) then provides the value of $\ln(C)$ which is used for calculating the linear coupling factor.

Using the cut-back technique, the propagation loss of the pump around 1550 nm was estimated to be ≈ 2.5 cm$^{-1} \pm 1.5\%$. Estimation of SH loss was complicated as the waveguide supported total internal reflection modes around SH wavelength in addition to the Bragg mode. As such, measurement of SH linear could not be carried out as it required preferential mode coupling into the Bragg mode using characterization techniques such as prism coupling.

Nonlinear characterization of the device was performed using a singly-resonant KTP OPO synchronously pumped by a mode-locked Ti:sapphire laser. A schematic of the apparatus is illustrated in Fig. 4.3(a). Pulses with a full-width at half-maximum bandwidth (FWHM) of $\approx$ 2-ps with a repetition rate of 76 MHz and an average power of 176 mW, measured before the front facet of the sample, were employed. Pulses centered around 1550 nm were end-fire coupled into a 2.4 mm long sample using a 40$\times$ objective lens. A cylindrical lens with a focal length of 150 mm was used before the input objective lens to laterally expand the beam to form an elliptical Gaussian beam as well as to separate the beam waists in orthogonal planes, as simulated in Figs. 4.3(b)-(d). This was intended

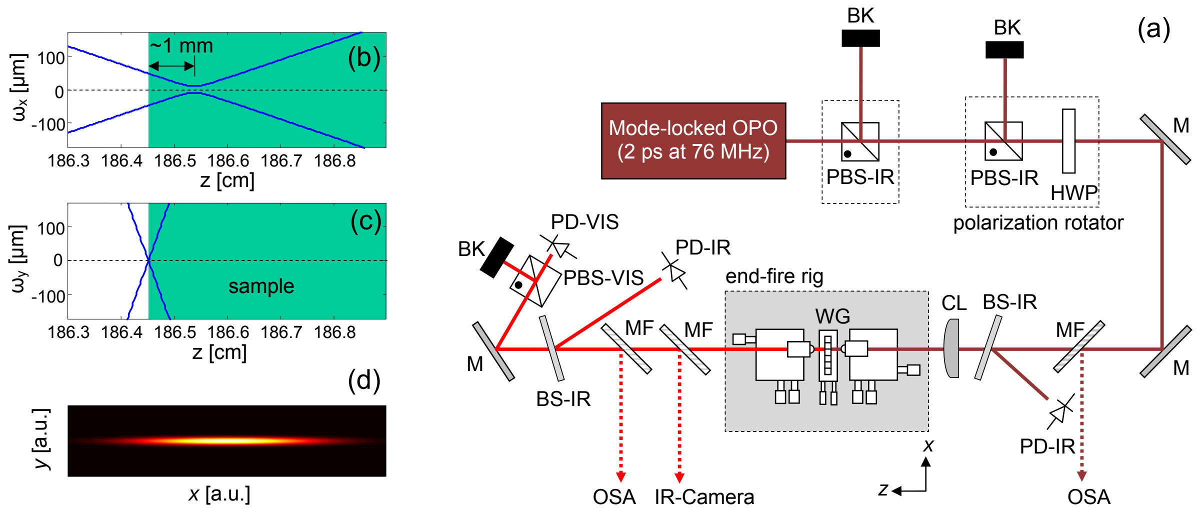

Figure 4.3: (a) Experimental setup for SHG in the slab waveguide of QtW-BRW-SH1. BK, beam-block; PBS-IR, polarization beam splitter for pump wavelength; HWP, half-waveplate; M, mirror; MF flip-out mirror; BS-IR, IR beam-sampler; PD-IR, InGaAs photodetector; CL, cylindrical lens; WG, nonlinear waveguide; PD-VIS, Germanium photodetector; PBS-VIS, polarization beam splitter for SH. Simulated cross section of the incident pump beam in (b) xz-plane and in (c) yz-plane. (d) Simulated intensity profile of the incident elliptical beam.

to improve the coupling efficiency into the slab mode and to maintain a minimum beam waist inside the sample for increasing peak pump power hence enhancing the nonlinear interaction. Through Gaussian beam modelling, it was predicted that the optimum configuration obtained using this setup provided a beam with a spot size of $2\omega_x = 37.5\ \mu$m at the sample's front facet. The beam waist in the horizontal plane was shifted by ≈ 1 mm inside the sample with a beam waist of $2\omega_{0x} = 27.8\ \mu$m. At the output of the sample the beam spot size was estimated to be $\approx 50\ \mu$m. In the transverse direction, the input waist was $2\omega_{0y} \approx 5\ \mu$m before the front facet and it remained confined inside the sample. At the output stage, the emerging pump mode we coupled into a IR photodetector to measure the pump transmission. Using Eq. (4.1b), the coupling efficiency into the slab was estimated to be $\approx 20\% \pm 5\%$.

Initially, second-harmonic power was monitored as a function of the wavelength of the TE-polarized pump to obtain the tuning curve of the process. The result is illustrated in

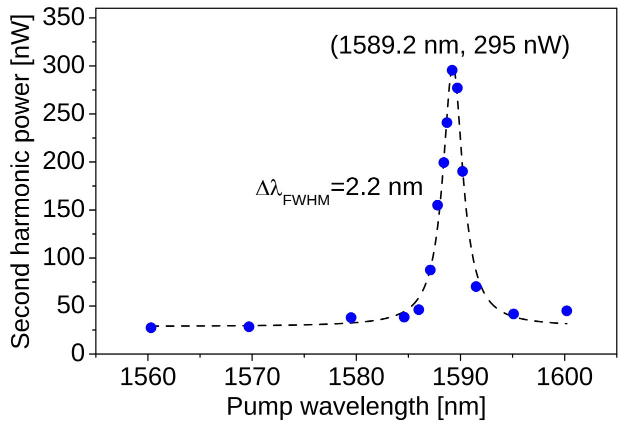

Figure 4.4: Second-harmonic power, measured outside the waveguide facet, as a function of pump wavelength for a fixed external pump power of 176 mW. The filled circles are the measured data while the dashed-line is the fit to a Lorentzian function [6].

Fig. 4.4 where the wavelength acceptance bandwidth of the SHG process was determined to be 2.2 nm. The pump power was fixed at an average value of 176 mW. Compensating for the facet reflection (29%) and the input coupling factor (20%) the internal pump power estimated right after the front facet of the waveguide was 25 mW. In obtaining the tuning curve of Fig. 4.4, no SH power could be detected when the input was switched to TM polarization. As can be seen from the figure, unphase-matched SH power with a minimum value of 27 nW could be detected throughout the range of wavelengths studied. However, for the pump wavelength centered at $\lambda_p = 1589.2$ nm, the SH power was increased to 295 nW, measured at the detector location, indicating an enhancement by more than an order of magnitude due to satisfying the phase-matching condition. Accounting for the transmission of the output objective lens (0.57) and the facet reflectivity (0.29), the internal SH power was estimated to be 729 nW. It should be noted that the slab QtW-BRW-SH1 sample was designed for phase-matching of a pump at 1550 nm. However, the measured phase-matching wavelength was at 1589.2 nm, indicating approximately 39 nm discrepancy between the theory and experiment. This mismatch in predicting phase-matching wavelength was mainly attributed to the less accurate model of Adachi [79] which was initially used for modeling the refractive indices of $Al_xGa_{1-x}As$ layers. Other

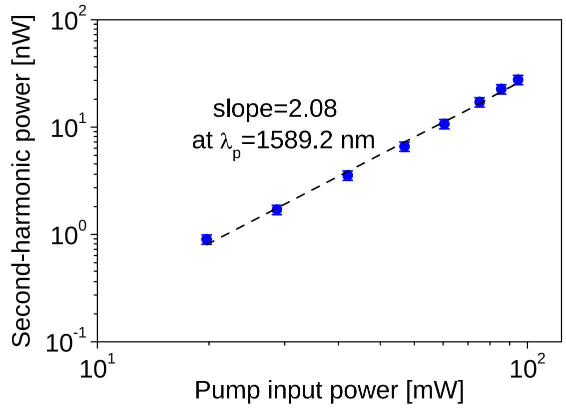

Figure 4.5: Dependency of second-harmonic power on pump power in log-log scale at phase-matching wavelength $\lambda_p = 1589.2$ nm. The filled circles are the measured data and the solid-line is a linear fit with a slope of 2.08 [6].

factors including variations of the layer thicknesses, particularly the core thickness, as well as the calibration of aluminum concentrations during wafer growth could additionally contribute to this wavelength mismatch.

The quadratic dependence between SH power and that of the pump was also examined. The result is shown in Fig. 4.5 where the powers of both harmonics are plotted in a log-log scale and fitted to a linear function. From the figure, the slope of the linear fit was estimated to be ≈ 2 confirming the parabolic power dependence. No sign of saturation could be detected for the pump power as high as 100 mW indicating that two-photon absorption (TPA) had negligible degrading effect in limiting the increase of SH power. During the design process, the effect of TPA was reduced by choosing the layer with lowest aluminum concentration in the structure as $Al_{0.2}Ga_{0.8}As$ with a half-bandgap more than 173 meV above the pump wavelength. Several factors contribute to uncertainties in estimating the SH power in Fig. 4.5. These include the collection efficiency and transmission of the output objective lens as well as the sensitivity of the Germanium photodetector used to monitor the SH power. The collection efficiency was not explicitly estimated for the characterized slab waveguide. However, our later cauterization of ridge BRWs denoted that, using a $40\times$ output objective lens, the collection efficiency was approximately 8%. The accurate measurement of the collection efficiency was not straightforward due to the fast divergence of the Bragg mode at the waveguide output facet. In the calculations of SH power in Fig. 4.5, the transmission of the output objective lens (Olympus PLN40×) was considered to be 0.57 around 780 nm. This value was obtained from the vendor of the lens without additional information regarding the error in their measurements. Using available datasheet, we consider an error of 1% in the transmission of the output objective lens. Error related to the sensitivity of the silicon photodetector (Newport 818-SL) was 1% obtained using the device manual. As a result, we consider an overall error of 10% in estimating the external SH power in Fig. 4.5. It should be noted that the errors discussed above are the systematic errors adhered to the uncertainties and tolerances of the equipments and the optics used in the experiment. Random or statistical errors which appear due to unknown or unpredictable changes in the experiment were not extracted here. Such errors could be determined using multiple measurements of the parameters of interest.

The power spectral density (PSD) for the TE-polarized pump at the unphase-matched wavelength of 1578.8 nm and at phase-matching wavelength of 1589.2 nm are shown in Fig. 4.6 (dashed-line). Also plotted are the spectra of their associated TM polarized SH signals (solid-line). From the figure, the FWHM spectral width of the unphase-matched SH was obtained to be 1.21 nm which was approximately half of that of the pump bandwidth of 2.88 nm. This was not the case for the phase-matched process. In this case, the FWHM spectral bandwidth of SH signal was measured to be 0.43 nm which was nearly an order of magnitude smaller than the corresponding pump bandwidth of 3.54 nm. This was considered as a strong evidence of the change in the nature of the SH signal and its transition from an unphase-matched to a phase-matched SHG process. The narrower spectral width suggested that the limitation of the phase-matching bandwidth was determined by the BRW structure.

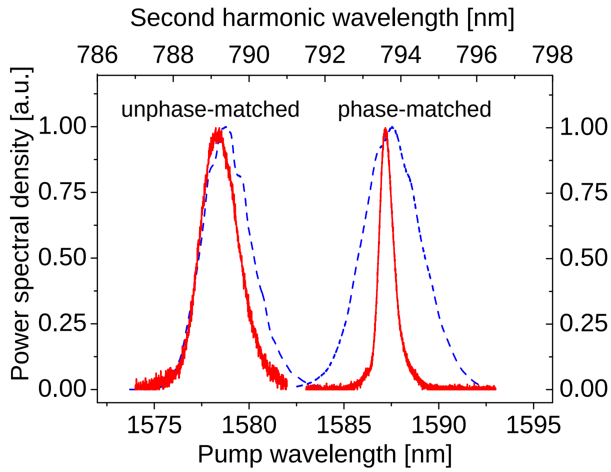

Figure 4.6: Power spectral densities of pump (dashed-line) and second-harmonic (solid-line) for the unphase-matched pump at $\lambda_p = 1578.8$ nm and for the phase-matched pump at $\lambda_p = 1589.2$ nm. The FWHM pump bandwidth was $\approx 2.9 - 3.4$ nm. The FWHM bandwidths of unphase-matched and phase-matched second-harmonic were 1.12 nm and 0.43 nm, respectively [6].

The performance of the SHG process can be quantitatively determined by estimating the conversion efficiency of the process. For the internal pump power of 25 mW and the SH power of 729 nW, we calculated the internal conversion efficiency to be $\eta = 0.12\ \%\mathrm{W}^{-1}$. Also, the conversion efficiency normalized to the device length was obtained to be $\eta_{\mathrm{norm}} = 2.0\ \%\mathrm{W}^{-1}\mathrm{cm}^{-2}$.

The initial illustration of SHG in BRWs suggested that different avenues could be undertaken for further enhancement of the nonlinear interaction. First, by employing ridge structures significant enhancement in the pump intensity can be expected. Second, using ridge structures enable using longer samples since the emerging light from the exit facet is spatially localized. Third, the BRW design offers the possibility of tuning the group velocity mismatch between the interacting waves, which additionally enhances the efficiency of the nonlinear interaction, particularly when ultrashort pump pulses are employed. Fourth, the laser pulses used exhibited a bandwidth larger than the wavelength acceptance bandwidth of the SHG process. Using a pulsed pump with better spectral match with SHG bandwidth increases the portion of the pump power utilized in the nonlinear interaction. The influence of pump bandwidth on the efficiency of SHG is further discussed in Chapter 6. We study the ridge BRWs for enhanced nonlinear interaction in the following section.

4.2 Phase-Matching SHG using Ridge BRW

Ridge waveguides are essential photonics elements in practical parametric devices. In an integrated devices where optical radiations should be received by and delivered to different components, the confinement of radiation in form of well-defined optical modes improves the coupling among the elements. Moreover, using ridge structures it is possible to enhance the confinement of interacting harmonics hence increasing the efficiency of a nonlinear interaction. As such, examination of the performance of ridge BRWs for phase-matching nonlinear processes is unavoidable. Ridge BRWs are implemented from a slab structure by patterning the waveguides through fabrication etch process. The resulted device posses additional parameters, namely the ridge width and the ridge depth, which can be used as extra designable parameters in tailoring modal dispersions hence altering the phase-matching condition. It should be stressed that the effect of forming a ridge structure from a slab one affects the functionality of the Bragg reflectors. As an example, consider a slab BRW with quarter-wave stacks designed at a certain operating wavelength. Making a ridge device from a slab structure results in changing the effective mode index of the slab which in turn makes the transverse Bragg reflectors non-quarter-wave. In fact, the level of deviation from the initial quarter-wave condition strongly depends on the physical geometry of the ridge including its width and depth.

Usually in practice, it is desired to limit the etch to a depth close to the core but not passing through it. A ridge depth which is too shallow and stops far away from the core cannot provide a tightly confined optical mode. On the other hand, a deeply etched waveguide exposes optical radiation to side wall roughness introduced during fabrication

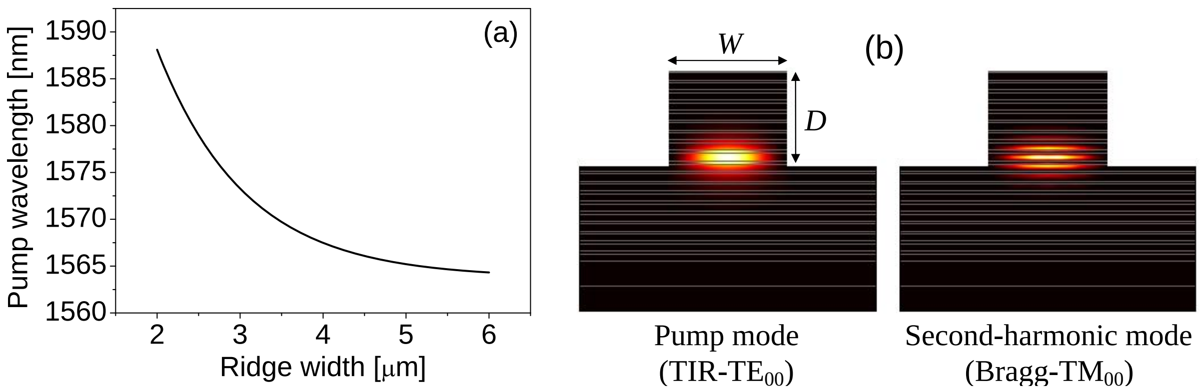

Figure 4.7: (a) Variation of phase-matched pump wavelength as a function of ridge width, W, simulated for QtW-BRW-SH1 design. The depth of the ridge was fixed at $D = 4.0$ μm. Decreasing W results in the shift of phase-matching wavelength toward longer wavelengths [7]. (b) Simulated intensity profiles of TE_{00} pump and TM_{00} second-harmonic modes for QtW-BRW-SH1 obtained for $W = 3.0$ μm and $D = 4.0$ μm.

hence increasing scattering losses.

The width of a ridge can be simply implemented through standard lithographic methods. Technically, one can fabricate ridge waveguides with widths ranging from a few hundreds of nanometers in making nanowires to a few tens or hundreds of micrometers in multimode waveguides. For nonlinear interactions, the geometry of the ridge has a considerable effect on phase-matching condition. To show this, in Fig. 4.7(a), variation of phase-matched pump wavelength, λ_p, as a function of ridge width is simulated for QtW-BRW-SH1 design. From the figure, it can be seen that as the width of the ridge is decreased, λ_p shifts towards longer wavelengths. The slope at which λ_p is shifted is larger for small ridge widths and it decreases as the 2-D structure tends toward the slab device. Also from Fig. 4.7(a), it can be seen that for QtW-BRW-SH1, λ_p can be tuned by more than 20 nm within the range of $1565 - 1585$ nm by simply using different waveguides with ridge widths in the range of $2.0 - 6.0$ μm. Exploiting this tuning mechanism in addition to the waveguide dispersion in BRWs offers a simple approach for controlling the bandwidth of generated harmonics in integrated parametric devices. This technique has been further elaborated in Chapter 7 where variation of ridge width is exploited for the bandwidth control of photon-pairs generated through spontaneous parametric down-conversion [13].

4.2.1 Pulsed characterization

To illustrate second-harmonic generation in ridge BRWs, the QtW-BRW-SH1 slab was used as the platform for the fabrication of a series of ridge structures [7]. Waveguides with different widths and depths were pattern using plasma etching. The characterized sample here had a ridge width of 3.0 μm, an etch depth of 4.0 μm and a length of 2.1 mm. The simulated intensity profiles of the pump and second-harmonic are shown in Fig. 4.7(b). Linear characterization of the waveguide was carried out for the total internal reflection pump mode using the Fabry-Pèrot method [81, 82, 83] with a tunable continuous-wave laser in telecommunication wavelength range. Propagation loss of the pump around 1555 nm was estimated to be 5.9 cm^{-1} which was higher than the loss value of $\approx$ 2.5 cm^{-1} measured in the slab structure owing to the side walls roughness. For non-linear characterization similar setup as shown in Fig. 4.3 was used with the exception that the cylindrical lens before the input objective was removed and circular Gaussian beam was coupled into the sample. Also, the sample was mounted on a temperature controlled manipulator. Before the SHG experiment, contribution of two-photon absorption in transmitted pump power was examined. It was observed that for pump with an average power more than 25 mW, measured before the front facet of the device, two-photon absorption was distinguishable. To estimate the input coupling factor, the pump power was reduced to approximately 21 mW where the transmitted power was measured to be 61.8 μW. Accounting for the linear loss (5.9 cm^{-1}) and the facet reflection (30%), the input coupling factor was determined to be $\approx$ 1.8%. It was believed that the error bar in estimating this input coupling factor was large due to the small core dimension of the waveguide which made light coupling into the device a challenging task.

Similar to the slab case, the SHG experiments were carried out for type-I interaction.

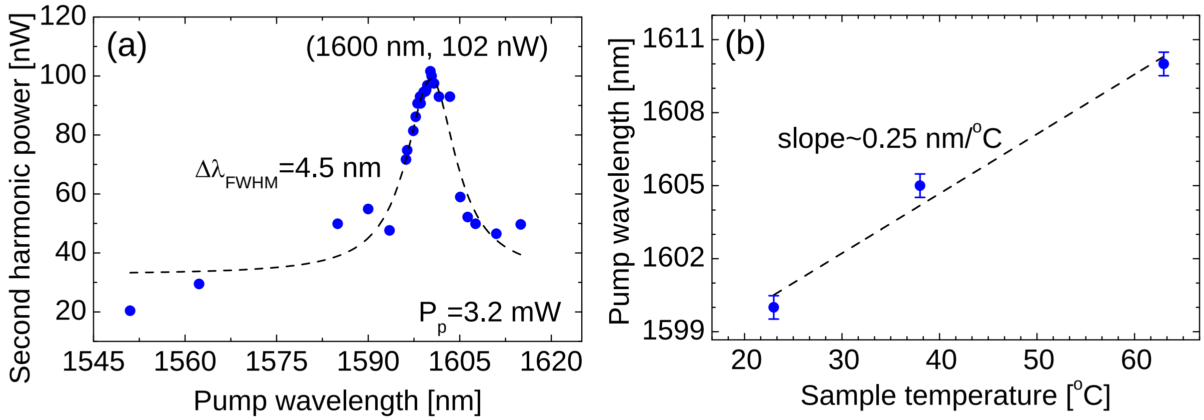

Figure 4.8: (a) Dependency of second-harmonic power as a function of pump wavelength for a fixed internal pump power of 3.2 mW. The filled circles are the measured data while the solid-line is the fit to a Lorentzian function. (b) Temperature tuning of phase-matching wavelength obtained experimentally (filled-circles). The dashed-line is a linear fit with a slope of 0.25 nm/°C [7].

The tuning curve of the process is illustrated in Fig. 4.8(a) where peak second-harmonic power of 102 nW was estimated at the pump wavelength of 1600.0 nm. The average power of the pump was fixed at 200 mW with an estimated internal value of 3.2 mW. In comparison to the slab case, the phase-matching wavelength was shifted by $\approx$ 12 nm toward longer wavelengths as a result of the ridge effect. This was in agreement with the simulation results shown in Fig. 4.7 where red-shift of λ_p with decreasing the ridge width was anticipated. The wavelength acceptance bandwidth of the process was measured to be $\Delta\lambda_{\mathrm{FWHM}} = 4.5$ nm, full-width at half-maximum. This value was almost half of the obtained value for the slab structure. It should be noticed that for the characterized ridge structure, the depth of the ridge was passing through the core. This made optical modes volatile to scattering by side wall roughness which was accounted for the broadening of the turning curve.

In another experiment we examined the temperature tuning of the phase-matching wavelength. Using a temperature controlled manipulator, the sample temperature was varied between 15-40 °C while peak second-harmonic power was monitored. The result is shown in Fig. 4.8(b) where the temperature tuning slope of 0.25 nm/°C was derived. The error bars in Fig. 4.8(b) denote the wavelength resolution setting of the optical spectrum analyzer used to determine the phase-matching wavelength for a given sample temperature. From the figure, it was understood that heating the sample, resulted in shifting the phase-matching wavelength toward longer wavelengths. During the experiment, it was observed that temperature variation significantly affected the pump beam coupling as a result of thermal expansion of the sample manipulator as well as variation in the thicknesses of dielectric layers. As such, for each measurement, the sample was appropriately manipulated to maintain optimum power coupling. It should be stressed that high pump power, such as the 200 mW external pump power used in this experiment, can additionally contribute to heating the sample hence shifting the phase-matching wavelength. Although we have not characterized the dependence of the phase-matching wavelength shift on the pump power, we observed that the increase of the pump power resulted in the red-shift of the phase-matching wavelength. In Fig. 4.8(b), the pump power was kept fixed for all the measured data points. As such, the estimated value of 0.25 nm/°C for the phase-matching wavelength detuning is ascribed to the external variation of the device temperature using the temperature controller.

The temperature tuning effect offers a simple while robust mechanism for aligning the phase-matching wavelength. In practice, this finds applications in integrated parametric devices where fine adjustment of phase-matching wavelength to a desired value is required. An example is integrated OPOs where tolerances in fabricating the diode laser pump and nonlinear frequency conversion element give rise to deviation from ideal phase-matching point. In such applications, temperature tuning of phase-matching wavelength can be utilized to compensate for such fabrication imperfections.

Considering the estimated internal pump power of 3.2 mW and the second-harmonic power of 102 nW, the conversion efficiency of the ridge QtW-BRW-SH1 sample was calculated to be $\eta = 0.99$ %W^{-1} or in normalized form $\eta_{\mathrm{norm}} = 22.6$ %W^{-1}cm^{-2}. In comparison to slab QtW-BRW-SH1 sample, the SHG efficiency in the ridge structure indicates an enhancement by approximately an order of magnitude. Although the loss in the later device was larger, the higher efficiency of the ridge structure was justifiable due to the enhanced pump intensity within the waveguide offered by the lateral confinement in the 2-D structure.

The successful demonstration of SHG in slab and ridge BRWs deserved additional design optimization to further enhance the nonlinear interaction. For this purpose, several routes were studied to design a new device with improved linear/nonlinear characteristics. First, it was intended to reduce the propagation loss of harmonics by selecting aluminum concentration of layers more than 0.24 ($x > 0.24$) to reduce material absorption by shifting the materials bandgap away from the operating wavelength. Second, loss reduction was aimed by limiting the ridge depth to a value before the core in order to limit the exposure of optical modes to side wall roughness. Third, leakage loss was curtailed by designing the Bragg mirrors with higher refractive index contrast bi-layers. Leakage loss could be further curtailed by including additional periods of quarter-wave bi-layers in the bottom Bragg reflector. However, the allowable overall growth thickness of the structure was limited to ≈ 9 μm using the metal-organic chemical vapour deposition (MOCVD) technique to avoid large surface morphology. This restriction was imposed by the induced stress between adjacent $Al_xGa_{1-x}As$ layers with different aluminum concentration and crystal lattice mismatch. Other growth techniques such as molecular beam epitaxy (MBE) allow fabrication of thicker devices. However MBE was not considered an option due to significantly increased fabrication cost. Finally, the design was further optimized for enhanced nonlinear overlap factor. The result was a new structure referred to as QtW-BRW-SH2. The geometry of the wafer along with the simulated modal indices of the pump and second-harmonic are summarized in Table 4.3.

Table 4.3: Type-I phase-matched QtW-BRW-SH2 design. The Bragg mirrors had 7 periods.

λ_p [nm]	(x_c, x_1, x_2)	(t_c, t_1, t_2) [nm]	$n_{\text{eff},\omega}^{(\text{TE})}$	$n_{\text{eff},2\omega}^{(\text{TM})}$
1550 nm	(0.40,0.25,0.75)	(301,124,392)	3.1099	3.1099

Ridges waveguides with widths ranging from $2.5 - 4.5$ μm with 0.5 μm steps and for a depth of ≈ 3.4 μm were fabricated using the QtW-BRW-SH2 wafer. The characterized device had a ridge width of 4.0 μm and a length of 2.0 mm. Linear properties were extracted using the Fabry-Pèrot method at low power of ≈ 2 mW around 1560 nm. For TE polarization, propagation loss of 7.78 cm^{-1}, input coupling efficiency of 50% and facet reflectivity of 29%, were the typical values measured. In comparison to the ridge QtW-BRW-SH1 sample, the new device indicated increased propagation losses. It was believed that this large propagation loss was introduced by plasma etch quality. Also, the measured value of 50% for the input coupling efficiency was higher than that reported for QtW-BRW-SH1. Accurate estimation of the coupling efficiency in the BRW was problematic due to the fact that linear characterization based on Fabry-Pèrot method provides good results when the device under test supports single mode propagation [83]. This was not valid for the characterized device where multi-modedness exists in both lateral and transverse directions. To get a better insight of the input coupling factor, we simulated this parameter by calculating the overlap integral between the field profiles of the pump beam and the excited pump mode. The simulation resulted in an overlap factor of 54%. Comparing the experimentally measured coupling factor with the simulated value, we considered 7% error in the measured value of the input coupling efficiency.

Nonlinear characterization of QtW-BRW-SH2 device was performed in the same setup as that used for the characterization of QtW-BRW-SH1. Figure 4.9(a) shows the tuning curve of the SHG process. The average power of the pump was fixed at 115 mW measured before the facet of the waveguide. This translated to an internal power of 41 mW inside

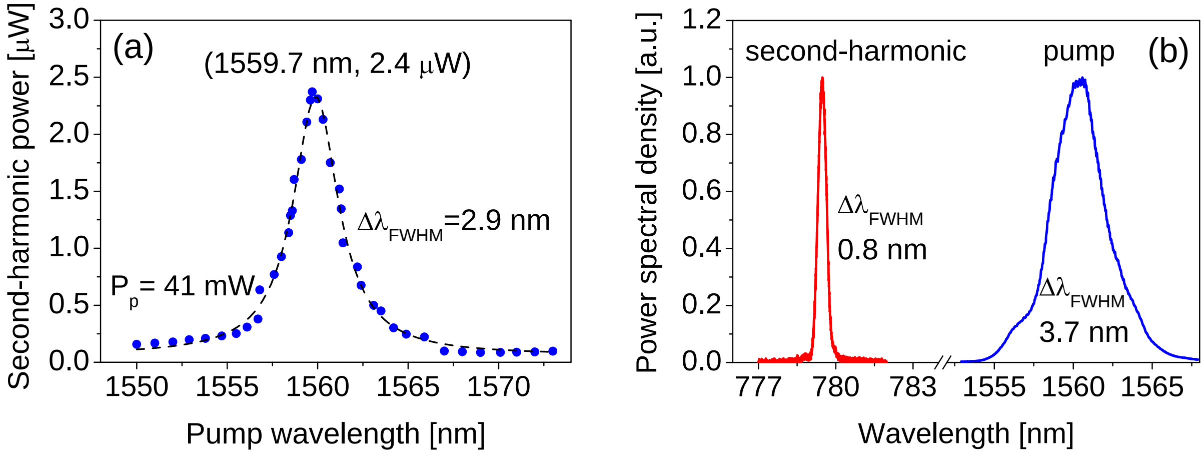

Figure 4.9: (a) Dependency of second-harmonic power on pump wavelength for a fixed internal pump power of 41 mW. The filled circles are the measured data while the solid-line is the fit to a Lorentzian function. (b) Power spectral density of pump and second-harmonic signal.

the sample. For the pump wavelength of 1559.7 nm where phase-matching condition was achieved, peak second-harmonic power of 2.4 μW was estimated before the exit facet of the sample. The pulsed conversion efficiency of the interaction was calculated to be $\eta = 0.14$ %W^{-1} or $\eta_{\text{norm}} = 3.6$ %W^{-1}cm^{-2}.

The power spectral density of the harmonics are illustrate in Fig. 4.9(b). From the figure, the spectral bandwidth of the SH is approximately five times smaller than that of the pump indicating that only part of the pump spectra was utilized in the nonlinear interaction.

4.2.2 Continuous-wave characterization

The results discussed so far involved characterization of BRWs in pulsed regime. It was also important to examine the performance of these devices in continuous-wave (CW) regime. Such examination is imperative for obtaining a better insight of the limitations and potentials of phase-matched BRWs for applications such as CW harmonic generation, high resolution spectroscopy and the generation of frequency anti-correlated photon-pairs through parametric down-conversion. Unlike pulsed nonlinear interactions where the

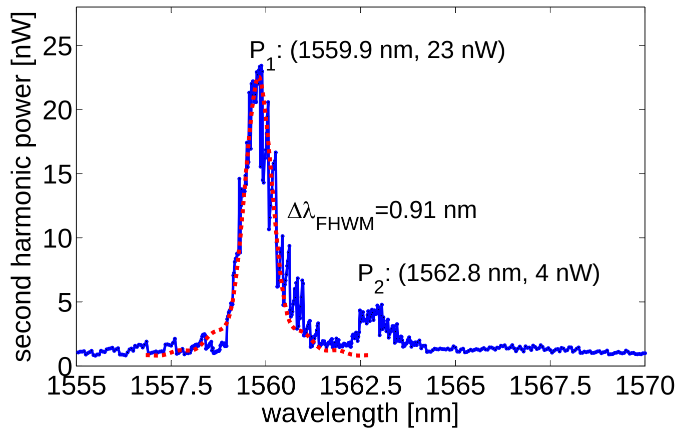

Figure 4.10: Second-harmonic power vs. tuning of fundamental wavelength. The solid-line is the experimental data and the dotted-line is a theoretical tuning curve obtained using beam propagation method. The theoretical curve is simulated by including the propagation loss of FH and an estimated loss of SH to match the full-width at half-maximum bandwidths of the two curves [8].

efficiency of the process is proportional to the peak powers of the involved harmonics, in CW regime, the efficiency is related to the associated average powers. As a result, CW interactions are expected to be much weaker in comparison to those in the pulsed regime.

We characterized the QtW-BRW-SH2 sample using a tunable single-frequency telecommunication laser [8]. Using a C-band Erbium-doped fiber-amplifier (EDFA), the pump power was amplified to $\approx$ 265 mW, which was measured before the front facet of the device. Taking into account 29% facet reflectivity and 50% end-fire coupling efficiency, one obtains an internal pump power of $\approx$ 94 mW at the input facet. Second-harmonic power was then monitored using a Germanium photodetector over the relevant wavelength range to obtain the tuning curve. Internal SH power was obtained by accounting for the transmission of the output objective lens (57%) and the facet reflectivity (29%). Figure 4.10 illustrates the resulted tuning curve in terms of the internal SH power. From the Fig. 4.10, the peak SH power of 23 nW was evident at the phase-matching wavelength of 1559.9 nm which was the same phase-matching wavelength observed during the pulsed characterization.

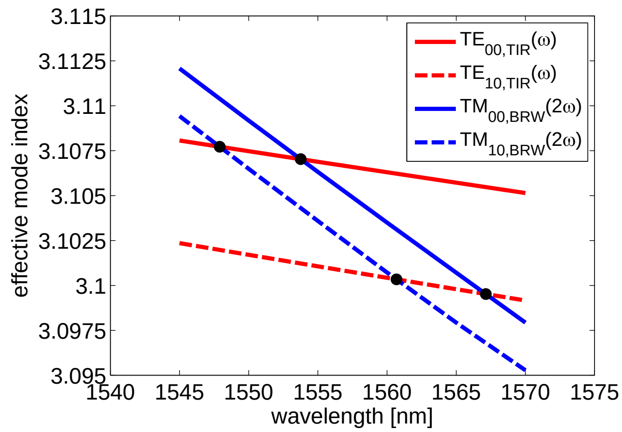

Figure 4.11: Calculations of the phase-matching conditions between higher order modes of the pump and SH. The filled-circles where two curves cross indicate wavelengths for which type-I phase-matching condition is satisfied. A discrepancy of $\approx$ 6 nm exists between the experimental phase-matching wavelength at 1559.9 nm in Fig. 4.10 and the simulated value where the curves of $\mathrm{TE}_{00}(\omega)$ and $\mathrm{TM}_{00}(2\omega)$ crosses [8].

The tuning curve illustrated in Fig. 4.10 indicated an additional SH resonance (P_2) at 1562.8 nm. This extra peak did not resemble the side lobe of a sinc^2 function and hence merits further attention. Initially, the experimental tuning curve was compared with a theoretical one in the ideal case, where the linear/nonlinear losses were ignored. From Eq. (2.24), it is known that $\mathcal{P}_{2\omega}(L) \propto \mathrm{sinc}^2(\Delta\beta L/2)$, where $\Delta\beta$ is the wavenumber mismatch defined in Eq. (2.20) and L is the sample length. From the theoretical tuning curve, the separation between the peak of the main lobe of the sinc^2 function and the immediate side lobes was found to be ≈ 1.0 nm. However, the feature at $\lambda_{\mathrm{P}_2} = 1562.8$ nm lies 2.9 nm away from the phase-matching point at 1559.9 nm, in contrast to the ideal case of 1.0 nm. We further investigated the possibility of having phase-matching among higher order modes of pump and second-harmonic. The simulations indicated modal phase-matching between higher order modes could indeed exist. For example, as illustrated in Fig. 4.11, phase-matching condition can be satisfied between $\mathrm{TE}_{00}(\omega)$ pump and $\mathrm{TM}_{10}(2\omega)$ second-harmonic with the peak SH power 5.8 nm below λ_{P_1}. Also $\mathrm{TE}_{10}(\omega)$ pum and $\mathrm{TM}_{10}(2\omega)$ second-harmonic can be phase-matched at 6.9 nm above λ_{P_1}. Phase-

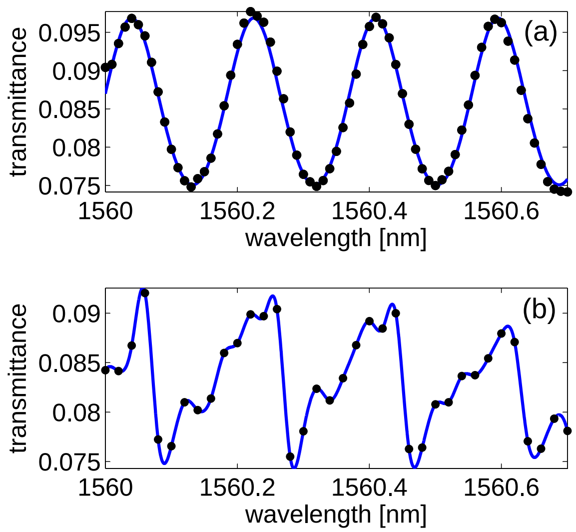

Figure 4.12: Transmittance of pump vs. wavelength at (a) input power level of 2 mW used for linear characterization and (b) input power level of 265 mW used for SHG experiment. In both figures circles are the measured data. The solid-line in (a) is the fit to the Airy's transmission of the pump and that in (b) is obtained using an interpolating fit. The non-uniform transmittance spectrum in (b) clearly denotes the effect of high order modes superposition in the detected signal at high input pump power [8].

matching between $\mathrm{TE}_{10}(\omega)$ and $\mathrm{TM}_{10}(2\omega)$ is more likely accountable for the SH feature at λ_{P_2}. Discrepancies between the calculated predictions and measurements could be due to fabrication tolerances.

To investigate the multimoded nature of the waveguide, we studied the Fabry-Pèrot transmissions of pump at low and high power. In Fig. 4.12(a) and (b), we have compared the pump transmittance at low power of 2 mW used for linear loss measurement and at high power of 265 mW used for second-harmonic generation, respectively. As can be seen from Fig. 4.12(a), the uniform Fabry-Pèrot fringes is a strong evidence of single mode propagation when 2 mW pump is launched in the waveguide. In contrast, the Fabry-Pèrot fringes in Fig. 4.12(b) exhibit a behaviour characteristic of multimode waveguides. The multimode behaviour at high pump power can be ascribed to two effects. First, higher order modes contain more power at higher pump and hence can contribute to the nonlinear conversion with output powers above the lowest detection limit. Second, increasing the pump power results in an increase of the core temperature where power confinement is the highest. This, in turn, increases the core refractive index [80, 85]. As a result, the waveguide acquires an enhanced index difference between the core and the lateral claddings, making it more multimoded. It is worth mentioning that although the characterized waveguide supported multi-mode propagation, it was believed that the Fabry-Pèrot method provided an accurate estimation of the linear loss value as the linear transmission spectrum of the device illustrated undistorted Fabry-Pèrot fringes at the wavelength range of interest at low pump power. Despite the fact that higher order modes were excited at the input facet of the waveguide at low power levels, they were filtered out from the detected output signal through loss discrimination mechanism.

The CW internal nonlinear conversion efficiency of the process was estimated to be $\approx 2.7 \times 10^{-4}$ %W^{-1} or $\eta_{\mathrm{norm}} \approx 6.8 \times 10^{-3}$ %W^{-1}cm^{-2}. It would be instructive to compare the CW conversion efficiency with that obtained in the pulsed pump setup. Such comparison indicates that the SHG process using pulsed pump is over 2.7 orders of magnitude more efficiency that in the CW regime. This enhancement is accredited to significantly larger peak intensity in the pulsed regime in comparison to the CW one. For the pump mode with an effective mode area of 4.6 μm^2, the peak intensity of the coupled power inside the waveguide in the pulsed system was 4.5 GW/cm^2, while that in the CW experiment was 2.0 MW/cm^2. The ratio of the pump peak intensity in the pulse system was found to be $\approx$ 3.3 orders of magnitude larger than that in the CW experiment. This ratio is within the same orders of magnitude of the ratio of the conversion efficiencies ($\approx$ 2.7). The fact that the ratio of the peak intensities is larger than the ratio of the conversion efficiencies suggests the increased losses in the pulsed system which is accounted for by the two-photon absorption of the pump signal.

From the tuning curve of Fig. 4.10, the wavelength acceptance bandwidth (FWHM) of the process was found to be $\approx$ 0.91 nm after filtering out the Fabry-Pèrot components of the pump from the spectrum. Ignoring harmonics losses, a theoretical estimation of the bandwidth can be obtained as [21]:

$$\Delta\lambda_{\mathrm{FWHM}} = \frac{0.4429\lambda}{L}\left|\frac{\partial n^{(\mathrm{TE})}_{\mathrm{eff},\omega}}{\partial\lambda} - \frac{1}{2}\frac{\partial n^{(\mathrm{TM})}_{\mathrm{eff},2\omega}}{\partial\lambda}\right| \tag{4.2}$$

From simulations, we obtained $\Delta\lambda_{\mathrm{FWHM}} = 0.78$ nm, indicating a factor of 1.16 difference between theoretical and the measured values. The increase in the process bandwidth suggests inhomogeneous broadening as a result of waveguide losses at interacting frequencies as well as fabrication imperfections [86]. Using beam propagation method [87] which includes harmonics losses, a SH loss value of 41 cm^{-1} was estimated by fitting the FWHM bandwidth of a normalized theoretical tuning curve to that of the experimental one (see dotted curve in Fig. 4.10). The estimated SH loss appeared to be larger than what was expected for this device. Several mechanisms can be accounted for this large loss value. First, in type-I phase-matching where second-harmonic signal is TM-polarized, substrate and cover leakage losses are more prominent as the direction of the electric field is dominantly normal to the interfaces formed by different layers. Second, the boundaries between adjacent layers can form rough interfaces which contribute to scattered losses. This source of loss is expected to be more significant in BRWs where the structure is composed of many dielectric layers. Finally, material absorption contributes to additional losses as the SH wavelength lies in proximity of the band-tail absorption of low aluminum concentration layers [88]. It would be instructive to compare the estimated SH loss with linear loss of the TM-polarized pump. Using Fabry-Pèrot technique, a loss value of 11.1 cm^{-1} was measured for the TM-polarized pump around 1560 nm where phase-matching was taking place. Taking into consideration the discussed sources of losses in BRWs, the estimated SH loss value of 41 cm^{-1} appeared as a rough estimation in the characterized device.

Chapter 5

Matching-Layer Enhanced BRW for $\chi^{(2)}$ Phase-Matching

Theoretical and experimental endeavours discussed so far in phase-matching $\chi^{(2)}$ interactions in BRWs have been aimed at utilizing quarter-wave TBRs where the bi-layers of Bragg reflectors are quarter-wave thick at second-harmonic wavelength. Operating at the quarter-wave condition is attractive. Its advantages include fastest exponential decay of Bragg mode in periodic claddings, simplicity of design equations, polarization degeneracy of the fundamental Bragg mode and versatile dispersion properties, to name a few. However, the quarter-wave condition also imposes constraints over the waveguide design which limits the efficiency of these devices for optical frequency mixing. Above all such constraints is the small core dimension of phase-matched QtW-BRWs which is dictated by the phase-matching condition. For example, for QtW-BRW-SH1 and QtW-BRW-SH2 samples discussed in Chapter 4, the thickness of the core was limited to 328 nm and 301 nm, respectively. Devices with such small core thicknesses severely suffer from degraded confinement factor and poor end-fire coupling efficiency. Moreover, the deterministic core thickness in phase-matched QtW-BRWs eradicates the flexibilities in tailoring waveguide dispersion while maintaining PM condition. The aforementioned disadvantages in using standard quarter-wave stacks suggest the necessity for investigating structures with more elaborated claddings.

A simple technique to relieve the constrain over the core dimension in QtW-BRWs is offered by using matching-layers (MLs) between the core and TBRs. In this technique, core thickness is regarded as an independent designable parameter while the characteristics of the matching layer including its material index and thickness are used to align the effective mode index such that the bi-layers of TBRs become quarter-wave-thick. In the resulting design, one can benefit from a phase-matched BRW with relaxed core dimension while taking advantage of all assets of quarter-wave stacks. The matching layer technique was initially proposed by Mizrahi *et al.* [89] to control the field profile and dispersion properties of air core planar and cylindrical BRWs for use in particle accelerators [67] and high power lasers. Here, we show that, in the context of optical frequency mixing, the inclusion of matching layer tremendously enhances the efficiency of nonlinear interactions in phase-matched BRWs.

5.1 Theoretical Analysis

A typical one-dimensional symmetric matching-layer enhanced BRW (ML-BRW) is illustrated in Fig. 5.1. The periodicity of the claddings extends in the x-direction, while the structure is homogeneous in the yz-plane. By convention, the TBRs consist of bi-layers with refractive indices n_1 and n_2 with associated thicknesses t_1 and t_2 and periodicity $\Lambda = t_1 + t_2$. The core has an index of refraction n_c and thickness t_c. The matching layer is located between the core and quarter-wave TBRs. The refractive index of the matching layer is n_m and the layer thickness is t_m. Modal analysis of ML-BRW using Bloch-Floquet formalism has been previously discussed in [1]. Our analysis here is based on the same approach used in Section 3.3.1 where modal dispersion equations were derived using the transition of overall electric field for TE and TM propagating modes at

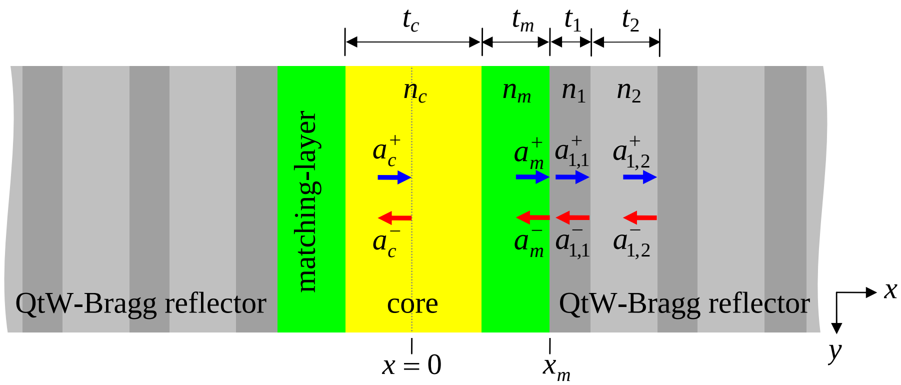

Figure 5.1: Schematic of a BRW containing matching layers with refractive index n_m and thickness t_m. The structure is symmetric with respect to the core center at $x = 0$. The propagation direction of optical modes is along the z-axis [1].

the interface between the matching-layer and quarter-wave TBRs.

For optical modes propagating in the $+z$ direction, the field distribution inside the core, matching-layer and the Bragg stacks are expressed as

$$\psi_j(x) = \begin{cases} a_c^+ \exp(ik_c^x x) + a_c^- \exp(-ik_c^x x) \\ 0 \le x \le t_c/2 \\ a_m^+ \exp[ik_m^x(x - \Delta_m)] + a_m^- \exp[-ik_m^x(x - \Delta_m)] \\ t_c/2 \le x \le \Delta_m \\ a_{j,1}^+ \exp[ik_1^x(x - \Delta_j + t_2)] + a_{j,1}^- \exp[-ik_1^x(x - \Delta_j + t_2)] \\ (j-1)\Lambda + \Delta_m \le x \le j\Lambda + \Delta_m - t_2 \\ a_{j,2}^+ \exp[ik_2^x(x - \Delta_j)] + a_{j,2}^- \exp[-ik_2^x(x - \Delta_j)] \\ j\Lambda + \Delta_m - t_2 \le x \le j\Lambda + \Delta_m, \end{cases} \tag{5.1}$$

where

$$\Delta_m = t_c/2 + t_m$$
$$\Delta_j = \Delta_m + j\Lambda \quad (j = 1, 2, \ldots)$$

In Eq. (5.1), the terms $a_{j,1}^+$ and $a_{j,1}^-$ are, the field amplitudes of the right and left travelling waves in the layer with refractive index n_1 of the j-th unit cell of TBRs, respectively, and $a_{j,2}^{+}$ and $a_{j,2}^{-}$ are those in the layer with refractive index n_2. The field amplitudes within the core are a_c^+ and a_c^-, and those within the matching layer are a_m^+ and a_m^-. For TE-polarization with non-zero field components (H_j^x, E_j^y, H_j^z), $\psi_j(x) = E_j^y$ while for TM-polarization with field components (E_j^x, H_j^y, E_j^z), $\psi_j(x) = H_j^y$. Here, only lowest order even Bragg mode will be considered which is of particular interest for nonlinear interactions. Without loss of generality, the field amplitude at the core center is taken to be unity: $a_c^+ + a_c^- = 1$. Using the symmetry of the structure and the fact that the field inside the core should be real, the right and left propagating field amplitudes of the core can be taken as $a_c^+ = a_c^- = 1/2$. The right and left propagating field amplitudes inside the matching layer can be simply derived by applying the appropriate boundary condition of $\psi_j(x)$ continuity at $x = t_c/2$ as For TE-polarization, the field amplitudes inside the matching-layer are obtained as

$$a_m^+ = \frac{1}{2}\cos(k_c\frac{t_c}{2})\exp(ik_m^x t_m) - i\frac{1}{2}f_{\mathrm{P}}\sin(k_c^x\frac{t_c}{2})\exp(+ik_m^x t_m) \tag{5.2a}$$
$$a_m^- = \frac{1}{2}\cos(k_c\frac{t_c}{2})\exp(ik_m^x t_m) + i\frac{1}{2}f_{\mathrm{P}}\sin(k_c^x\frac{t_c}{2})\exp(-ik_m^x t_m), \tag{5.2b}$$

where

$$f_{\mathrm{P}} = \begin{cases} \dfrac{k_c^x}{k_m^x} & \text{for TE polarization} \\ \dfrac{n_m^2 k_c^x}{n_c^2 k_m^x} & \text{for TM polarization} \end{cases} \tag{5.3}$$

The quarter-wave stack in ML-BRW causes the propagating mode to experiences either a perfect electric-wall or a perfect magnetic-wall at the interface of the matching-layer with periodic claddings. For the former case, the total electric field at the interface vanishes to form a node. This is satisfied when the amplitudes of the right and left propagating plane waves within the matching layer cancel each other out, which requires that $a_m^+ + a_m^- = 0$. For the latter case where the quarter-wave stack operates as a magnetic-wall, the total electric field has a maximum/minimum at the matching-layer interface with periodic claddings. In this case $a_m^+ - a_m^- = 0$. Using these unique characteristics of quarter-wave stacks, the dispersion equations for TE-polarization are obtained as

$$\cot\left(k_c^x t_c/2\right)\cot\left(k_m^x t_m\right) = +\frac{k_c^x}{k_m^x} \quad \left(k_m^x < k_1^x\right) \tag{5.4a}$$

$$\cot\left(k_c^x t_c/2\right)\tan\left(k_m^x t_m\right) = -\frac{k_c^x}{k_m^x} \quad \left(k_m^x > k_1^x\right) \tag{5.4b}$$

Similarly, for TM-polarization

$$\cot\left(k_c^x t_c/2\right)\cot\left(k_m^x t_m\right) = +\frac{n_m^2 k_c^x}{n_c^2 k_m^x} \quad \left(n_1^2 k_m^x < n_m^2 k_1^x\right) \tag{5.5a}$$

$$\cot\left(k_c^x t_c/2\right)\tan\left(k_m^x t_m\right) = -\frac{n_m^2 k_c^x}{n_c^2 k_m^x} \quad \left(n_1^2 k_m^x > n_m^2 k_1^x\right) \tag{5.5b}$$

Equation (5.4) can be solved for t_m to obtained the required thickness of the matching-layer which makes the the bi-layers of Bragg reflectors quarter-wave thick. For TE-polarization we get

$$t_m = \frac{1}{k_m^x}\left[\operatorname{acot}\Big(+\frac{k_c^x}{k_m^x}\tan(k_c^x t_c/2)\Big) + p\pi\right] \quad \left(k_m^x < k_1^x\right) \tag{5.6a}$$

$$t_m = \frac{1}{k_m^x}\left[\operatorname{atan}\Big(-\frac{k_c^x}{k_m^x}\tan(k_c^x t_c/2)\Big) + p\pi\right] \quad \left(k_m^x > k_1^x\right) \tag{5.6b}$$

where $p = 0, 1, 2, \ldots$ is the matching-layer thickness scaling factor. Similarly, using Eq. (5.5), for TM-polarization we have

$$t_m = \frac{1}{k_m^x}\left[\operatorname{acot}\Big(+\frac{n_m^2 k_c^x}{n_c^2 k_m^x}\tan(k_c^x t_c/2)\Big) + p\pi\right] \quad \left(n_1^2 k_m^x < n_m^2 k_1^x\right) \tag{5.7a}$$

$$t_m = \frac{1}{k_m^x}\left[\operatorname{atan}\Big(-\frac{n_m^2 k_c^x}{n_c^2 k_m^x}\tan(k_c^x t_c/2)\Big) + p\pi\right] \quad \left(n_1^2 k_m^x > n_m^2 k_1^x\right) \tag{5.7b}$$

The above equations are the generic dispersion relations and design equations of ML-BRW. In what follows, these equations will be used to compare the performance of ML-BRWs to conventional QtW-BRWs for SHG as a case study. We are going to study modal properties of typical ML-BRWs while PM condition is maintained for SHG with a pump at $\lambda_p = 1540.0$ nm. We examine linear and nonlinear parameters including effective mode index, core thickness, nonlinear overlap factor (ξ), effective nonlinear coefficient (d_{eff}), nonlinear conversion efficiency (η), group velocity mismatch (GVM), group velocity dispersion (GVD) and the wavelength acceptance bandwidth ($\Delta\lambda_{\text{FWHM}}$) of typical devices.

In order to illustrate the improvements offered by ML-BRW over its conventional counterpart, an optimized type-I phase-matched QtW-BRW will be used as a reference point for comparison. This reference structure is referred to as QtW-BRW$_{\text{opt}}$. The design of QtW-BRW$_{\text{opt}}$ focused on maximizing the nonlinear overlap factor, ξ, while operating at the QtW point. Previous simulations indicated that ξ can be maximized by increasing the contrast between the material index of Bragg reflectors bi-layers [25]. As such, this was the route we have undertaken to define QtW-BRW$_{\text{opt}}$ with maximal nonlinear overlap factor. In order to avoid two-photon absorption (TPA) in the structure, $Al_xGa_{1-x}As$ layers with lowest aluminum concentration was taken to be $Al_{0.20}Ga_{0.80}As$. This maintains the half-bandgap of the layers in the structure above the photon energy of the pump [32] around 1540.0 nm. Layers with highest aluminum concentration was selected as $Al_{0.80}Ga_{0.20}As$ to minimize the oxidation of the exposed sample areas. Furthermore, the bi-layers of the Bragg reflectors were arranged such that the core was sandwiched between the layer with higher refractive index. This was essential for excitation of the lowest order even Bragg mode where the field amplitude peaked in the middle of the core. The choice of placing the low index layer on the sides of the core was also feasible. However, this configuration results in the excitation of lowest odd Bragg mode, where the field vanishes at the core centre while it peaks, with opposite signs, at the core interface with the TBRs. In a QtW-BRW, since the pump mode also has a maximum at the core center, the later configuration would reduce the modal nonlinear overlap between the two harmonics. Given these limitations in the TBR materials and their arrangement, the choice of core thickness and aluminum concentration are dictated by the quarter-wave condition. This further highlights the limitations of optimizing such structures while being governed by the range of refractive indices offered by a given material system and while operating at the quarter-wave point. The core was considered as the only designable parameter during optimization. For maximal nonlinear overlap factor, the optimum core material was found to be $Al_{0.62}Ga_{0.38}As$. The complete parameter set of the structure is summarized in Table 5.1.

Table 5.1: Simulated parameters of QtW-BRW$_{\text{opt}}$ [1]

Parameter	Value
λ_p [nm]	1540.0
(x_c, x_1, x_2)	$(0.62, 0.20, 0.80)$
(t_c, t_1, t_2) [nm]	$(386, 109, 333)$
$(n^{(\mathrm{TE})}_{\mathrm{eff},\omega}, n^{(\mathrm{TM})}_{\mathrm{eff},2\omega})$	$(3.0692, 3.0692)$
ξ [m^{-1}]	39.74
d_{eff} [pm/V]	92
η_{norm} [%W^{-1}cm^{-2}]	5.80×10^{-5}
GVM [ps/mm]	2.55
$(\mathrm{GVD}^{(\mathrm{TE})}_{\omega}, \mathrm{GVD}^{(\mathrm{TM})}_{2\omega})$ [fs$^2/\mu$m]	$(1.09, 5.26)$
$\Delta\lambda_{\mathrm{FWHM}}$ [nm]	0.57
Number of periods in Bragg mirrors	10

The design algorithm of phase-matched ML-BRWs can be summarized as:

1. For a given nonlinear interaction, determine the type of the optical modes (Bragg mode or TIR mode) and their polarizations involved in the frequency mixing process.

2. Choose the materials of core, n_c, the bi-layers of the Bragg reflectors, n_1 and n_2, and the matching-layer, n_m, considering material constraints such as absorption,

oxidation, ...

3. Choose the core thickness.

4. Given the refractive index profile of the structure at the wavelength of each harmonic, determine the range of the mode indices, $[n_{\text{eff},1}, n_{\text{eff},2}]$, within which phase-matching condition might be satisfied.

5. For $n_{\text{eff}} \in [n_{\text{eff},1}, n_{\text{eff},2}]$, given the polarization of the Bragg mode, calculate the matching-layer thickness, t_m, using Eq. (5.6) or Eq. (5.7).

6. Determine the thicknesses of the Bragg reflectors bi-layers, $t_1 and t_2$, using Eq. (3.28).

7. Given the number of periods of bi-layers within each Bragg reflector, construct the structure using (n_c, n_m, n_1, n_2) and (t_c, t_m, t_1, t_2).

8. Solve for TE- and TM-polarized TIR modes for the harmonics propagate as bounded modes.

9. Given the dispersion of Bragg mode and TIR modes as a function of n_{eff}, determine the value of n_{eff} for which phase-matching condition is satisfied.

10. Determine the nonlinear properties of the design including nonlinear overlap factor, nonlinear conversion efficiency and structure effective second-order nonlinearity.

11. Use perturbations in waveguide parameters, particularly core thickness and the indices of the core and matching-layer, and go to step 1 to search for a locally optimum design.

12. Using a 2-D mode solver, investigate the effect of waveguide ridge width and depth on nonlinear properties such as the detuning of the phase-matching wavelength from the nominal 1-D case.

To investigate the properties of ML-BRW systematically, we scanned the parameter space of x_m and x_c, while noting the waveguide salient characteristics. Using the above algorithm, the properties of a range of phase-matched ML-BRW structures were examined. The parameters of these structures that are pertinent to efficient PM will be discussed next.

Initial simulations indicated that maximized nonlinear overlap factor between the modes, hence maximized nonlinear conversion efficiency, could be obtained when (x_c, x_1, x_2) were chosen to have lower index of refraction in comparison to x_m. The effect of this choice is an improved confinement of the TIR mode at the pump frequency within the core. The fact that the core thickness of ML-BRW is a pre-determined parameter demonstrates the flexibility this class of structures offers in comparison to conventional QtW-BRWs. This is additionally important for reducing the insertion loss associated with end-fire coupling in free-space optics settings. For the structures studied in this work, a locally optimum core thickness of $t_c = 490$ nm was found for maximizing the nonlinear overlap factor. This choice of core thickness implied $\approx 27\%$ increase in t_c in comparison to the core thickness of BRW_{opt}. Also, in the design of the matching-layer, Eq. (5.7) was used with the scaling factor of $p = 1$. This provided significant enhancement in the parameters of interest for SHG when compared to those obtained for $p = 0$. The range of values spanned for x_m and x_c was $0.20 < x_m < 0.35$ and $0.45 < x_c < 0.65$. This ensured that there exist Bragg modes where phase-matching condition could be satisfied, while the other layers adhere to the restrictions on the aluminum content including oxidation and material absorption. The TBRs consist of $Al_{0.25}Ga_{0.75}As/Al_{0.70}Ga_{0.30}As$ bi-layers ($x_1 = 0.25$, $x_2 = 0.70$) with 6 periods to provide stacks reflectivity close to unity as well as keeping the overall structure thickness below 9 μm as a fabrication constraint. It should be stressed that in ML-BRWs one can consider matching-layers as part of the core. In this case, the overall thickness of the three-layer core is significantly increased in comparison to the single-layer core in QtW-BRWs. To illustrate this, in Fig. 5.2(a), we have plotted the dependency of matching-layer thickness on (x_c, x_m). From the figure, it can be seen that t_m varies in the range of $313 - 427$ nm denoting an increase of $\approx 190 - 248\%$ in the thickness of three-layer core in ML-BRW in comparison to the

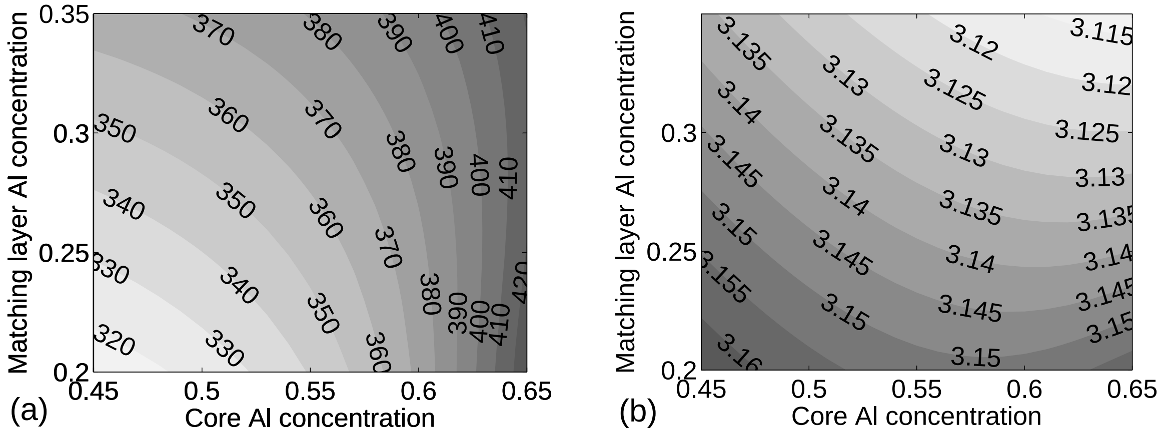

Figure 5.2: (a) Matching-layer thickness [nm] and (b) type-I phase-matched effective mode index as functions of (x_c, x_m).

single-layer core thickness of QtW-BRW$_{\text{opt}}$. Also in Fig. 5.2(b), the effective mode index of phase-matched ML-BRWs are plotted. For the entire sweeping range of (x_c, x_m), the calculated mode indices at pump wavelength was higher than n_c, the index of core. This justified the argument that in ML-BRWs, the real core of the structures is the three-layer core consisting of the center layer and the two matching-layers on the sides.

5.1.1 Nonlinear properties

Performance of a SHG device is quantitatively characterized by the nonlinear conversion efficiency which is expressed by Eq. (2.27). For a given device length, the nonlinear conversion efficiency can be improved by enhancing the nonlinear overlap factor, ξ, structure effective nonlinear coefficient, d_{eff}, and increasing the device length, L. The dependency of conversion efficiency on all these three parameters are quadratic. Theoretically, it might appear that the most simple route for enhancing η is increasing the device length. However, practically, this is not the case once degrading effects of linear/nonlinear propagation losses and dispersion effects are accounted for. In Figs. 5.3, the contour plots of ξ, d_{eff} and η_{norm} as functions of (x_c, x_m) are shown. From Fig.5.3(c), it can be seen that noticeable improvement can readily be achieved for larger x_c with smaller x_m. For

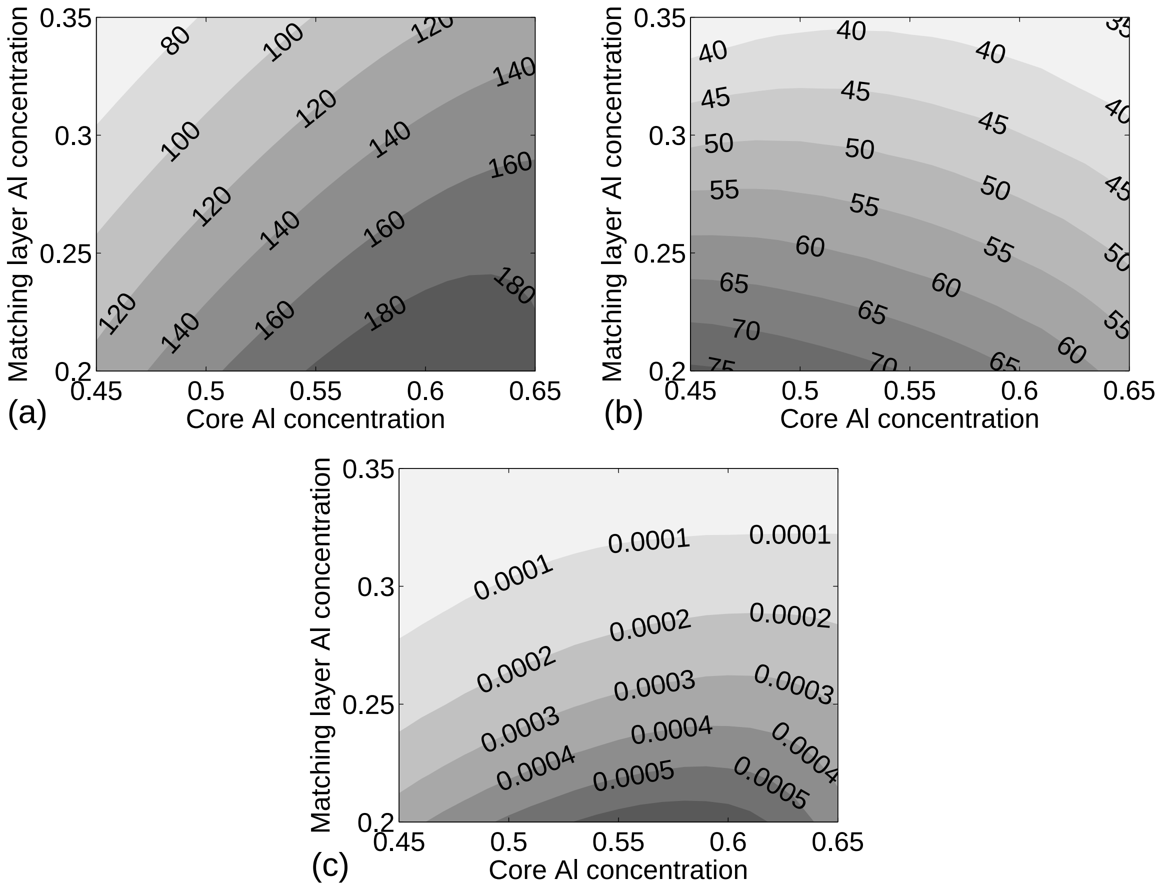

Figure 5.3: (a) Nonlinear overlap factor, ξ [m^{-1}], (b) structure effective nonlinear coefficient, d_{eff} [pm/V] and (c) normalized nonlinear conversion efficiency, η_{norm} [$\%\mathrm{W}^{-1}\mathrm{cm}^{-2}$], as functions of (x_c, x_m). Maximum conversion efficiency is simulated for the design with $(x_c, x_m) = (0.58, 0.20)$ with $\xi = 197.0\ \mathrm{m}^{-1}$, $d_{\mathrm{eff}} = 76$ pm/V and $\eta_{\mathrm{norm}} = 6.73 \times 10^{-4}\ \%\mathrm{W}^{-1}\mathrm{cm}^{-2}$.

$(x_c, x_m) = (0.58, 0.20)$, η_{norm} increases from $5.80 \times 10^{-5}\ \%\mathrm{W}^{-1}\mathrm{cm}^{-2}$ for QtW-BRW$_{\mathrm{opt}}$ to $6.73 \times 10^{-4}\ \%\mathrm{W}^{-1}\mathrm{cm}^{-2}$ for the optimum ML-BRW. In order to better understand how the matching-layer affects the nonlinear coupling efficiency, the normalized fields of the two designs were plotted as illustrated in Fig. 5.4. The enhanced η_{norm} in ML-BRW can be attributed to three factors. First, the inclusion of the matching-layer assists in increasing the core dimension, where the overlap between the harmonics is maximum. Second, within the first period of the Bragg stacks where the fields of the harmonics are at opposite phases in QtW-BRW$_{\mathrm{opt}}$, ML-BRW benefits from weaker out of phase field

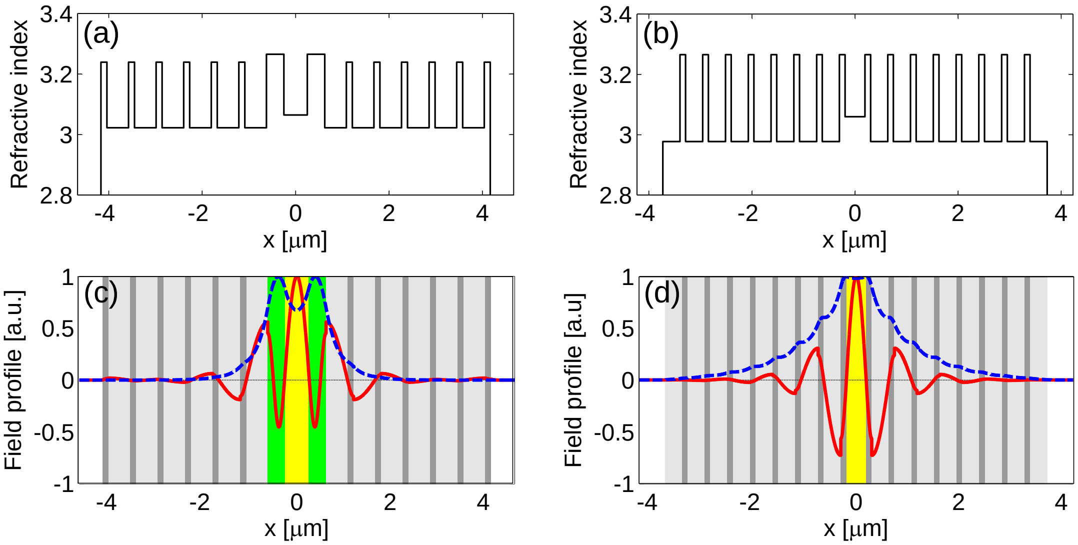

Figure 5.4: (a) Index profile of ML-BRW and (b) that of QtW-BRW$_{\text{opt}}$ at pump wavelength of 1540 nm. (c) Normalized profiles of ML-BRW and (d) QtW-BRW$_{\text{opt}}$. The dashed-line in (c) and (d) are the profiles of TE-polarization, $E_{\omega}^{y}(x)$, and the solid-lines are the profiles of TM-polarization, $E_{2\omega}^{x}(x)$ [1].

amplitudes. Third, the inclusion of the matching layer results in the sign change of the Bragg mode within the n_1 layer beside the matching-layer which gives rise to constructive overlap between the profiles.

A useful figure of merit which represents the matching-layer control over the field profiles of the harmonics is the confinement factor, Γ. It is defined as the ratio between the power within the core to the overall mode power. In calculating Γ for ML-BRW, the confined power in the complex three-layer core including the matching layers beside n_c should be considered. For QtW-BRW$_{\text{opt}}$, for the pump mode, the confinement factor was calculated to be $\approx 0.31\%$ and it increased to $\approx 0.86\%$ for ML-BRW. For the same structures, the enhancement in Γ was less pronounced for second-harmonic modes. In this case, the confinement factor was increased from $\approx 0.43\%$ for For QtW-BRW$_{\text{opt}}$ to $\approx 0.64\%$ for ML-BRW.

To further investigate the effect of core thickness on nonlinear parameters, the dependencies of ξ, d_{eff} and η_{norm} on t_c was simulated as illustrated in Figs. 5.5(a)-(c). From the

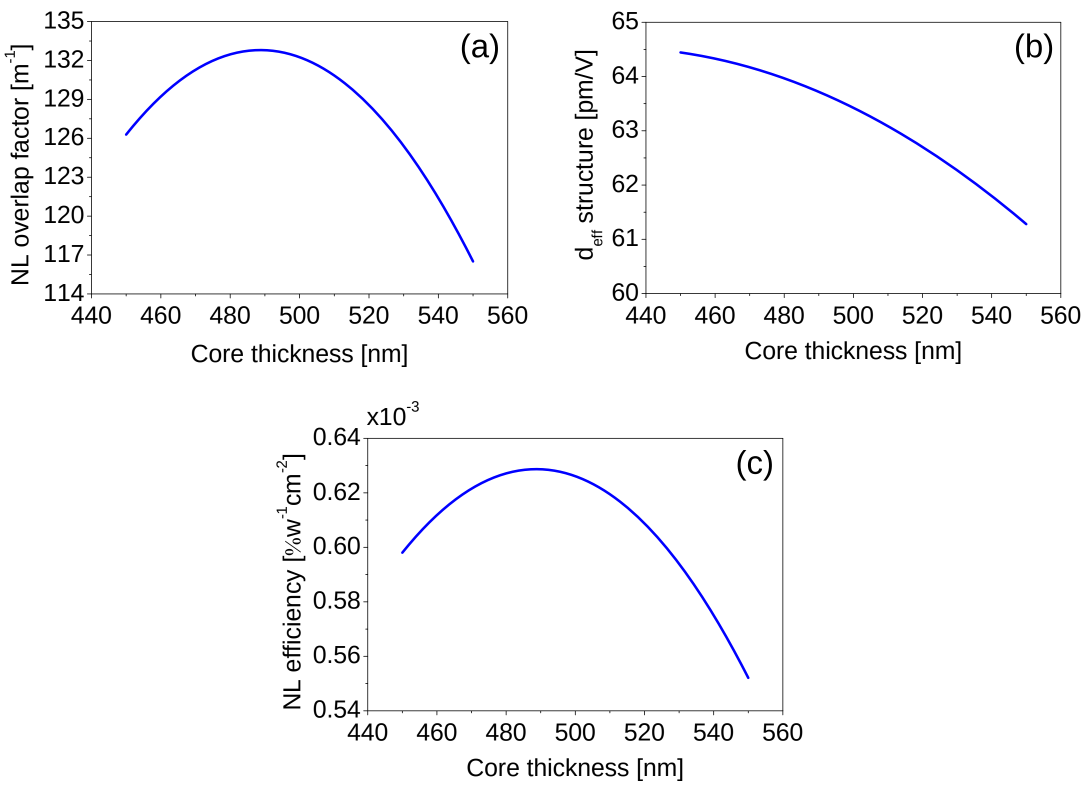

Figure 5.5: (a) Nonlinear overlap factor, ξ [m^{-1}], (b) structure effective nonlinear coefficient, d_{eff} [pm/V] and (c) normalized nonlinear conversion efficiency, η_{norm} [%W^{-1}cm^{-2}], as functions of core thickness with $(x_c, x_m) = (0.58, 0.20)$ and $(x_1, x_2) = (0.70, 0.25)$ where an optimized ML-BRW was simulated.

figures, the peak nonlinear conversion efficiency was associated with $t_c = 490$ nm which justifies the initial choice of the core thickness. The existence of an optimum (η, t_c) pair can also be important from a practical perspective. Due to strong dependence of modal dispersion on core thickness in BRWs, operating at an optimum core thickness results in reducing the sensitivity of the design to unavoidable fabrication tolerances during growth. It should be noticed that the design of $(x_c, x_m, t_c) = (0.58, 0.20, 490 \text{ nm})$ is not necessarily a global optimum; it is a representative design with substantial advantages over its conventional QtW-BRW counterpart. A global optimum might be attained through advanced nonlinear optimization algorithms.

5.1.2 Group velocity and group velocity mismatch

In parametric processes involving ultrashort pulses, dispersion effects should be carefully regarded in the nonlinear interaction. Of particular importance is the temporal pulse walk-off between the harmonics which substantially degrades the nonlinear conversion efficiency. By definition, the group velocity of an optical signal is

$$v_g = \frac{\partial \omega}{\partial \beta} = c \left| n_{\text{eff}} - \lambda \frac{\partial n_{\text{eff}}}{\partial \lambda} \right|^{-1} \tag{5.8}$$

The two signals which initially overlap in time lose this overlap due to the mismatch between their group velocities. Quantitatively, this can be expressed by the group velocity mismatched (GVM) between the harmonics. For perfect phase-matching in a type-I interaction, GVM is given as

$$\text{GVM} = \left| \frac{1}{v_{g,\omega}} - \frac{1}{v_{g,2\omega}} \right| = \frac{\lambda}{c} \left| \frac{1}{2} \frac{\partial n^{(\text{TM})}_{\text{eff},2\omega}}{\partial \lambda} - \frac{\partial n^{(\text{TE})}_{\text{eff},\omega}}{\partial \lambda} \right| \tag{5.9}$$

The effect of GVM becomes particularly pronounced in parametric processes, where some of the interacting harmonics are in proximity to the waveguide material bandgap. However, the control of waveguide dispersion offered by BRWs serves to mitigate the effect of GVM on reducing the effective interaction length. The contour plots of group velocities of FH and SH as well as the group velocity mismatch for ML-BRW are illustrated in Figs. 5.6(a)-(c). From Fig. 5.6(c), a minimum GVM of ≈ 2.2 ps/mm was obtained for $(x_c, x_m) = (0.65, 0.20)$. In comparison with QtW-BRW$_{\text{opt}}$, the new structure benefits from $\approx 14\%$ reduction in GVM. The comparison between the two design reveals that the enhancement offered by ML-BRW in reducing GVM is not significant. However, this reduction is associated with slight increase in the effective interaction length of a waveguide. Considering GVM as the primary parameter for design optimization, one can expect to further curtail the GVM in ML-BRWs by optimized choice of layers constituents

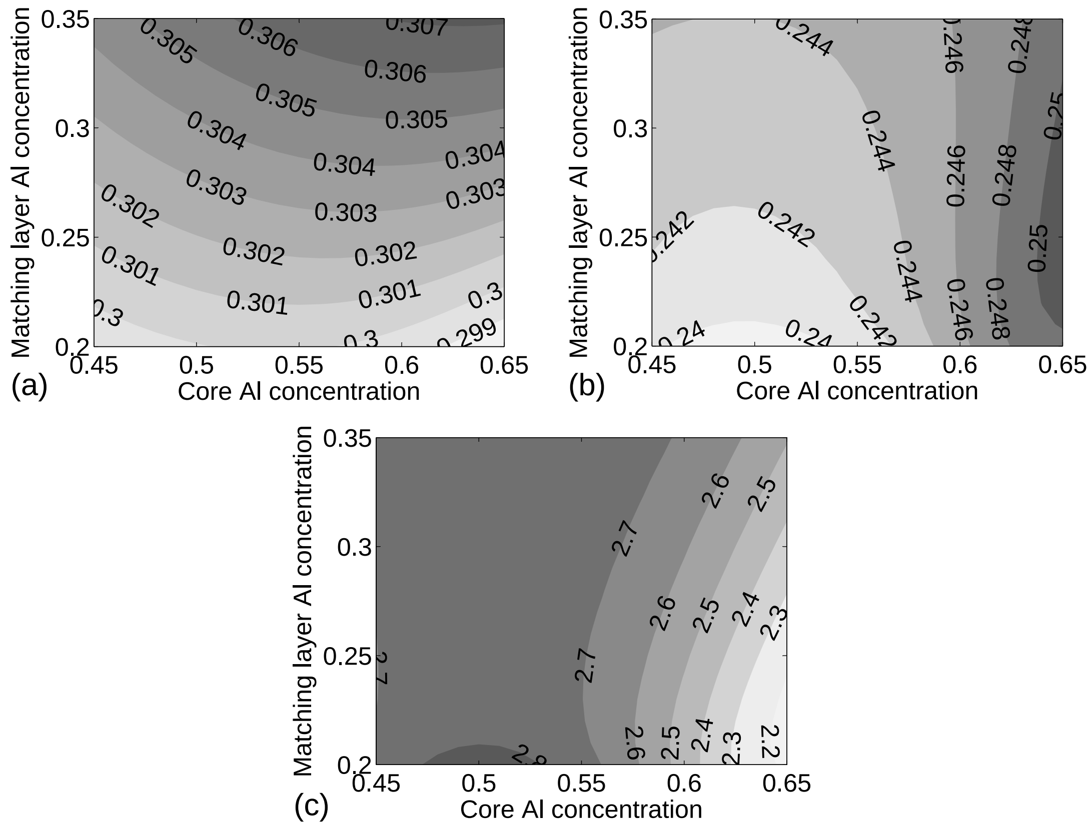

Figure 5.6: (a) Group velocity of TE-polarized pump and (b) TM-polarized second-harmonic as functions of (x_c, x_m) normalized to speed of light in vacuum. (c) Group velocity mismatch [ps/mm] between pump and second-harmonic modes.

and their thicknesses.

5.1.3 Group velocity dispersion

Tailoring the group velocity dispersion (GVD) is essential in many applications involving ultra short pulses. Certain waveguide designs demand minimized GVD at the operating wavelength to reduce the pulse distortion along the propagation distance. In applications concerned with temporal solitons in materials with nonlinear refractive indices, GVD determines the threshold intensity for soliton formation [90]. In quantum optics, tailoring the group velocity dispersion plays an important role in controlling the frequency correlation of photon-pairs generated through spontaneous parametric down-conversion (SPDC) [91]. Dispersion properties of QtW-BRWs have been comprehensively analyzed

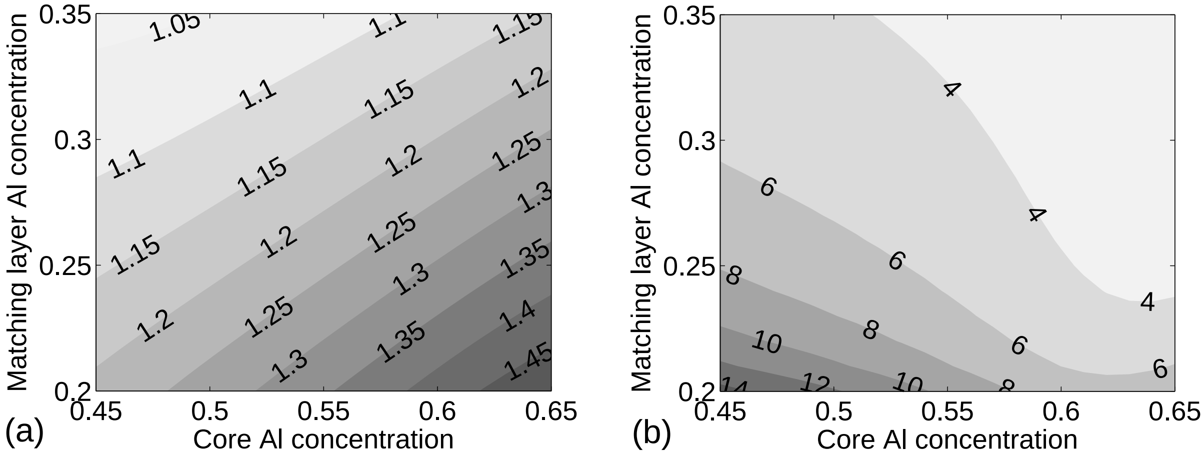

Figure 5.7: (a) Group velocity dispersion [fs^2/μm] of TE-polarized pump and (b) TM-polarized second-harmonic as functions of (x_c, x_m).

in [44] using perturbation method. Here, we present the numerical modelling of GVD for ML-BRWs. By definition

$$\text{GVD} = \frac{\partial^2 \beta}{\partial \omega^2} = \frac{\lambda^3}{2\pi c^2} \frac{\partial^2 n_{\text{eff}}}{\partial \lambda^2} \tag{5.10}$$

Plots of GVD versus (x_c, x_m) for TE-polarized pump and TM-polarized SH are shown in Figs. 5.7(a) and (b), respectively. At pump frequency GVD decreases with the increase of x_m. For the choice of (x_c, x_m) where nonlinear coupling efficiency is high, the GVD of the FH is larger compared to that of QtW-BRW$_{\text{opt}}$ and reaches a minimum value of ≈ 1 fs^2/μm. However, the GVD at 2ω exhibits stronger variations with respect to the design parameters. A minimum of ≈ 2.1 fs^2/μm is obtained for $(x_c, x_m) = (0.65, 0.35)$.

In the picosecond regime, the effect of group velocity dispersion on SH pulses is negligible, and becomes severe when optical pulses as short as few tens of femtoseconds are employed [92, 93]. To overcome the problem of pulse distortion, usually the interaction length of the device is reduced at the expense of NL interaction efficiency. The capability of ML-BRWs to provide reduced GVD at the SH can offer significant advantages for SHG with ultrashort pulses without sacrificing the effective interaction length. It should be noted that the parameter space, namely (x_c, x_m), where GVD at 2ω is small can be tailored to partly overlap with that providing large nonlinear efficiency. The capability

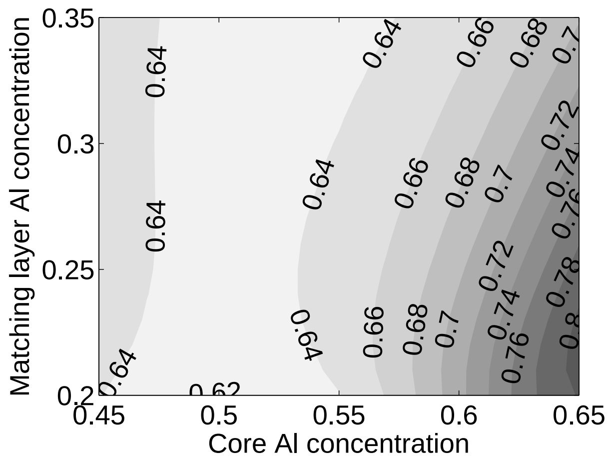

Figure 5.8: Phase-matching bandwidth, λ_{FWHM} [nm], of ML-BRW as a function of (x_c, x_m) for device length of $L = 2$ mm.

of enhancing the effective nonlinearity while reducing the GVD is remarkable which has not been investigated in other phase-matching schemes.

5.1.4 Wavelength acceptance bandwidth

As the last studied parameter, we examined the dependency of phase-matching bandwidth ($\Delta\lambda_{\mathrm{FWHM}}$), defined in Eq. (4.2), on (x_c, x_m). The simulated result is shown in Fig. 5.8 for device lengths of $L = 2$ mm. From the figure, the theoretical bandwidth of ML-BRW varies between $0.62 - 0.81$ nm indicating increased spectral bandwidth compare to the value of 0.57 nm simulated for QtW-BRW$_{\mathrm{opt}}$. From Fig. 5.8, it is interesting to note that $\Delta\lambda_{\mathrm{FWHM}}$ shows opposite behaviour with respect to the increase of x_c and x_m. For a fixed core aluminum concentration, the increase of x_m, dominantly, results in a decrease in phase-matching bandwidth while for a fixed x_m, bandwidth increases with the increase of x_c. This suggests that for the simulated ML-BRWs, the effect of matching-layer with high refractive index on modal dispersion of pump and second-harmonic is more pronounced in comparison to the low index core.

5.2 Device description and fabrication

The simulated structures in the previous section was used as the basis for the design and fabrication of a new wafer referred to as ML-BRW-SH3. The wafer was grown on nominally undoped GaAs [100] substrate using metal-organic chemical vapour deposition (MOCVD). It had a 200 nm $Al_{0.85}Ga_{0.15}As$ buffer grown on the substrate and was capped by 50 nm GaAs. The bi-layers of this new design were taken as $(x_1, x_2) = (0.70, 0.25)$ with associated thicknesses of $(t_1, t_2) = (461, 129)$ nm. The core was $Al_{0.61}Ga_{0.39}As$ with 500 nm thickness. This was in contrast to the simulated optimized ML-BRW discussed in the previous section where $(x_c, t_x) = (0.58, 490\text{ nm})$. The reason behind the slight offset between optimum simulated design and the fabricated device was due to our initial choice of the order of n_1 and n_2 layers [1]. The matching-layer was 375 nm thick $Al_{0.20}Ga_{0.80}As$. The TBRs were symmetric with respect to the core and consisted of 6 periods. The wafer was designed for type-I SHG for a pump at 1540.0 nm. However, simulations denoted that type-II interaction was also supported for a pump at 1546.5 nm. Table-5.2 summarizes the theoretical predictions of the nonlinear parameters of interest for ML-BRW-SH3.

Table 5.2: ML-BRW-SH3 nonlinear parameters for type-I and type-II SHG

Parameter	Type-I	Type-II
λ_p [nm]	1540.0	1546.5
ξ [m]$^{-1}$	197.3	220.8
d_{eff} [pm/V]	37.1	39.6
η_{norm} [%W^{-1}cm^{-2}]	2.15×10^{-4}	3.05×10^{-4}

5.3 Characterization results

In this section, we discuss the characterization of ML-BRW-SH3 sample for various nonlinear processes including second-harmonic generation, sum-frequency generation and

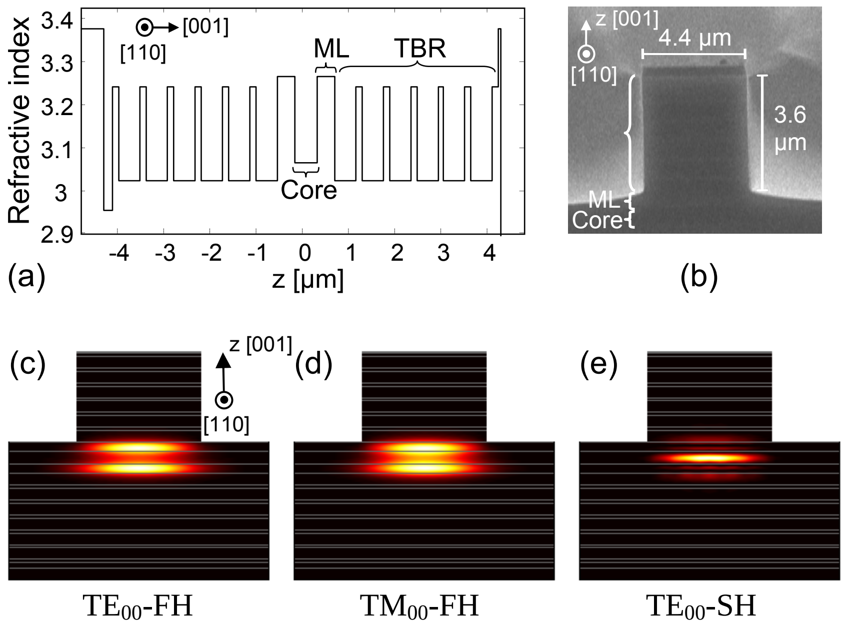

Figure 5.9: (a) Index profile of ML-BRW-SH3 wave. (b) Cross section SEM of a characterized waveguide [9]. (c) Simulated intensity profile of TE-polarized pump, (d) TM-polarized pump and (e) TE-polarized SH.

difference-frequency generation. We also discuss the observation of a novel type-0 SHG in these samples where the both pump and SH poses TM-polarization state.

5.3.1 Second-harmonic generation

Ridge waveguides with variable widths ranging between 2.8−4.8 μm were patterned along [110] direction through dry etching the ML-BRW-SH3 wafer to a depth of 3.6 μm. The waveguides had a length of 2.17 mm. The index profile of the fabricated device along with the cross section scanning-electron-microscope micrograph (SEM) of the characterized waveguide are shown in Figs. 5.9(a) and (b), respectively. Also shown in Figs. 5.9(c)-(e) are the simulated intensity profiles of the harmonics involved in a type-II SHG process.

The waveguide's linear properties were characterized using Fabry-Pèrot method with a single-mode tunable laser source in the wavelength range of 1550 nm – 1560 nm. Best device performance was obtained for a waveguide with a ridge size of 4.4 μm. For this waveguide, loss values for TE- and TM-polarization states were measured to be 2.0 cm^{-1}

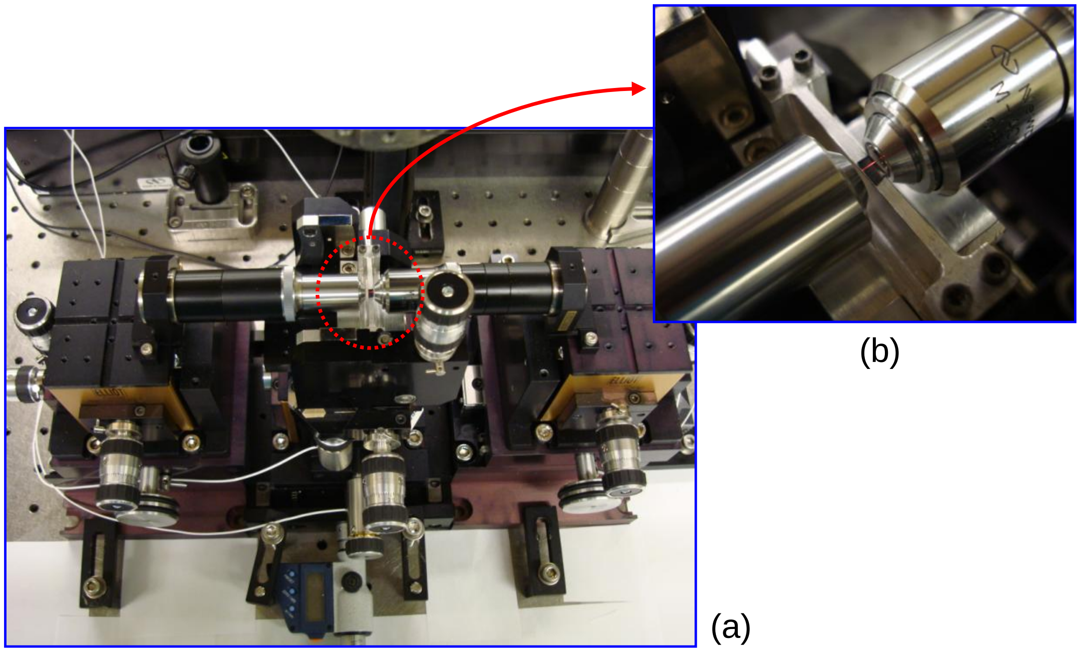

Figure 5.10: (a) End-fire rig setup with ML-BRW-SH3 sample in an SHG experiment. (b) Zoomed device mounted on sample manipulator. The red light on top of the device was the second-harmonic signal scattered from the top surface at phase-matching wavelength of 1555 nm in a type-II interaction. It was disappeared by detuning the wavelength from the phase-matched point.

and 2.2 cm^{-1}, respectively. Fresnel reflection at the waveguide facets were estimated to be 29%. Linear coupling efficiency due to spatial overlap between the pump beam and the excited fundamental mode was measured to be 49%. It is worth mentioning that estimation of linear characteristics of BRWs can be problematic as a result of the support for multi-mode propagation. However, the clear Fabray-Pèrot fringes of the transmitted spectrum ensured the detection of only the lowest order mode of the pump (FH) as desired. Although, higher order modes were likely excited at the front facet, they were filtered out from the transmitted spectrum through loss discrimination mechanism.

Characterization of waveguide nonlinear properties was carried out in an end-fire rig setup (see Fig. 5.10), using a mode-locked Ti:Sapphire pumped OPO. The employed pulses had a temporal width of 1.8 picosecond with a repetition rate of 76 MHz. The pulses were nearly transform limited with a temporal-spectral bandwidth product of

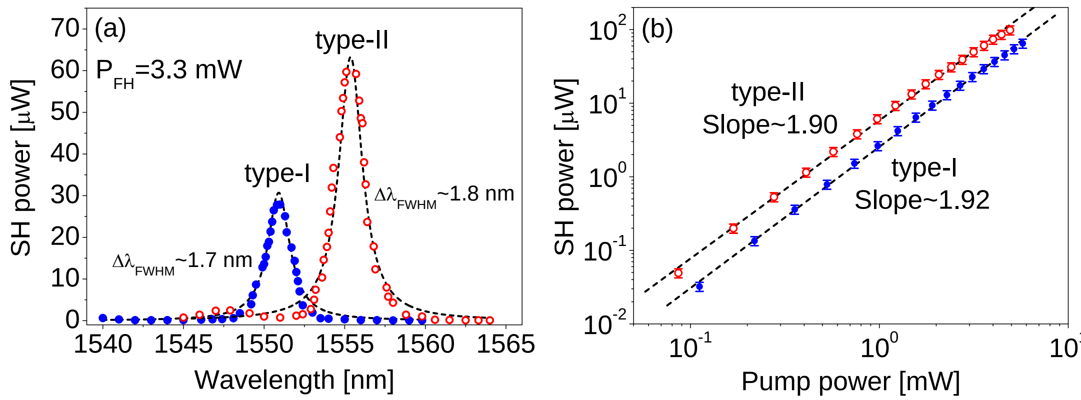

Figure 5.11: (a) Measured second-harmonic power as a function of pump wavelength in type-I (filled-circle) and type-II (circle) interactions for a fixed internal pump power of 3.3 mW. The dashed-lines are the best fits to Lorentzian function. (b) Dependency of SH power on that of the pump plotted on a log-log scale [9].

$\Delta\tau\Delta\nu = 0.53$. A 40× diode objective lens was used at the input stage for coupling light into the waveguide. The emerging signal was collected using a 40× objective lens. A long wavelength pass filter was used in the output stage to separate the pump from the second harmonic. We examined both type-I and type-II nonlinear interactions. To investigate the polarization state of the frequency doubled signal, a polarization beam splitter was used before a silicon detector. Initially, we obtained the SHG tuning curve by monitoring second-harmonic power as a function of the pump wavelength for a fixed pump power of 9.6 mW, measured before the input objective lens. Accounting for the transmission of the input objective lens (98%), facet reflection and end-fire coupling factor, the internal pump power, $\mathcal{P}_{\mathrm{FH}}$, estimated after the waveguide front facet was obtained to be 3.3 mW. For type-I and type-II interactions, 16 μW and 33 μW peak SH power respectively was measured at the detector. Taking into consideration the transmission of the output objective lens (81%), reflection of long wavelength pass filter (96%) and the facet reflection, we estimate internal SH powers to be 28 μW and 60 μW for type-I and type-II PM. The resulted tuning curves are shown in Fig. 5.11(a) where the associated phase-matching wavelengths were measured at 1551 nm and 1555 nm.

In Fig. 5.11(b), we have plotted the SH power versus pump power on a log-log scale at phase-matching wavelengths where the dashed-lines show fit of measured data to a linear function. The approximate slope values of two well correspond to the quadratic dependence between the powers of SH and that of the pump in an ideal lossless process. Multi-photon absorption, particularly at high pump power, is accounted for the deviation of the harmonics power relation from a perfect quadratic relation. The error bars in Fig. 5.11(b) were obtained by accounting for 1% error in the transmission of the output objective lens, 1% error in the reflectivity of the waveguide facet, 8% error in the collection efficiency of the output objective lens and 1% error related to the sensitivity of the silicon photodetector (Newport 818-SL) used for measuring the SH power. As a result, we consider an overall error of 11% in estimating the generated SH power before the exit facet of the device.

The normalized internal conversion efficiencies for type-I and type-II PM was estimated to be 5.30×10^3 $\%\mathrm{W}^{-1}\mathrm{cm}^{-2}$ and 1.14×10^4 $\%\mathrm{W}^{-1}\mathrm{cm}^{-2}$, respectively. In comparison to QtW-BRW-SH1 sample discussed in section 4.2.1, the estimated values of conversion efficiencies of ML-BRW-SH3 denote an enhancement by over an order of magnitude which highlights the improvement offered by the matching layer structure. Also, it would be informative to compare the efficiency of our device with the work in [33] where, to the best of our knowledge, record-high pulsed SHG conversion was measured in a type-I birefringent phase-matched (BPM) GaAs/Al_2O_3 waveguide. In the referenced work, 200 femtosecond pulses were used in a 1 mm long waveguide. Second-harmonic power of 650 μW was then detected for a pump with 5 mW internal power. We extracted the pulse conversion efficiency of [33] to be 2.6×10^5 $\%\mathrm{W}^{-1}\mathrm{cm}^{-2}$. For ML-BRW-SH3 device, type-I and type-II interactions are less efficient by approximately 1.7 and 1.4 orders of magnitude, respectively, when compared with the BPM sample. However, a more meaningful comparison could be obtained by mapping the value of η_{norm} in [33] into the picosecond regime. Such conversion is derived by calculating the ratio of the pulse duty cycles in [33] (1.8×10^{-5}) to that in our experiment (1.37×10^{-4}). As a result, the fem-

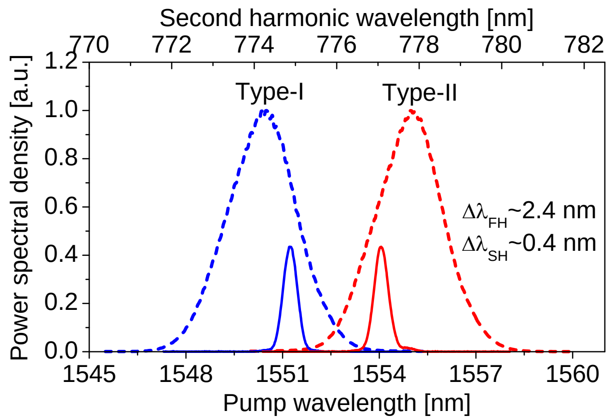

Figure 5.12: Spectra of the pump (dashed-line) with a FWHM bandwidth of ≈ 2.4 nm and the corresponding SH spectra (solid-line) in type-I and type-II interactions [9].

tosecond conversion efficiency in [33] should be scaled down with a calculated factor of 0.13 to an equivalent picosecond value of 3.38×10^4 %W^{-1}cm^{-2}. This compensated value is within the same order of magnitude as our measured type-II interaction. It should be stressed that the actual conversion efficiencies of ML-BRW-SH3 were indeed larger than our estimated values here. This was justified by the fact that in our calculations of η_{norm}, the collection efficiency of the output objective lens was considered to be 100%. Due to the fast divergence of BRW modes from the output facet, only part of the generated SH power could be collected during the experiment.

The power spectral densities of the harmonics at the phase-matching wavelengths are illustrated in Fig. 5.12. From the figure, the full-width at half-maximum bandwidths of the pump (dashed-line) and SH (solid-line) were measured to be 2.4 nm and 0.4 nm, respectively. The narrower SH spectrum clearly indicates that only portion of the pump spectrum was utilized during the nonlinear interaction. Further improvements in nonlinear conversion can be expected through incorporating a pump signal with closer spectrum match with that of SH.

We further characterized linear/nonlinear properties of waveguides with ridge sizes 2.8 μm, 3.6 μm and 4.8 μm. Of particular interest was to investigate any correlation between linear propagation loss (α_ω) and normalized conversion efficiency. In Fig. 5.13,

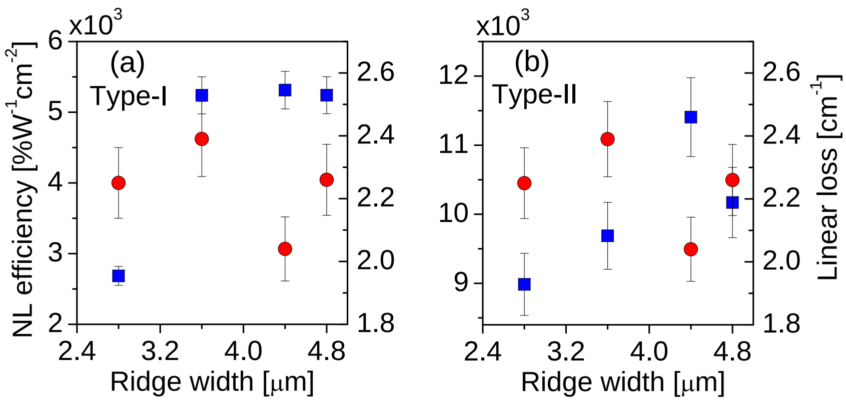

Figure 5.13: Normalized conversion efficiency (squares) and linear propagation loss (circles) as a function of waveguide ridge width for (a) type-I and (b) type-II phase-matching [9].

we have plotted α_ω and η_{norm} as functions of the ridge size for both type-I and type-II PM. From the figure, a clear correlation between α_ω and η_{norm} could not be established for either nonlinear interactions. For example, consider type-I process in waveguides with 2.8 μm and 4.8 μm ridge widths. Although, in this case, the linear loss of both waveguides were approximately the same (2.3 cm^{-1}), the conversion efficiency of the former was almost half of that of the latter waveguide. This behaviour indicates that the NL efficiency is not sensitive to observed variations of the fundamental propagation loss. Rather, we speculate the SH loss played a more significant role on the NL efficiency. For the waveguide with 4.4 μm ridge size, using the Lorentzian fit as depicted in Fig. 5.11, SH loss values of 43 cm^{-1} and 46 cm^{-1} were obtained for type-I and type-II processes, respectively.

It should be noted that the above values for the propagation losses of second-harmonic are believed to be an over estimation of the real loss values. This statement is supported by the fact that in fitting the measured second-harmonic powers to the Lorentzian fits in Fig. 5.11(a) we did not include the spectral width of the 2-ps pulsed pump. The major source of the second-harmonic loss is considered to be the side-wall scattering loss due to fabrication imperfections. The substrate leakage loss is believed to be negligible as a result of the high reflectivity of the bottom Bragg mirror, which separates the core from the substrate. Also, linear absorption at the second-harmonic frequency can be neglected as the energy of the second-harmonic photons around 775 nm (1.599 ev) was well below the bandgap of the lowest aluminum content layer, $Al_{0.20}Ga_{0.80}As$, with a bandgap of 740 nm (1.677 ev). Multi-photon absorption of the second-harmonic was also negligible due to the low level of the generated SH power. As will be discussed in the next chapter, for high pump operation, material absorption at both pump and second-harmonic frequencies can be further curtailed by incorporating $Al_xGa_{1-x}As$ elements with higher aluminum content and larger bandgaps.

5.3.2 Sum-frequency generation

Sum-frequency generation is attractive for some nonlinear applications such as photon up-conversion to near-infrared regime (600–800 nm) where efficient, low noise, single-photon detectors are present [94, 95]. Another application is high-speed optical sampling [96, 97] using ultra-short pulses for waveform measurements and demultiplexing of time-domain multiplexed optical signals [98].

The SFG process involves the interaction of a pump at wavelength λ_p, a signal at λ_s and a sum-frequency (SF) at λ_{SF}. Conservation of energy requires that $\omega_{\mathrm{SF}} = \omega_p + \omega_s$. For maximizing power transfer among the interacting frequencies, the phase-matching condition should be satisfied: $n_{\mathrm{SF}}/\lambda_{\mathrm{SF}} = n_p/\lambda_p + n_s/\lambda_s$ where n_p, n_s and n_{SF} are the associated mode indices of the pump, signal and SF respectively. In phase-matched BRW, the pump and signal propagate as bounded modes based on total internal reflection while the SF propagate as a Bragg mode. In order to better understand the potentials of BRWs for sum-frequency generation, we characterized the ML-BRW-SH3 sample for degenerate CW sum-frequency generation. Detailed discussion is presented here.

The apparatus used to characterize SFG consisted of two CW tuneable continuous-wave (CW) laser sources in the telecommunication wavelength range. A drawing of the

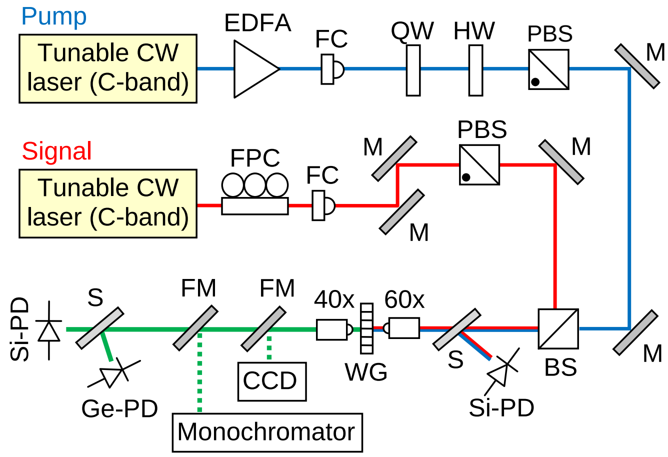

Figure 5.14: Schematic of SFG experimental setup. FPC: fibre polarization controller; FC: fiber collimator; QW: quarter-wave plate; HW: half-wave plate; PBS: polarization beam splitter; BS: beam splitter; S: beam sampler; M: mirror; WG: waveguide; FM: flip mount; Si-PD: Silicon photodetector; Ge-PD: Germanium photodetector.

setup is shown in Fig. 5.14. The Pump and signal were focused into the waveguide using a 60× objective lens. Emerging harmonics from the waveguide output were collected using a 40× microscope objective lens. An infrared camera was used for optimizing the coupled beams into the waveguide. The average powers of pump and signal at input/output stages were recorded using InGaAs photodetectors. The average power of the SF signal was monitored using a silicon photodetector. A polarization beam splitter in a TM rejection configuration was used at the output stage to eliminate any existing TM component in the SF signal, hence confirming type-II SFG, as well as rejecting the parasitic second-harmonic of TE-signal generated in type-I interaction.

The SFG tuning curve, which was obtained by monitoring the power of sum-frequency signal as a function of signal wavelength, is illustrated in Fig. 5.15(a) and is compared with simulations in Fig. 5.15(b). In obtaining the experimental curve, the average power of pump and signal were 1.98 mW and 1.00 mW, respectively, measured before the waveguide front facet. Accounting for facet reflection and input coupling factor, internal pump and signal powers were estimated to be 0.69 mW and 0.35 mW, respectively, and maintained fixed over the wavelength sweeping range. From Fig. 5.15(a), an internal

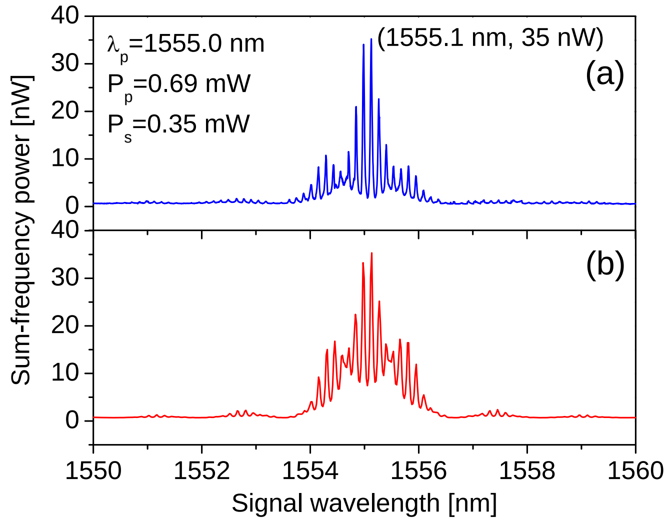

Figure 5.15: Sum-frequency power as a function of signal wavelength obtained (a) experimentally and (b) numerically with $\alpha_{\mathrm{SF}} = 5.0\ \mathrm{cm}^{-1}$. The simulated curves are normalized to the measured peak SF power to facilitate the comparison between the two models [10].

peak SF power of 35 nW was estimated at 1555.1 nm. The fast oscillating component on the spectrum indicates the resonance effects of the waveguide cavity at both signal and sum-frequency wavelengths. These resonance effects were also considered in the simulated curves of Fig. 5.15(b) where the single-pass sum-frequency power in Eq. (2.41) was multiplied by the Airy function transmission of the signal and sum-frequency waves. Simulations indicated that the envelope and fringes of calculated and measured curves agreed best for values of α_{SF} less than 6 cm^{-1}. However, due to the aforementioned challenges in measuring α_{SF}, these values could not be confirmed experimentally. It should be noted that the low powers used in the experiment ensured the elimination of any third order nonlinear effects and attest to the efficiency of the process.

The quantum efficiency of the device, η_q, is expressed as $\eta_q = \mathcal{P}_{\mathrm{SF}}\lambda_{\mathrm{SF}}/(\mathcal{P}_s\lambda_s)$ [95]. The measured quantum efficiency as a function of the pump power is illustrated in Fig. 5.16 for a signal with 0.35 mW internal power. One can see that η_q is proportional to the pump power at low power levels. The error bars in Fig. 5.16 were obtained by accounting for the uncertainties involved in estimating the internal SF power and the internal signal

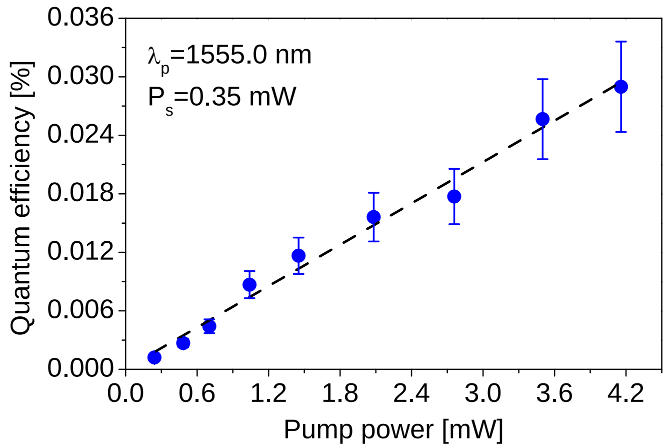

Figure 5.16: Quantum efficiency as a function of pump power. Squares are the measured data while the dashed line is the linear fit [10].

power. For the internal SF power we considered 1% error in the transmission of the output objective lens, 1% error in the reflectivity of the waveguide facet, 8% error in the collection efficiency of the output objective lens and 1% error related to the sensitivity of the silicon photodetector (Newport 818-SL) used for monitoring the SF power. For the internal signal power, we considered 1% error in input coupling factor, 1% error in reflectivity of the waveguide input facet and 2% error due to the calibration uncertainty of the germanium photodector (Newport 818-IR) used for measuring the incident signal power. As a result, we consider an overall error of 15% in estimating the generated SH power before the exit facet of the device.

Using the fit in Fig. 5.16 (dashed-line) and a simulated square of spatial nonlinear overlap coefficient of $\xi^2 = 1.14 \times 10^{-2}$ μm^{-2}, an extracted value of κ was obtained to be $\approx 1.60 \times 10^{-3}$ W$^{-1/2}$. The corresponding effective nonlinear coefficient of the structure was then calculated to be $\approx$ 40 pm/V. Using the model proposed in [3, 99], the theoretical d_{eff} of the waveguide was calculated to be 39 pm/V. The close agreement between simulated and extracted d_{eff} is remarkable since the measured d_{eff} is expected be smaller than the calculated value, given that the expression of η_q does not take into account the propagation losses. Using Eq. (2.41), at zero detuning where $\Delta k = 0$ and

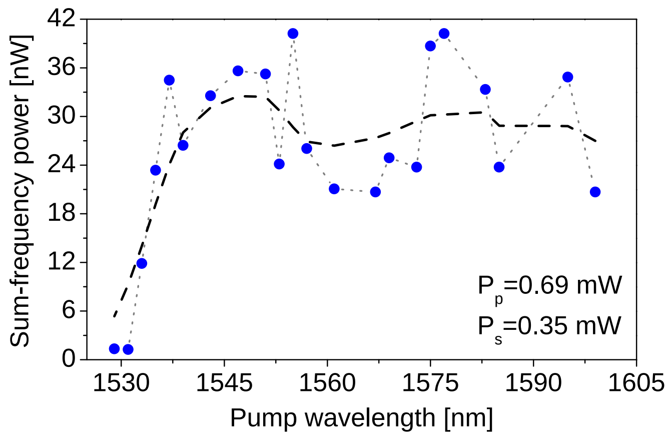

Figure 5.17: Sum-frequency power as a function of pump wavelength. Squares are the measured data while the solid-line is a smoothed fit to remove cavity resonance effect. The pump wavelength range limited the tunability on the long wavelength side [10].

neglecting propagation losses, the required pump power for a quantum efficiency of 100% is given by $\mathcal{P}_p^{100} = \lambda_s/(\xi^2\kappa_{\mathrm{SF}}^2 L^2\lambda_{\mathrm{SF}})$. The device maximum quantum efficiency was measured to be 0.029% which was obtained for a pump and signal with internal powers of 4.15 mW and 0.35 mW, respectively. The associated SF power was estimated to be ≈ 0.20 μW. While 100% quantum efficiency can be obtained with a pump power as high as 14 W, other factors such as heat generation, two photon absorption may further increase this power value. Considering the short waveguide length, this leaves ample room for decreasing $\mathcal{P}_p^{100}$ provided that propagation losses can be minimized. For example, $\mathcal{P}_p^{100}$ can be reduced to 690 mW for a device length of 10 mm. Another figure of merit which demonstrates the device performance is the normalized SFG efficiency defined as $\eta_{\mathrm{norm}} = \mathcal{P}_{\mathrm{SF}}/(\mathcal{P}_p\mathcal{P}_s L^2)$. For our device, η_{norm} was estimated to be 298 %W^{-1}cm^{-2} [100]. This suggests that the devices described here have the potential to detect single photons with further optimization, particularly by means of reducing propagation losses.

A salient parameter for a SFG device is the process phase-matching bandwidth. Here, the bandwidth was determined by detuning the pump wavelength from the degenerate wavelength of 1555.0 nm while the signal wavelength was tuned to maximize the generated SF power. The result is shown in Fig. 5.17. The large variation in the data was caused by the modulation of Fabry-Pérot resonance of signal and SF wave. Broad band phase-matching was obtained around 1550 nm with a full-width at half-maximum (FWHM) bandwidth exceeding 60 nm. The real bandwidth is believed to exceed 60 nm as the available pump source tunability prevented extending the wavelength beyond 1600 nm. This feature can be further enhanced through appropriate design of the BRW properties to provide a widely spaced signal and pump with respect to the sum-frequency [13]. This will, in turn, facilitate the spectral filtering of parasitic second-harmonic of pump/signal from the sum-frequency wave required for practical up-conversion applications in integrated single-photon detection [94, 95].

5.3.3 Difference-frequency generation

Tunable coherent infrared sources are indispensible for a vast number of optical applications including infrared spectroscopy, environmental monitoring, chemical sensing, medical diagnostics and terahertz generation. Difference-frequency generation (DFG) is an attractive technique for the generation of widely tunable coherent infrared radiation, where no appropriate laser medium exists [23, 101, 102]. Crystals such as $LiNbO_3$ and KTP have been the most popular choice of nonlinear optical materials for DFG devices. However, such crystals exhibit large material absorption above 5 μm [103], which limits their operating range for infrared generation.

To date, several techniques have been successfully reported for PM of DFG processes in AlGaAs such as form-birefringence phase-matching [104, 34] and quasi-phase-matching [37, 105]. In this section, we report type-II DFG in AlGaAs BRWs. The results discussed here are attractive for the implementation of monolithic integrated parametric devices such as self-pumped DFG devices and integrated photon-pair sources, where a nonlinear element for optical frequency mixing and a diode laser pump are fabricated on the same chip.

The DFG process involves the interaction of a pump at wavelength λ_p, a signal at

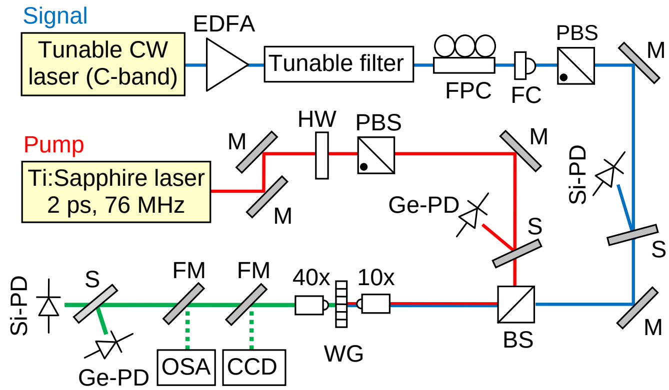

Figure 5.18: A schematic of the experimental setup. FPC: fiber polarization controller; FC: fiber collimator; HW: half wave plate; PBS: polarization beam splitter; BS: beam splitter; S: beam sampler; M: mirror; WG: waveguide; FM: flip mount; Si-PD: Silicon photodetector; Ge-PD: Germanium photodetector [11].

λ_s and a difference-frequency (DF) at λ_{DF}. Conservation of energy requires that $\omega_{\mathrm{DF}} = \omega_p - \omega_s$. For maximizing power transfer among the interacting frequencies, the PM condition should be satisfied: $n_{\mathrm{DF}}/\lambda_{\mathrm{DF}} = n_p/\lambda_p - n_s/\lambda_s$ where n_p, n_s and n_{DF} are the associated mode indices of the pump, signal and DF respectively. Using BRW phase-matching, the pump propagates as a Bragg mode while signal and DF propagate as a bounded mode based on total internal reflection (TIR).

Device characterization was carried out in an end-fire coupling setup, which is schematically shown in Fig. 5.18. A 2-ps mode-locked Ti:Sapphire laser operating at 778 nm with a repetition rate of 76 MHz was used as the pump source. The signal was taken from a tunable continuous-wave C-band tunable laser amplified by an erbium-doped fibre amplifier (EDFA). A tunable fibre grating filter was cascaded with the EDFA to suppress the amplified spontaneous emission (ASE) of the EDFA below the level of the DF. Tunability of the signal wavelength was limited to the $1532 - 1572$ nm range. The range was imposed by the C-band operation of the EDFA and the fibre grating filter. Absolute measurement of DF power requires separation of signal and idler waves at the waveguide output and use of phase-sensitive detection setup. However, due to proximity of signal and DF wavelengths, the beams could not be separated spatially neither by using

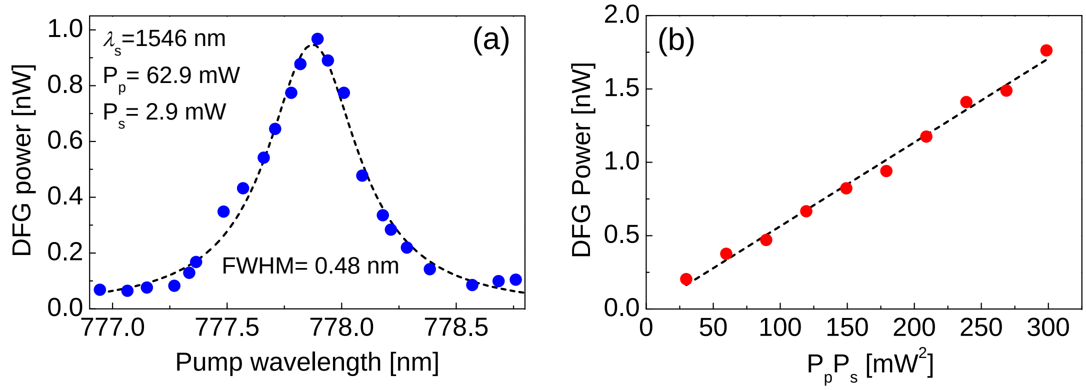

Figure 5.19: (a) The variation of the difference frequency power as a function of pump wavelength for $\lambda_s = 1545.9$ nm. Filled-circles are the measured data and the solid-line is the Lorentzian fit. (b) Dependence of $\mathcal{P}_{\rm DF}$ on $\mathcal{P}_p\mathcal{P}_s$. Open-circles are the measured data while the dashed-line is a linear fit [11].

dispersion prism or by employing narrow-line spectral filters. However, a rough estimation of the DF power was obtained by carefully calibrating an optical spectrum analyzer (OSA) and integrating the area underneath of the DF spectra recorded by the OSA. We examined type-II DFG, where TE-polarized pump and signal generated TM-polarized difference-frequecy.

The dependence of the DF power ($\mathcal{P}_{\rm DF}$) on the pump wavelength, for $\lambda_s = 1545.9$ nm, is shown in Fig. 5.19(a), where filled-circles show the measured data and the solid-line is a Lorentzian fit. In obtaining the tuning curve of Fig. 5.19(a), the average powers of pump ($\mathcal{P}_p$) and signal ($\mathcal{P}_s$), measured before the front facet of the waveguide, were fixed to 62.9 mW and 2.9 mW, respectively. From the experiment, peak DF power of 0.95 nW, estimated right after the output facet of the waveguide, was obtained for the phase-matched pump at 777.8 nm. The bandwidth of the process was found to be 0.48 nm.

In a DFG process, $\mathcal{P}_{\rm DF}$ is proportional to the product of pump and signal powers. We verified this relation for a fixed pump power of 62.9 mW and a signal sweeping power range of $0.48 - 4.8$ mW. The result is shown in Fig. 5.19(b), where the linear fit (dashed-line) agrees well with the relation $\mathcal{P}_{\rm DF} \propto \mathcal{P}_p\mathcal{P}_s$. The slope of the linear fit in the

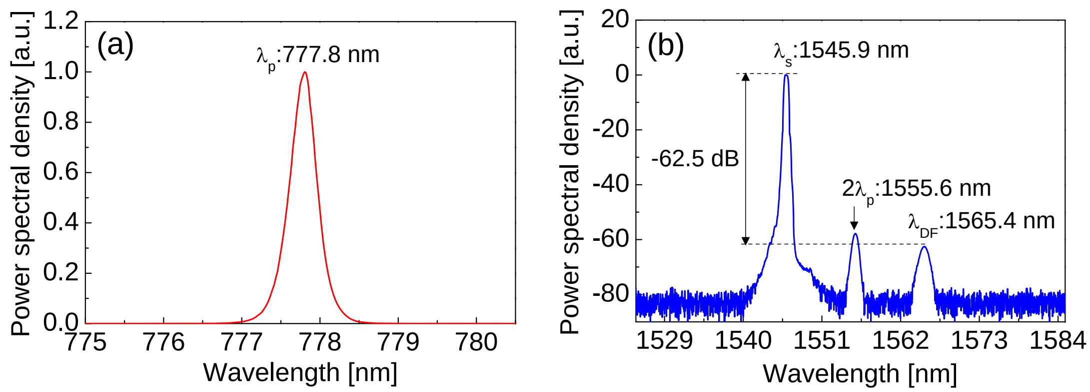

Figure 5.20: (a) The power spectral density (PSD) of pump and (b) those of the signal at 1545.9 nm and the converted difference-frequency. The central peak at 1555.6 nm is the second-order diffraction of the pump from the OSA internal grating [11].

graph of $\mathcal{P}_{\mathrm{DF}}$ vs. $\mathcal{P}_p\mathcal{P}_s$ provides an estimation of the DFG conversion efficiency defined as $\eta = \mathcal{P}_{\mathrm{DF}}/(\mathcal{P}_p\mathcal{P}_s)$. For the characterized device, η was found to be $\approx 5.7 \times 10^{-4}$ %W^{-1}. This efficiency is equal to 2.5×10^{-2} %W^{-1}cm^{-2} when normalized to the device length. It should be noted that the DFG efficiency reported here is the external value. The internal efficiency, which requires the estimation of the internal powers inside the device, is considerably larger than the external one. The internal pump power could not be determined due to difficulties in extracting the linear coupling factor defined as the spatial overlap between the incident pump beam and the excited pump, which is a Bragg mode. Simulations indicate that this coupling efficiency is likely to be a few percent, which implies that a low pump power level is likely to be responsible for the output powers measured. Several routes can be undertaken to further enhance the DFG process in BRWs. These include preferential coupling to Bragg mode using prism coupling, grating-assisted coupling or using spatial light modulators (SLM). In a more advanced scheme, these coupling techniques can be completely avoided by developing self-pump DFG devices, where a Bragg laser pump [106] with phase-matched cavity is fabricated on the same wafer platform.

A typical spectrum taken at the waveguide output is shown in Fig. 5.20(b). The pump central wavelength was set at 777.8 nm as shown in Fig. 5.20(a) with a signal at

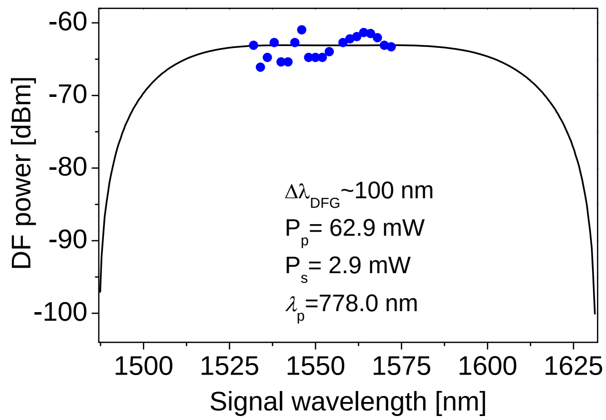

Figure 5.21: The difference-frequency power as a function of signal wavelength for the degenerate pump at 778.0 nm. The filled-circles are the measured data and the solid-line is the simulated data [11].

1545.9 nm. The DF wavelength was measured to be 1565.4 nm. From the spectra, the power level of DF was -62.5 dB lower than that of the signal. The spectra in Fig. 5.20(b) denotes an additional spectral feature at $\approx 2\lambda_p = 1555.6$ nm, which is due to the second-order diffraction of the pump from the grating of the OSA. This was confirmed by the fact that the spectral feature at 1555.6 nm remained unchanged while the DF spectra shifted in the opposite direction to the signal during tuning.

A key parameter for a DFG device is the PM bandwidth, $\Delta\lambda_{\mathrm{DFG}}$, which determines the useful spectral range within which frequency conversion is efficient. To determine $\Delta\lambda_{\mathrm{DFG}}$, the pump was set at the degenerate wavelength of 778.0 nm, while the generated DF power was monitored and recorded versus signal wavelength in the range of $1532 - 1572$ nm. The result is illustrated in Fig. 5.21, where the filled-circles show the measured data and the solid-line is the theoretical curve obtained from numerical simulations. It should to be mentioned that the PM peak wavelength obtained from simulation is about 6 nm shorter than that from the experiment. This discrepancy is attributed to the uncertainties of the fabrication processes of the device. The spread in the measured power levels was attributed to the cavity resonance effects. The C-band operation of signal laser and the fibre grating filter limited the tunability of the signal beyond the

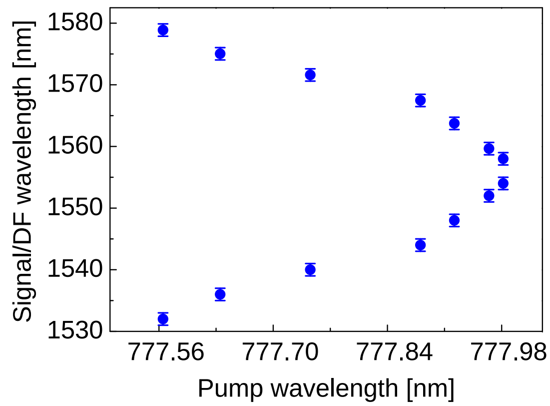

Figure 5.22: Signal and difference frequency wavelength as a function of pump wavelength [11].

range of $1532-1572$ nm. As such, complete measurement of $\Delta\lambda_{\mathrm{DFG}}$ could not be carried out. However, numerical simulations denoted that $\Delta\lambda_{\mathrm{DFG}}$ was approximately 100 nm for this device.

Figure 5.22 shows the DFG tuning curve, which was obtained by detuning the pump wavelength from degeneracy while tracking the wavelengths of signal and DF for maximal DF power. It was observed that a detuning of the pump wavelength by 0.4 nm resulted in a span of ≈ 40 nm between signal and DF wavelengths. Further detuning of the pump from degeneracy was expected to offer broader separation between signal and DF. This could not be confirmed experimentally due to the aforementioned constrains in tuning the signal wavelength.

5.3.4 Type-0 $\mathrm{TM}_{\omega} \rightarrow \mathrm{TM}_{2\omega}$ second-harmonic generation

For photonics applications, the GaAs/$\mathrm{Al}_x\mathrm{Ga}_{1-x}$As system is grown along the [001] crystal axis with equivalent [110] cleavage planes. Two types of nonlinear interactions arise from the non-vanishing susceptibility tensor elements; namely type-I interaction where a TE_{ω}-polarized pump (FH) generates $\mathrm{TM}_{2\omega}$-polarized second-harmonic (SH) and type-II interaction where a pump with hybrid $\mathrm{TE}_{\omega}+\mathrm{TM}_{\omega}$ polarization state generates $\mathrm{TE}_{2\omega}$-polarized SH. Type-II is known to be more efficient as it benefits from enhanced effective second-order nonlinearity. Beside the relative efficiency of the aforementioned interactions, their polarization selection rules govern the applications within which they are used. For instance, in generating photon-pairs with frequency correlation properties using spontaneous parametric down-conversion process (SPDC), the phase-matching type governs the polarization state of the down-converted photons. Adding other types of interactions to the existing ones $TE_\omega \rightarrow TM_{2\omega}$ and $TE_\omega + TM_\omega \rightarrow TE_{2\omega}$ would further enrich the capabilities and functionality obtained by using the $\chi^{(2)}$ nonlinearity of the GaAs/$Al_xGa_{1-x}As$ system in classical and non-classical applications.

In this section, we demonstrate a second modality of type-0 second order nonlinear interaction in $Al_xGa_{1-x}As$ Bragg reflection waveguides. Here, a TM_ω-polarized pump is used to generate second-harmonic with $TM_{2\omega}$ polarization state. The $TM_\omega \rightarrow TM_{2\omega}$ interaction has been previously reported in asymmetric multiple-quantum-well waveguides, where the broken symmetry of the structure along the z-axis induced a weak susceptibility tensor element $\chi^{(2)}_{zzz}$ [41, 107]. However, the source of $TM_\omega \rightarrow TM_{2\omega}$ interaction observed in ML-BRW-SH3 was not due to the appearance of $\chi^{(2)}_{zzz}$ element. Rather; it was ascribed to the characteristic of TM_ω propagating mode where the pump electric field has a non-zero component along the propagation direction. It is further demonstrated that the observed $TM_\omega \rightarrow TM_{2\omega}$ interaction can simultaneously take place with $TE_\omega \rightarrow TM_{2\omega}$ and $TE_\omega + TM_\omega \rightarrow TE_{2\omega}$ processes within a narrow spectral range and with comparable efficiency. This in turn allows one to utilize all three interactions provided that the phase-matching wavelengths of all three processes reside within the pump spectral bandwidth; a condition that can be easily met by using femtosecond pulses.

In order to understand the origin of this newly observed $TM_\omega \rightarrow TM_{2\omega}$ interaction, light needs to be shed on second order nonlinear interactions in $Al_xGa_{1-x}As$ semiconductors. In bulk $Al_xGa_{1-x}As$ waveguides, second order nonlinear interactions rely on accessing the nonlinear tensor element $\chi^{(2)}_{xyz}$. Customarily, this is achieved in waveguides with [001] fabrication growth direction and [110] cleaved facets. A schematic of a representative

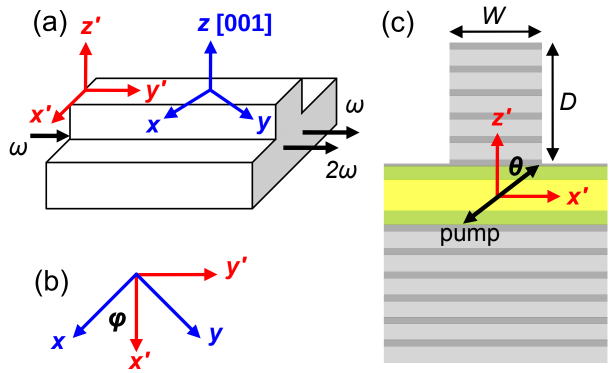

Figure 5.23: (a) Schematic of BRW where $x'y'z'$ and xyz indicating the laboratory and crystal frames, respectively. (b) Orientation of the reference frames with an orientation angle of $\phi = \pi/4$. (c) Cross section of the characterized BRW with the ridge width W and the ridge depth D [2].

BRW is shown in Fig. 5.23. The coordinate systems xyz and $x'y'z'$, in Fig. 5.23(a), denote the crystal and laboratory frames respectively. The two reference frames are rotated by an angle of $\phi = \pi/4$ with respect to the crystal z-axis (see Fig. 5.23(b)). For simplicity, the analysis is performed in the $x'y'z'$ coordinates, where the major fields of TE and TM propagating modes are aligned with the axes of the reference frame. It is further assumed that the scalar wave equations are justified in expressing the propagation of the optical modes. The modal electric field at ω_σ frequency, $\omega_\sigma \in \{\omega, 2\omega\}$, can be written as:

$$E^{\mu'}_{\omega_\sigma}(x', y', z') = A_{\omega_\sigma}(y') E^{\mu'}_{\omega_\sigma}(x', z') \exp(i k_{\omega_\sigma} y') \tag{5.11}$$

where $\mu' \in \{x', y', z'\}$, A_{ω_σ} is the slowly varying amplitude, $E^{\mu'}_{\omega_\sigma}(x', y')$ is the normalized spatial profile and k_{ω_σ} is the propagation constant. The μ'-th component of the second-order electric polarization vector at 2ω writes [18]:

$$[P^{(2)}_{2\omega}]_{\mu'} = \epsilon_0 K \chi^{(2)}_{\mu'\alpha'\beta'} E^{\alpha'}_{\omega} E^{\beta'}_{\omega} \tag{5.12}$$

where $\{\alpha', \beta'\} \in \{x', y', z'\}$ and K is the degeneracy factor, which is equal to 1/2 for SHG. The elements of the susceptibility tensor in the laboratory and crystal frames are related to each other using the transformation [18]:

$$\chi^{(2)}_{\mu'\alpha'\beta'} = R_{\mu'\mu}(\phi)R_{\alpha'\alpha}(\phi)R_{\beta'\beta}(\phi)\chi^{(2)}_{\mu\alpha\beta} \tag{5.13}$$

where $R_{j'j}(\phi)$ are the elements of the rotation matrix with the rotation angle of $\phi = \pi/4$. Using Eq. (5.13), the non-zero elements of $\chi^{(2)}$ tensor in the $x'y'z'$ frame are:

$$\chi^{(2)}_{x'x'z'} = \chi^{(2)}_{x'z'x'} = \chi^{(2)}_{z'x'z'} = +\chi^{(2)}_{xyz} \tag{5.14}$$

$$\chi^{(2)}_{y'y'z'} = \chi^{(2)}_{y'z'y'} = \chi^{(2)}_{z'y'y'} = -\chi^{(2)}_{xyz} \tag{5.15}$$

Unlike in the crystal frame, where $\chi^{(2)}_{\mu\alpha\beta}$ is non-zero for $\mu \neq \alpha \neq \beta$, from Eq. (5.14) and (5.15), it can be seen that in the laboratory coordinate system, $\chi^{(2)}_{\mu'\alpha'\beta'}$ is non-zero for those permutations of x', y' and z' for which two indices are identical while the third index is z'.

For a pump mode with TM$_\omega$ polarization state, the major field components are $(H^{x'}_\omega, E^{y'}_\omega, E^{z'}_\omega)$. Using Eqs. (5.12)–(5.15), the elements of the electric polarization vector in the laboratory frame are:

$$[P^{(2)}_{2\omega}]_{x'} = 0 \tag{5.16}$$

$$[P^{(2)}_{2\omega}]_{y'} = -\epsilon_0\chi^{(2)}_{xyz}E^{z'}_\omega E^{y'}_\omega \tag{5.17}$$

$$[P^{(2)}_{2\omega}]_{z'} = -\frac{\epsilon_0}{2}\chi^{(2)}_{xyz}E^{y'}_\omega E^{y'}_\omega \tag{5.18}$$

From Eq. (5.16), the existence of the non-vanishing electric polarization component, $[P^{(2)}_{2\omega}]_{z'}$, can support second-harmonic generation with TM$_{2\omega}$ polarization state. This SH signal does not lend itself to the broken symmetry along the crystal z-axis hence the appearance of $\chi^{(2)}_{zzz}$ tensor element, rather it originates from the pump electric field component, $E^{y'}_\omega$, which lies along the propagation direction. In order to obtain a better understanding about the strength of the second-harmonic in type-0 interaction, it would

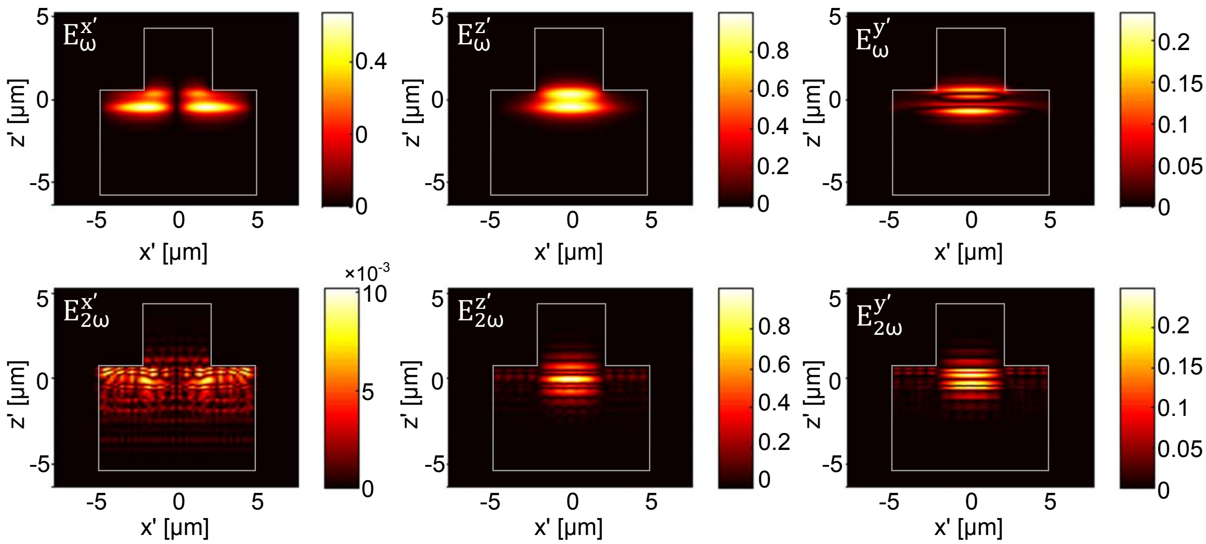

Figure 5.24: Simulated magnitudes of electric field components [V/m] for TM$_\omega$ propagating pump at ω frequency ($E_\omega^{x'}, E_\omega^{y'}, E_\omega^{z'}$) and those for TM$_{2\omega}$ propagating second-harmonic at 2ω frequency ($E_{2\omega}^{x'}, E_{2\omega}^{y'}, E_{2\omega}^{z'}$) [2]. The fields profiles were simulated using Lumerical mode solver [12].

be instructive to examine the electric field components of TM modes of both harmonics in a typical BRW. In Fig. 5.24, we have plotted the magnitudes of electric field components for TM$_\omega$ polarized pump ($E_\omega^{x'}, E_\omega^{y'}, E_\omega^{z'}$) and those for TM$_{2\omega}$ polarized second-harmonic ($E_{2\omega}^{x'}, E_{2\omega}^{y'}, E_{2\omega}^{z'}$) obtained using a full-vectorial mode solver [12] for the structure reported in [9]. From the figure, it can be deduced that the magnitude of the pump electric field along the propagation direction, $E_\omega^{y'}$, is approximately within the same order of magnitude of that of the dominate field component $E_\omega^{z'}$. As such, it is to observe the generated TM$_{2\omega}$ polarized SH with noticeable strength in the type-0 interaction. Also in Fig. 5.24, it is interesting to compare the magnitudes of the field components $E_\omega^{x'}$ and $E_\omega^{z'}$. It can be seen that $|E_\omega^{x'}|$ is comparable with $|E_\omega^{z'}|$ which can be attributed to the hybrid nature of optical modes in ridge waveguides. Unlike slab waveguides where propagating modes can be split into pure TE and pure TM modes, such decomposition in ridge structures is viable only as an approximation which is adopted for analytical simplifications. The same comparison at 2ω frequency reveals the fact that $|E_{2\omega}^{x'}|$ is almost two orders of magnitudes smaller than $|E^{z'}_{2\omega}|$. The fact that $|E^{x'}_{\omega}|$ is comparable with $|E^{z'}_{\omega}|$ while $|E^{x'}_{2\omega}|$ is significantly smaller than $|E^{z'}_{2\omega}|$ can be explained by the difference between the wavelengths of pump and second-harmonic with respect to the waveguide dimensions. Due to the longer wavelength at the pump frequency, the physical dimensions of the ridge which offer the lateral confinement become more comparable to the optical mode. As a result, the effect of 2-D confinement of the ridge in rendering the propagating mode polarization hybrid nature is more pronounced at the pump wavelength.

The expected smaller magnitude of generated second-harmonic in $\mathrm{TM}_{\omega}\rightarrow\mathrm{TM}_{2\omega}$ interaction may justify why type-0 process in $\mathrm{Al}_x\mathrm{Ga}_{1-x}\mathrm{As}$ has not received attention in the literature. However, the characterization results discussed in the next section, demonstrates that even with the existence of a small $E^{y'}_{\omega}$ component, a strong second-harmonic can be generated thanks to the high efficiency of the nonlinear interaction in Bragg reflection waveguides.

The characterized waveguide was the same sample as ML-BRW-SH3 discussed earlier. Nonlinear characterization was carried out in an end-fire rig setup using pulses with a temporal width of 2 picosecond. The pulses were nearly transform limited with a temporal-spectral bandwidth product of $\Delta\tau\Delta\nu = 0.53$ and with a repetition rate of 76 MHz. Optical pulses were coupled to the waveguide using a 40$\times$ diode objective lens ($T \approx 0.98$). The emerging signal was collected using a 60$\times$ objective lens ($T \approx 0.78$). The average power of the pump was recorded using InGaAs photodetectors. Second-harmonic power was monitored using a Si photodectector. A polarization beam splitter with TM passing configuration was used after the output objective lens to examine the polarization of generated SH.

The scan of SH power as a function of pump wavelength is illustrated in Fig. 5.25(a). We have also plotted the tuning curves of $\mathrm{TE}_{\omega}\rightarrow\mathrm{TM}_{2\omega}$ and $\mathrm{TE}_{\omega}+\mathrm{TM}_{\omega}\rightarrow\mathrm{TE}_{2\omega}$ discussed in previous section for comparison. For the current samples, the internal pump power ($\mathcal{P}_{\omega}$), after accounting for facet reflection (29%) and input coupling factor (49%), was

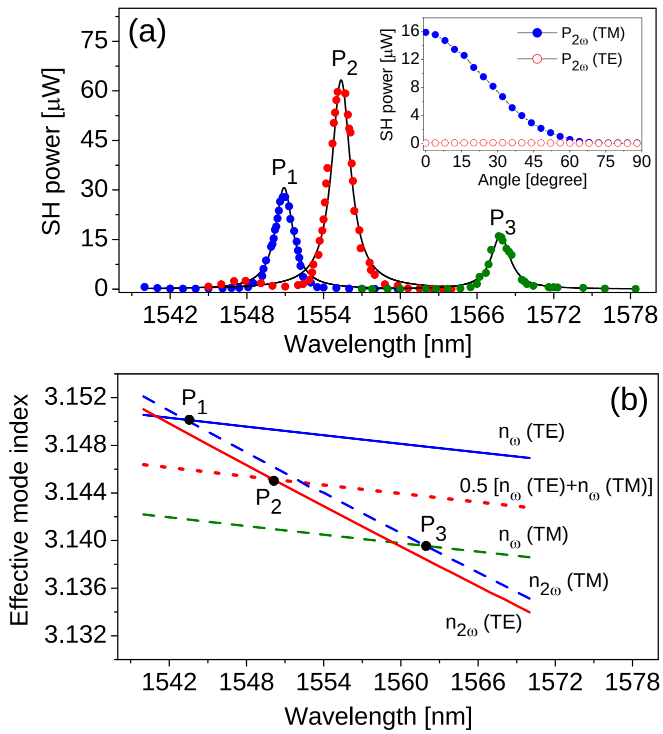

Figure 5.25: (a) Second-harmonic power as a function of pump wavelength measured for three phase-matching scheme. The solid lines are the Lorentzian fit to the measured date. (Inset) Variation of $TE_{2\omega}$ and $TM_{2\omega}$ components of SH power as a function of pump angle θ. (b) Simulated modal dispersion of fundamental modes of pump and SH. The crossing points P_1, P_2 and P_3 denote the phase-matching point for $TE_\omega \rightarrow TM_{2\omega}$, $TE_\omega + TM_\omega \rightarrow TE_{2\omega}$ and $TE_\omega \rightarrow TM_{2\omega}$ processes, respectively [2].

estimated to be 3.3 mW, which was the same pump power in the previous section. Peak internal SH power ($\mathcal{P}_{2\omega}$) estimated before the end facet of the waveguide was obtained to be 16 μW at 1567.8 nm. We calculated the normalized conversion efficiency to be 2.84×10^3 %$W^{-1}cm^{-2}$ for the sample with $L = 2.2$ mm length. The inset of Fig. 5.25(a) shows the variation of $TE_{2\omega}$ and $TM_{2\omega}$ components of SH power as a function of θ, the angle between the pump field and the z-axis of the crystal, when there was no polarization beam splitter at the output stage. As expected, the magnitude of $TE_{2\omega}$ component was practically zero, while the $TM_{2\omega}$ component was decreasing with the increase of θ.

The estimated value of η is noteworthy when compared with the efficiencies of type-I $TE_\omega \rightarrow TM_{2\omega}$ and type-II $TE_\omega + TM_\omega \rightarrow TE_{2\omega}$ interactions. A summary of the performance

Table 5.3: Summary of the three phase-matched second-harmonic generation [2]

Nonlinear Process	θ [rad]	$\mathcal{P}_{\omega}$ [mW]	$\mathcal{P}_{2\omega}$ [μW]	η_{norm} [%W^{-1}cm^{-2}]
Type-0: $\mathrm{TM}_{\omega}\rightarrow\mathrm{TM}_{2\omega}$	0	3.3	16	2.84×10^{3}
Type-I: $\mathrm{TE}_{\omega}\rightarrow\mathrm{TM}_{2\omega}$	$\pi/2$	3.3	28	5.30×10^{3}
Type-II: $\mathrm{TE}_{\omega}+\mathrm{TM}_{\omega}\rightarrow\mathrm{TE}_{2\omega}$	$\pi/4$	3.3	60	1.14×10^{4}

of the three processes is provided in Table 5.3. From the table it is apparent that the efficiency of the new type-I interaction is within the same order of magnitude as those of the $\mathrm{TE}_{\omega}\rightarrow\mathrm{TM}_{2\omega}$ and $\mathrm{TE}_{\omega}+\mathrm{TM}_{\omega}\rightarrow\mathrm{TE}_{2\omega}$ processes. To the best of our knowledge, this unique behaviour of BRWs in simultaneously offering high efficient nonlinear interaction with additional degrees of freedom in controlling the polarization of the generated harmonic is not offered by any other waveguide structure and by any other phase-matching scheme.

We further simulated the modal dispersion of the pump and second-harmonic for both orthogonal polarizations to justify the experimental phase-matching points in Fig. 5.25(a). The result is shown in Fig. 5.25(b) where P_1, P_2 and P_3 denote the phase-matching points associated with $\mathrm{TE}_{\omega}\rightarrow\mathrm{TM}_{2\omega}$, $\mathrm{TE}_{\omega}+\mathrm{TM}_{\omega}\rightarrow\mathrm{TE}_{2\omega}$ and $\mathrm{TM}_{\omega}\rightarrow\mathrm{TM}_{2\omega}$ interactions, respectively. In comparing Fig. 5.25(a) and (b), the slight mismatch of less than 7 nm between the measured phase-matching wavelengths and those predicted from simulation was inevitable due to inaccuracies of the refractive index model of $\mathrm{Al}_x\mathrm{Ga}_{1-x}\mathrm{As}$ elements [80] as well as the fabrication tolerances and imperfections.

The $\mathrm{TM}_{\omega}\rightarrow\mathrm{TM}_{2\omega}$ interaction could also exist due to the hybrid nature of modes in 2-D guided structures. We investigated the waveguide influence on polarization coupling by examining the ratio between the TM_{ω} to TE_{ω} components of the pump power before and after the waveguide. The measurement resulted in a ratio of ≈ 100 before the input objective lens while it was ≈ 21 after the output objective lens. Such a small TE_{ω} component cannot be entirely accounted for initiating a type-I $\mathrm{TE}_{\omega}\rightarrow\mathrm{TM}_{2\omega}$ interaction with the level of SH power observed here given the field ratios alone. This confirms that the observed nonlinear process was indeed type-0 $TM_{\omega} \rightarrow TM_{2\omega}$ interaction.

It can be clearly seen from Figs. 5.25(a) and (b) that the phase-matching wavelengths of all three nonlinear interactions can be semi-independently tailored by waveguide design and in this case they reside within a spectral range as small as 17 nm. This spectral range lies within the spectral bandwidth of typical ultra-short pulses with temporal widths of a few hundred femtoseconds. The close proximity of the observed phase-matching wavelengths allows one to generate a second-harmonic signal with mixed polarization states. In practice, such flexibility might be potential for certain polarization sensitive applications. An example is an integrated source of photon-pairs, where controlling the polarization entanglement is the major interest. This can be carried out in SPDC processes with ultrashort pump where the pump spectral bandwidth is large enough to cover the phase-matching wavelengths of all three type-0, type-I and type-II interactions. Generating photon-pairs with different polarization-entangled states can then be carried out in a single device by simply rotating the polarization angle of the incident pump beam (the angle θ in Fig. 5.23). Polarization insensitive applications can also benefit from the current design. For example wideband SHG can be expected using a mixed TE_{ω}/TM_{ω} polarized femtosecond pump while both polarization components of the SH are maintained.

The modal birefringence of a propagating mode at frequency ω_σ is defined as $\Delta n_{\omega_\sigma} = n_{\omega_\sigma}^{(TE)} - n_{\omega_\sigma}^{(TM)}$ where $n_{\omega_\sigma}^{(TE)}$ and $n_{\omega_\sigma}^{(TM)}$ are the effective indices of TE_{ω_σ} and TM_{ω_σ} modes. In this work, the small birefringence of the second harmonic mode played a pivotal role in obtaining the phase-matching wavelengths of all three nonlinear interactions discussed above in proximity of each other. For slab BRWs with quarter-wave Bragg mirrors, it is known that $\Delta n_{2\omega} = 0$ denoting the fact that the lowest order Bragg modes are polarization degenerate [71]. Departing from the quarter-wave condition by either imposing the lateral confinement of a ridge structure or incorporating non-quarter-wave layers in transverse Bragg reflectors (TBRs) results in polarization non-degeneracy of fundamen-

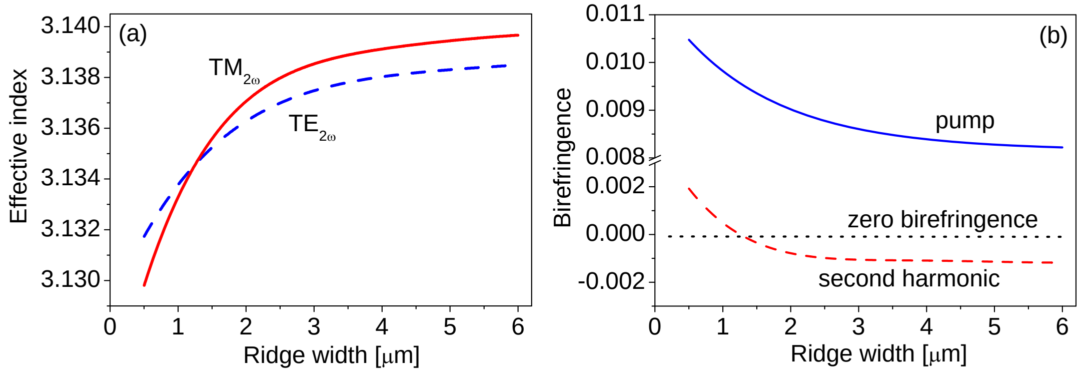

Figure 5.26: Modal birefringence of pump (solid-line) and second-harmonic (dashed-line) as functions of the waveguide ridge width. The crossing point between the modal birefringence of SH with the zero birefringence (dotted-line) indicates a design for which the SH mode is polarization degenerate [2].

tal Bragg mode. To further clarify this in Fig. 5.26, we simulated the birefringence at ω (dashed-line) and at 2ω (solid-line) as functions of the waveguide ridge width W. From the figure, the simulated birefringence at 2ω is less than an order of magnitudes smaller than that of the pump for the entire simulated values of W. Moreover, there exists a waveguide design for which the polarization degeneracy condition can be fulfilled. The small modal birefringence in addition to the existence of polarization degenerate design is a unique characteristic of Bragg reflection waveguides which deserve further investigation and can find potential applications involved with linear and nonlinear effects.

The quadratic dependence of SH power versus pump power was examined. The result is illustrated in Fig. 5.27(a) where the two powers are plotted on a log-log scale at the phase-matching wavelength of 1567.8 nm. At pump power levels below 10 mW the slope of a linear fit (solid-line) was obtained to be 1.92. For pump power above 10 mW clear deviation from a perfect quadratic dependence can be observed which were solely ascribed to third-order nonlinear effects mainly two-photon absorption. The uncertainties involved in estimating the SH power in Fig. 5.27(a) was similar to those discussed earlier for type-I and type-II SHG (see Fig. 5.11(b)). We considered an overall error of 11% in estimating the internal SH power in type-0 process. Also, Figure 5.27(b) shows a

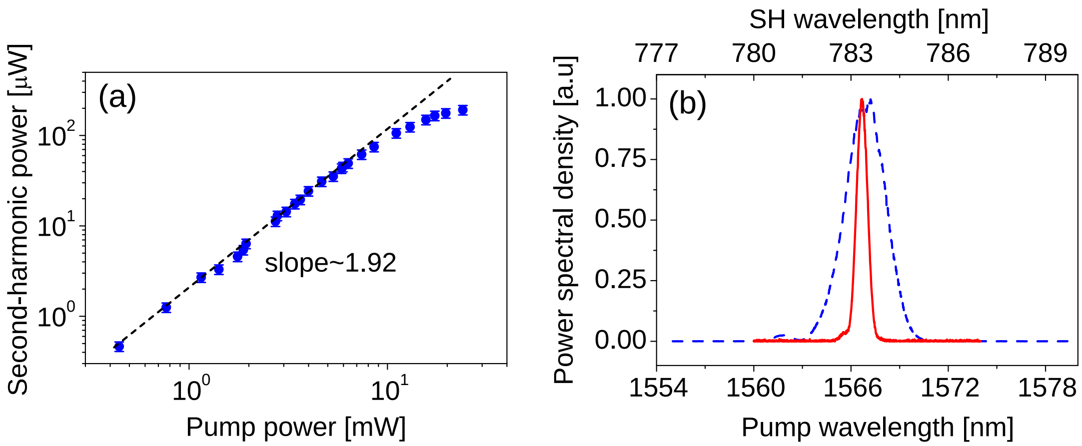

Figure 5.27: (a) Second-harmonic power as a function of pump plotted on a log-log scale. (b) Normalized spectra of pump (dashed-line) and SH (solid-line) [2].

comparative illustration of the normalized spectra of the pump and that of the SH. The full-width at half-maximum bandwidth of the pump was measured as 2.4 nm, while that of the SH was 0.4 nm.

Chapter 6

Generalized Matching-Layer Enhanced BRWs

In this chapter, matching-layer enhanced BRWs are further generalized to structures where a given number of dielectric layers, referred to as intermediate layers, are located between the core and the matching-layer. It is assumed that these intermediate layers do not establish any particular periodicity either in material indices or layer thicknesses. The inclusion of such intermediate layers offers additional designable parameters which can be incorporated in the design optimization. Validation of the benefits of the new design in enhancing nonlinear interaction cannot be carried out without including the contribution of the effective $\chi^{(2)}$ nonlinearity for a given structure. In this regard, an extracted model for the d_{eff} of bulk $\text{Al}_x\text{Ga}_{1-x}\text{As}$ elements is presented which was used for the theoretical estimation of the structure d_{eff}.

6.1 Theoretical analysis

Here, we develop the analytical tools required for the analysis of generalized ML-BRWs. Our approach for deriving the modal dispersion equations and the matching-layer thickness rely on applying the proper boundary conditions of field transitions at the interface

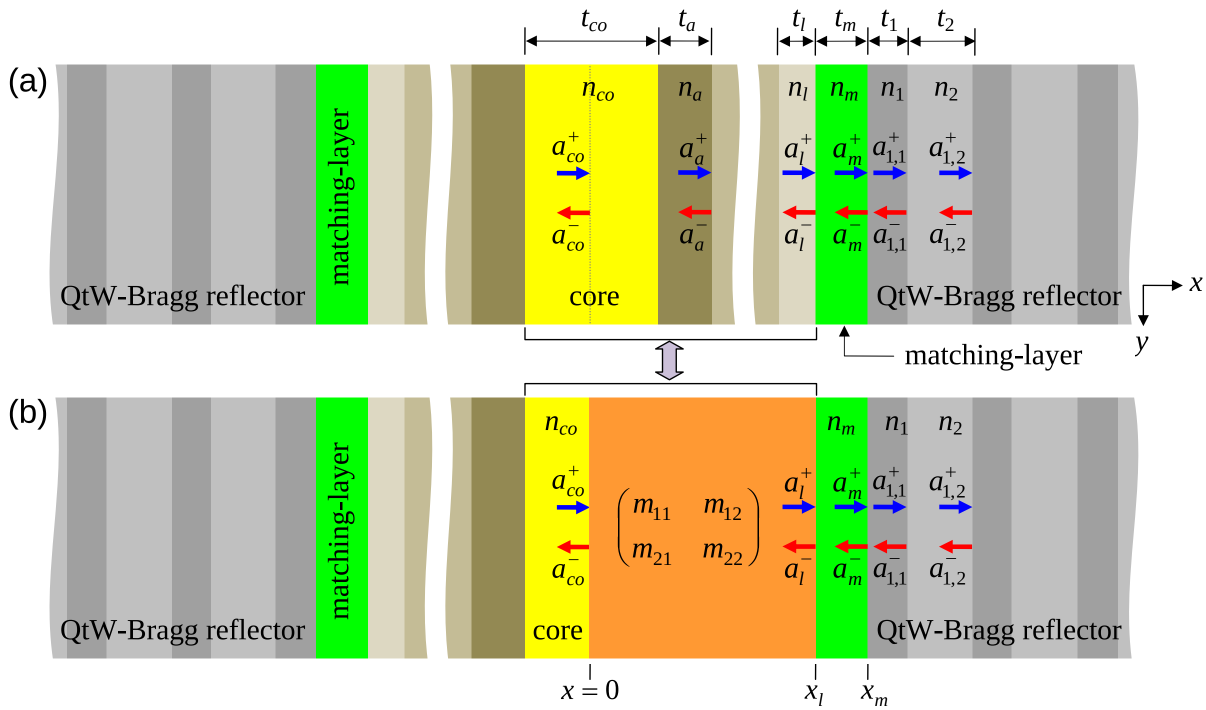

Figure 6.1: (a) Schematic of a generalized symmetric matching-layer enhanced BRW. The ML separates the core and the intermediate layers indicated by the refractive indices $n_a, n_b, \ldots, n_l$ from the quarter-wave Bragg reflectors. (b) A conceptual structure where half of the core and the intermediate layers in (a) are represented by an $ABCD$ matrix with elements $m_{11}, \ldots, m_{22}$.

between the matching-layer and quarter-wave Bragg reflector. We consider two general designs namely a symmetric ML-BRW where the structure poses geometrical symmetry with respect to the core center and a mono-stack ML-BRW where one of the periodic claddings in a symmetric structure is replaced by a single layer or by aperiodic dielectric layers.

6.1.1 Generalized symmetric ML-BRW

The schematic of a generalized symmetric ML-BRW is shown in Fig. 6.1(a). In this structure, the core is surrounded, on either side, by intermediate layers with refractive indices $n_a, n_b, \ldots, n_l$ and associated thicknesses $t_a, t_b, \ldots, t_l$. The matching-layer with a refractive index of n_m and a thickness of t_m separates the core and $n_a, n_b, \ldots, n_l$ layers from the quarter-wave Bragg reflector. To design the structure, it is assumed that all material indices are given a priori. Also, the core thickness and the thicknesses $t_a, t_b, \ldots, t_l$ are assumed to be given. The thickness of the matching-layer is then used for aligning the effective mode index of the structure such that the bi-layers of the Bragg reflectors satisfy the quarter-wave condition of Eq. (3.26).

The modal dispersion equations and the required matching-layer thickness of generalized symmetric ML-BRW can be obtained by applying the boundary condition of transition of overall electric field amplitude at the interface between the ML and quarter-wave TBR. The overall field amplitude, formed by the superposition of the right and left propagating plane waves, either vanishes or peaks at the interface between the quarter-wave TBR and the matching-layer, the immediate layer before the TBR. Referring to Fig. 6.1, this condition requires that $a_m^+ + a_m^- = 0$ or $a_m^+ - a_m^- = 0$ in order to obtain a node or anti-node at $x = x_m$. For a given number of intermediate layers between core and matching-layer, extraction of the modal dispersion equation and matching-layer thickness requires multiplying the transfer matrices of half-core propagation and the intermediate layers and solving for t_m. Although theoretically feasible, one can find this approach mathematically tedious which requires different solutions for different numbers of given intermediate layers. Here, a generic solution is proposed which can be utilized for any given number of intermediate layers between the core and the matching-layer.

We start by relating the right and left propagating field amplitudes of the core $(a_{\text{co}}^+, a_{\text{co}}^-)$ to those of the layer prior to the matching-layer (a_l^+, a_l^-). This relation is expressed as

$$\begin{bmatrix} a_l^+ \\ a_l^- \end{bmatrix} = \mathbf{M} \begin{bmatrix} a_{\text{co}}^+ \\ a_{\text{co}}^- \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix} \begin{bmatrix} a_{\text{co}}^+ \\ a_{\text{co}}^- \end{bmatrix} \tag{6.1}$$

where the matrix $\mathbf{M}$ with elements $m_{11}, \ldots, m_{22}$ is an $ABCD$ matrix which replaces half of the core and the intermediate layers $n_a, n_b, \ldots, n_l$ as shown in Fig. 6.1(b). Here, we are interested in the lowest order even Bragg mode for which the superposition of right and left propagating fields within the core is maximum. Due to the symmetry of the structure one can assume that $a_{\text{co}}^{+} = a_{\text{co}}^{-}$. Without loss of generality, the overall field amplitude at the core center is taken to be unity such that $a_{\text{co}}^{+} = a_{\text{co}}^{-} = 1/2$. using transfer matrices of intermediate layers, the field amplitudes in the layer before the matching-layer, the l-th layer, are expressed as

$$\begin{aligned} a_l^{+} &= (m_{11} + m_{12})/2 \\ a_l^{-} &= (m_{21} + m_{22})/2 \end{aligned} \tag{6.2}$$

Given a_l^{+} and a_l^{-}, the field amplitudes in the ML can be calculated using the single transfer matrix as

$$\begin{bmatrix} a_m^{+} \\ a_m^{-} \end{bmatrix} = \frac{1}{4} \begin{bmatrix} (1+f_{\text{P}})\exp(-i\phi_m) & (1-f_{\text{P}})\exp(-i\phi_m) \\ (1-f_{\text{P}})\exp(+i\phi_m) & (1+f_{\text{P}})\exp(+i\phi_m) \end{bmatrix} \begin{bmatrix} m_{11}+m_{12} \\ m_{21}+m_{22} \end{bmatrix} \tag{6.3}$$

where $\text{P} = \{\text{TE}, \text{TM}\}$ is the subscript used for indicating the polarization state of the propagating field, $\phi_m = k_m^x t_m$ is the phase accumulated in the matching-layer and

$$f_{\text{P}} = \begin{cases} \dfrac{k_l^x}{k_m^x} & \text{for TE polarization} \\ \dfrac{n_m^2 k_l^x}{n_l^2 k_m^x} & \text{for TM polarization} \end{cases} \tag{6.4}$$

Using Eq. (6.3), the matching-layer field amplitudes are derived as

$$a_m^{+} = \frac{1}{4}[(m_{11}+m_{12})(1+f_{\text{P}})\exp(-i\phi_m) + (m_{21}+m_{22})(1-f_{\text{P}})\exp(-i\phi_m)] \tag{6.5a}$$

$$a_m^{-} = \frac{1}{4}[(m_{11}+m_{12})(1-f_{\text{P}})\exp(+i\phi_m) + (m_{21}+m_{22})(1+f_{\text{P}})\exp(+i\phi_m)] \tag{6.5b}$$

For TE-propagating mode, the electric field vanishes ($a_m^{+} + a_m^{-} = 0$) at $x = x_m$ for $k_m^x < k_1^x$ and it peaks ($a_m^{+} - a_m^{-} = 0$) for $k_m^x > k_1^x$. Using Eqs. (6.5), the modal dispersion for TE mode is obtained as

$$\begin{aligned}
\cot(k_m^x t_m) &= +if_{\mathrm{TE}}\frac{m_{11}+m_{12}-m_{21}-m_{22}}{m_{11}+m_{12}+m_{21}+m_{22}} \quad (k_m^x < k_1^x) \\
\tan(k_m^x t_m) &= -if_{\mathrm{TE}}\frac{m_{11}+m_{12}-m_{21}-m_{22}}{m_{11}+m_{12}+m_{21}+m_{22}} \quad (k_m^x > k_1^x)
\end{aligned} \tag{6.6}$$

Equations (6.6) can be solved to calculate the thickness of the matching-layer as

$$\begin{aligned}
t_m &= \tfrac{1}{k_m^x}\big[\mathrm{acot}\big(+if_{\mathrm{TE}}\frac{m_{11}+m_{12}-m_{21}-m_{22}}{m_{11}+m_{12}+m_{21}+m_{22}}\big)+p\pi\big] \quad (k_m^x < k_1^x) \\
t_m &= \tfrac{1}{k_m^x}\big[\mathrm{atan}\big(-if_{\mathrm{TE}}\frac{m_{11}+m_{12}-m_{21}-m_{22}}{m_{11}+m_{12}+m_{21}+m_{22}}\big)+p\pi\big] \quad (k_m^x > k_1^x)
\end{aligned} \tag{6.7}$$

where $p = 0, 1, 2, \ldots$ is the matching-layer thickness scaling factor.

For TM propagation, the electric field vanishes at $x = x_m$ for $n_1^2 k_m^x < n_m^2 k_1^x$ and it peaks for $n_1^2 k_m^x > n_m^2 k_1^x$. Using Eqs. (6.5), modal dispersion of TM propagating mode is obtained as

$$\begin{aligned}
\cot(k_m^x t_m) &= +if_{\mathrm{TM}}\frac{m_{11}+m_{12}-m_{21}-m_{22}}{m_{11}+m_{12}+m_{21}+m_{22}} \quad (n_1^2 k_m^x < n_m^2 k_1^x) \\
\tan(k_m^x t_m) &= -if_{\mathrm{TM}}\frac{m_{11}+m_{12}-m_{21}-m_{22}}{m_{11}+m_{12}+m_{21}+m_{22}} \quad (n_1^2 k_m^x > n_m^2 k_1^x)
\end{aligned} \tag{6.8}$$

Solving Eq. (6.8) for t_m, the thickness of the matching-layer is given as

$$\begin{aligned}
t_m &= \tfrac{1}{k_m^x}\big[\mathrm{acot}\big(+if_{\mathrm{TM}}\frac{m_{11}+m_{12}-m_{21}-m_{22}}{m_{11}+m_{12}+m_{21}+m_{22}}\big)+p\pi\big] \quad (n_1^2 k_m^x < n_m^2 k_1^x) \\
t_m &= \tfrac{1}{k_m^x}\big[\mathrm{atan}\big(-if_{\mathrm{TM}}\frac{m_{11}+m_{12}-m_{21}-m_{22}}{m_{11}+m_{12}+m_{21}+m_{22}}\big)+p\pi\big] \quad (n_1^2 k_m^x > n_m^2 k_1^x)
\end{aligned} \tag{6.9}$$

For the case that the structure is composed of lossless dielectrics, the above equations can be further simplified. In order to show this, we remind the reader that when the waveguide layers are lossless, the elements of the matrix $\mathbf{M}$ in Eq. (6.1) inherit the property that $m_{22} = m_{11}^*$ and $m_{21} = m_{12}^*$. Let's define the complex number ζ as

$$\begin{aligned}
\zeta = |\zeta|\exp(i\phi_\zeta) &= m_{11}+m_{12} \\
\tan(\phi_\zeta) = \frac{\mathrm{Im}[\zeta]}{\mathrm{Re}[\zeta]} &= \frac{\mathrm{Im}[m_{11}+m_{12}]}{\mathrm{Re}[m_{11}+m_{12}]}
\end{aligned} \tag{6.10}$$

For TE-polarization, Eq. (6.6) can then be simplified as

$$\begin{aligned}\cot(k_m^x t_m) &= +if_{\mathrm{TE}}\tan(\phi_\zeta) \quad (k_m^x < k_1^x)\\ \tan(k_m^x t_m) &= -if_{\mathrm{TE}}\tan(\phi_\zeta) \quad (k_m^x > k_1^x)\end{aligned} \tag{6.11}$$

with the matching-layer thickness expressed as

$$\begin{aligned}t_m &= \frac{1}{k_m^x}\left[\mathrm{acot}\big(-f_{\mathrm{TE}}\tan(\phi_\zeta)\big) + p\pi\right] \quad (k_m^x < k_1^x)\\ t_m &= \frac{1}{k_m^x}\left[\mathrm{atan}\big(+f_{\mathrm{TE}}\tan(\phi_\zeta)\big) + p\pi\right] \quad (k_m^x > k_1^x)\end{aligned} \tag{6.12}$$

Similarly, for TM-propagation Eq. (6.8)

$$\begin{aligned}\cot(k_m^x t_m) &= +if_{\mathrm{TM}}\tan(\phi_\zeta) \quad (n_1^2 k_m^x < n_m^2 k_1^x)\\ \tan(k_m^x t_m) &= -if_{\mathrm{TM}}\tan(\phi_\zeta) \quad (n_1^2 k_m^x > n_m^2 k_1^x)\end{aligned} \tag{6.13}$$

with the matching-layer thickness given by

$$\begin{aligned}t_m &= \frac{1}{k_m^x}\left[\mathrm{acot}\big(-f_{\mathrm{TM}}\tan(\phi_\zeta)\big) + p\pi\right] \quad (n_1^2 k_m^x < n_m^2 k_1^x)\\ t_m &= \frac{1}{k_m^x}\left[\mathrm{atan}\big(+f_{\mathrm{TM}}\tan(\phi_\zeta)\big) + p\pi\right] \quad (n_1^2 k_m^x > n_m^2 k_1^x)\end{aligned} \tag{6.14}$$

Equations (6.6)–(6.9) serve as the design equations for generalized symmetric matching-layer enhanced BRWs with any given number of intermediate layers $n_a, \ldots, n_l$. The physical parameters of the intermediate layers including their refractive indices and thicknesses offer extra degrees of freedom for design optimization. Although in our analysis we assumed that the phase in each of these intermediate layers do not establish any particular relations with each other, in practice we can impose particular phase relations among them. For example, one can design a double grating mirror where the intermediate layers form a periodic structure with quarter-wave bi-layers with high reflectivity at a given wavelength of λ_1 while the TBRs are designed to have high reflection at a different wavelength of λ_2. The matching-layer then operates as a phase tuning layer between the two

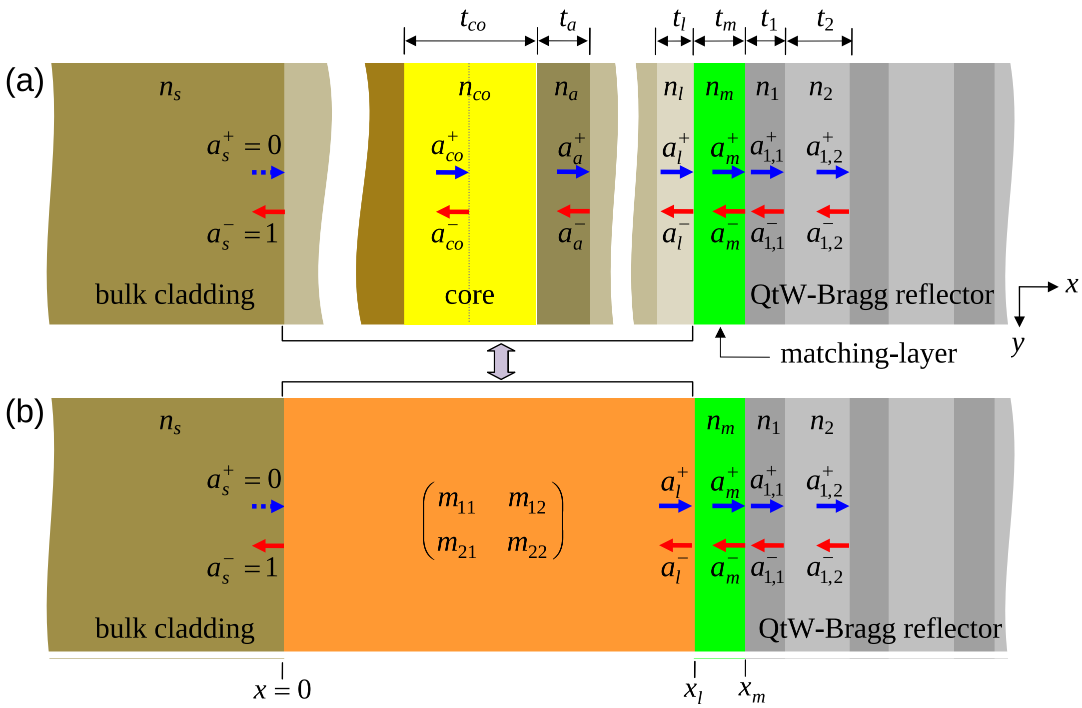

Figure 6.2: (a) Schematic of a generalized mono-stack matching-layer enhanced BRW where one of the periodic claddings of a symmetric BRW is replaced by a single layer cladding or aperiodic multilayer cladding. The matching-layer separates the single quarter-wave Bragg reflector front the rest of the structure. (b) A conceptual structure where all the layers between the bulk cladding and the matching-layer in (a) are replaced by an $ABCD$ matrix with elements $m_{11}, \ldots, m_{22}$.

quarter-wave periodic structures.

6.1.2 Generalized mono-stack ML-BRW

Generalized matching-layer enhanced BRWs can be extended to mono-stack structures where one of the periodic claddings, usually the top cladding, is replaced by either a single layer cladding or a multilayer cladding. In this section, we derive the modal dispersion equations and the analytical form of matching-layer thickness in this class of BRWs. A drawing of the generalized mono-stack ML-BRW is shown in Fig. 6.2(a). In this design, the core is surrounded on one side by a multilayer cladding which is assumed to have no periodicity and on the other side by intermediate layers with material indices

$n_a, n_b, \ldots, n_l$ and respective thicknesses $t_a, t_b, \ldots, t_l$ followed by the matching-layer. The ML with an index of n_m and a thickness of t_m is followed by a quarter-wave TBR.

The propgation of the Bragg mode in mono-stack ML-BRW relies on total internal reflection at $x = 0$ where the bulk cladding is located and Bragg reflection, at $x = x_m$, from the quarter-wave TBR. The total-internal reflection at $x = 0$ requires the existence of evanescent field for $x < 0$ which can be satisfied only if a left propagating plane wave exist in the bulk cladding. As such, the field amplitudes in the bulk cladding are taken as $a_s^+ = 0, a_s^- = 1$. Similar to the analysis discussed in Chapter 5, we replace the dielectric layers between the interfaces at $x = 0$ and $x = x_l$ by the matrix $\mathbf{M}$ with the $ABCD$ elements of $m_{11}, \ldots, m_{22}$. The field amplitudes in the l-th intermediate layer, the layer before the ML, are then related to those of the bulk cladding according to

$$\begin{bmatrix} a_l^+ \\ a_l^- \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix} \begin{bmatrix} a_s^+ \\ a_s^- \end{bmatrix} \tag{6.15}$$

Using the assumptions that $a_s^+ = 0$ and $a_s^- = 1$, the above equation simplifies to

$$\begin{aligned} a_l^+ &= m_{12} \\ a_l^- &= m_{22} \end{aligned} \tag{6.16}$$

The field amplitudes in the matching-layer are related to a_l^+ and a_l^- by a single translation matrix. Using Eq. (6.16), we obtain

$$\begin{bmatrix} a_m^+ \\ a_m^- \end{bmatrix} = \frac{1}{2} \begin{bmatrix} (1 + f_{\mathrm{P}}) \exp(-i\phi_m) & (1 - f_{\mathrm{P}}) \exp(-i\phi_m) \\ (1 - f_{\mathrm{P}}) \exp(+i\phi_m) & (1 + f_{\mathrm{P}}) \exp(+i\phi_m) \end{bmatrix} \begin{bmatrix} m_{12} \\ m_{22} \end{bmatrix} \tag{6.17}$$

where f_{P} is given in Eq. (6.4) and $\phi_m = k_m^x t_m$. The field amplitudes in the matching-layer are then expressed by

$$a_m^+ = \frac{1}{2}\left[(m_{12}+m_{22}) + (m_{12}-m_{22})f_{\mathrm{P}}\right]\exp(-i\phi_m) \tag{6.18a}$$

$$a_m^- = \frac{1}{2}\left[(m_{12}+m_{22}) - (m_{12}-m_{22})f_{\mathrm{P}}\right]\exp(+i\phi_m) \tag{6.18b}$$

Similar to the analysis of generalized symmetric ML-BRW discussed in the previous section, for each orthogonal polarization, the modal dispersion equations are derived by forcing the overall field at $x = x_m$ to be either zero ($a_m^+ + a_m^- = 0$) or maximum ($a_m^+ - a_m^- = 0$). From Eqs. (6.18) for TE-propagating mode, dispersion equations are derived as

$$\begin{aligned} \cot(k_m^x t_m) &= +if_{\mathrm{TE}}\frac{m_{12}-m_{22}}{m_{11}+m_{22}} \quad (k_m^x < k_1^x) \\ \tan(k_m^x t_m) &= -if_{\mathrm{TE}}\frac{m_{12}-m_{22}}{m_{11}+m_{22}} \quad (k_m^x > k_1^x) \end{aligned} \tag{6.19}$$

Equations (6.19) can be solved to calculate the required thicknesses of the matching-layer as

$$\begin{aligned} t_m &= \tfrac{1}{k_m^x}\big[\mathrm{acot}\big(+if_{\mathrm{TE}}\frac{m_{12}-m_{22}}{m_{12}+m_{22}}\big) + p\pi\big] \quad (k_m^x < k_1^x) \\ t_m &= \tfrac{1}{k_m^x}\big[\mathrm{atan}\big(-if_{\mathrm{TE}}\frac{m_{12}-m_{22}}{m_{12}+m_{22}}\big) + p\pi\big] \quad (k_m^x > k_1^x) \end{aligned} \tag{6.20}$$

Similarly, for TM propagating mode

$$\begin{aligned} \cot(k_m^x t_m) &= +if_{\mathrm{TM}}\frac{m_{12}-m_{22}}{m_{12}+m_{22}} \quad (n_1^2 k_m^x < n_m^2 k_1^x) \\ \tan(k_m^x t_m) &= -if_{\mathrm{TM}}\frac{m_{12}-m_{22}}{m_{12}+m_{22}} \quad (n_1^2 k_m^x > n_m^2 k_1^x) \end{aligned} \tag{6.21}$$

with the matching-layer thickness given by

$$\begin{aligned} t_m &= \tfrac{1}{k_m^x}\big[\mathrm{acot}\big(+if_{\mathrm{TM}}\frac{m_{12}-m_{22}}{m_{12}+m_{22}}\big) + p\pi\big] \quad (n_1^2 k_m^x < n_m^2 k_1^x) \\ t_m &= \tfrac{1}{k_m^x}\big[\mathrm{atan}\big(-if_{\mathrm{TM}}\frac{m_{12}-m_{22}}{m_{12}+m_{22}}\big) + p\pi\big] \quad (n_1^2 k_m^x > n_m^2 k_1^x) \end{aligned} \tag{6.22}$$

Equations (6.19)–(6.22) are the design equations of the generalized mono-stack ML-BRWs.

The design algorithm of generalized symmetric ML-BRWs and mono-stack BRWs are very similar to that of the original ML-BRW discussed in the previous chapter. The only difference is the fact that the parameters of the intermediate layers, including their thicknesses and material indices, should be considered as given parameters at the begining of the design. Examination of mono-stack BRWs for nonlinear interactions is an ongoing research subject. At the time of writing this dissertation, no such device has been fabricated. However, initial simulations have indicated that this class of BRWs can alleviate the fabrication process by reducing the overall thickness of the structure while supporting nonlinear interactions with efficiency comparable to symmetric ML-BRWs. In the next section, we demonstrate a generalized symmetric ML-BRW with a single intermediate layer between the core and the matching-layer.

6.2 Device description and fabrication

In Chapter 5 we demonstrated notable SHG efficiency enhancement offered by matching-layer enhanced BRW. The demonstrated design aimed at improving nonlinear overlap factor, ξ, to enhance efficiency of the nonlinear interaction. The discussed design of ML-BRW-SH3 incorporated $Al_xGa_{1-x}As$ layers with Al concentrations as low as $x = 0.20$ with an associated bandgap of ≈ 1.67 eV. While using layers with low Al concentrations were theoretically found to be highly effective for improving the nonlinear overlap factor, experimental data denoted that the efficiency of SHG in ML-BRW-SH3 was restrained to internal pump powers below ≈ 9 mW. For pump powers above 9 mW, saturation of second-harmonic power was clearly observed, which was understood as a result of degrading effects of $\boldsymbol{\chi}^{(3)}$ nonlinearities such as two-photon absorption (2PA) and self-phase modulation (SPM). It is well known that third order effects are more pronounced for low aluminum concentration $Al_xGa_{1-x}As$ elements due to smaller bandgaps [108, 109]. In order to overcome the saturation of SH power at high pump powers, we considered employing materials with larger bandgaps such as high aluminum concentration $Al_xGa_{1-x}As$ while constraining the minimum Al concentration within the structure to $x = 0.35$ with a bandgap of ≈ 1.87 eV. The choice of $Al_xGa_{1-x}As$ elements with $x > 0.35$ introduced the trade-off of reducing the nonlinear conversion efficiency, η, since both the effective nonlinear coefficient of the structure and the nonlinear overlap factor were reduced. Following the design technique of ML-BRW presented in the previous chapter, it was unfeasible to compensate for this reduction of nonlinear conversion efficiency in using high aluminum content layers. However, using the design idea of generalized symmetric ML-BRW discussed in this chapter, it was possible to mainly compensate for the reduction of nonlinear conversion efficiency while incorporating high aluminum concentration layers.

Theoretical calculations of nonlinear conversion efficiency requires an approximation for the structure d_{eff}. While the values of d_{eff} for bulk $Al_xGa_{1-x}As$ can be theoretically estimated using Miller's rule [110], the literature lacks a comprehensive model which provides the dependency of d_{eff} as a function of aluminum concentration, operation wavelength and temperature. In [3], Ohasi has reported the experimentally measured dependence of the ratio $d_{\text{eff}}^{\text{Al}_x\text{Ga}_{1-x}\text{As}}/d_{\text{eff}}^{\text{GaAs}}$ as a function of Al concentration obtained using Nd:YAG laser operating at 1064 nm. We adopt the reported data in [3] to extract a numerical model for estimating d_{eff} of bulk $Al_xGa_{1-x}As$. Figure 6.3(a) illustrates the extracted data of $d_{\text{eff}}^{\text{Al}_x\text{Ga}_{1-x}\text{As}}/d_{\text{eff}}^{\text{GaAs}}$ from [3] along with the fifth-order polynomial fit

$$d_{\text{eff}}^{\text{Al}_x\text{Ga}_{1-x}\text{As}}/d_{\text{eff}}^{\text{GaAs}} = \sum_{m=0}^{5} c_m x^m \tag{6.23}$$

which we used as the theoretical model for the ratio of $d_{\text{eff}}^{\text{Al}_x\text{Ga}_{1-x}\text{As}}/d_{\text{eff}}^{\text{GaAs}}$. The coefficients of the polynomial fits for each graph in Fig. 6.3(a) are given in Table 6.1.

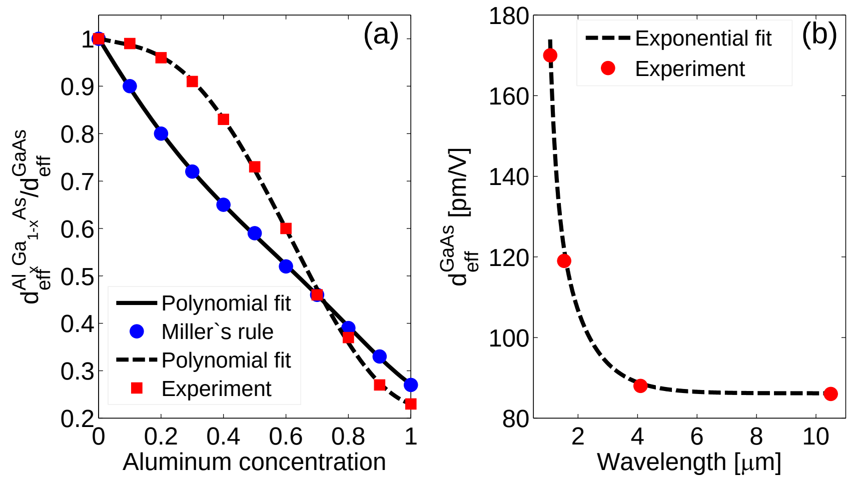

Figure 6.3: (a) Dependency of $d_{\text{eff}}^{\text{Al}_x\text{Ga}_{1-x}\text{As}}/d_{\text{eff}}^{\text{GaAs}}$ on aluminum molar concentration extracted using the data reported in [3]. The solid- and dashed-line show fifth-order polynomial fits used to obtain a numerical model for estimating the effective nonlinearity. (b) Variation of $d_{\text{eff}}^{\text{GaAs}}$ as a function of wavelength obtained using the reported experimental data in [4] (filed-circle) and an extracted exponential decay fit (dashed-line).

Table 6.1: Coefficients of fifth-order polynomials $d_{\text{eff}}^{\text{Al}_x\text{Ga}_{1-x}\text{As}}/d_{\text{eff}}^{\text{GaAs}} = \sum_{m=0}^{5} c_m x^m$ extracted from the reported data in [3].

Model	c_0	c_1	c_2	c_3	c_4	c_5
Miller's rule	1.000	-1.077	0.202	1.661	-2.797	1.282
Experiment	1.000	-0.176	0.870	-6.011	6.469	-1.923

Equation (6.24) requires a numerical value for $d_{\text{eff}}^{\text{GaAs}}$. Although the fits in Fig. 6.3(a) were obtained for the wavelength of 1064 nm, we assumed that the slopes of the fits more or less follows the same trends at other wavelengths. To apply Eq. (6.24), one needs a numeric for the value of $d_{\text{eff}}^{\text{GaAs}}$ at a given wavelength. In our model, we estimated $d_{\text{eff}}^{\text{GaAs}}$ using the discrete values reported in [4]. Using a shape-preserving interpolation fit followed by an exponential decay fit, as illustrated in Fig. 6.3, we derived an extracted model for the values of $d_{\text{eff}}^{\text{GaAs}}$ reported in [4] in form of

$$d_{\text{eff}}^{\text{GaAs}}(\lambda) = d_0 + d_1 \exp(-\lambda/\lambda_1) + d_2 \exp(-\lambda/\lambda_2) + d_3 \exp(-\lambda/\lambda_3) \tag{6.24}$$

where $d_0, \ldots, d_3$ and $\lambda_1, \ldots, \lambda_3$ are the fitting constants summarized in Table 6.2.

Table 6.2: Constants of the exponential decay fit used for modeling the dependency of $d_{\text{eff}}^{\text{GaAs}}$ on wavelength obtained using the reported data in [4]

d_0 [pm/V]	d_1 [pm/V]	λ_1 [nm]	d_2 [pm/V]	λ_2 [nm]	d_3 [pm/V]	λ_3 [nm]
86	4058	194	4081	194	152	993

Using the model of d_{eff} for bulk $\text{Al}_x\text{Ga}_{1-x}\text{As}$ an estimation for the effective nonlinear coefficient of the BRW structures could be obtained. A new wafer, referred to as ML-BRW-SH4, was then designed for Type-II SHG where the nonlinear conversion efficiency was taken as the objective function of an optimization problem. The motivation behind the design of ML-BRW-SH4 was to investigate the effect of employing high aluminum content $\text{Al}_x\text{Ga}_{1-x}\text{As}$ layers on the efficiency of nonlinear interaction, particularly at high pump power. Table 6.3 provides a theoretical comparison between the salient nonlinear parameters of the previously fabricated wafer of ML-BRW-SH3 and the new wafer of ML-BRW-SH4. The simulations were carried out for the slab structures. From the table, it can be seen that while ML-BRW-SH4 incorporated materials with high aluminum fractions, its conversion efficiency is within the same order of magnitude as that of ML-BRW-SH3. Using the table, we expect that the efficiency of ML-BRW-SH4 to be approximately one-third of the efficiency of ML-BRW-SH3, provided that propagation losses in the two devices are comparable. The theoretical estimations of η_{norm} in Table 6.3 did not take into account $\boldsymbol{\chi}^{(3)}$ nonlinear effects when a pump with high power is employed. As will be discussed in the next section, ML-BRW-SH4 was indeed a superior design in comparison to ML-BRW-SH3 under high pump condition due to the

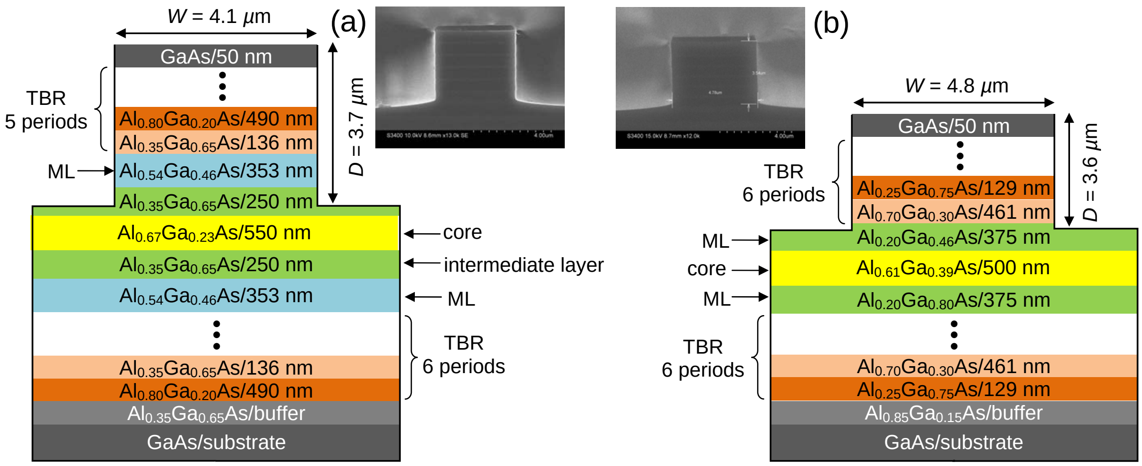

Figure 6.4: (a) Schematic of ML-BRW-SH4 waveguide and the cross section SEM. (b) Schematic of ML-BRW-SH3 waveguide and the cross section SEM.

employment of $Al_xGa_{1-x}As$ layers with larger bandgaps.

Table 6.3: Simulated nonlinear parameters of ML-BRW-SH3 and ML-BRW-SH4 designs for Type-II SHG

Design	λ_p [nm]	ξ [m]$^{-1}$	d_{eff} [pm/V]	η_{norm} [%W^{-1}cm^{-2}]
ML-BRW-SH3	1546.5	221	40	3.0×10^{-4}
ML-BRW-SH4	1540.0	133	38	1.1×10^{-4}

6.3 Characterization results

The wafer of ML-BRW-SH4 was grown on a GaAs [001] substrate using metal-organic chemical vapour deposition (MOCVD). A drawing of the waveguide cross section is shown in Fig. 6.4(a). Also in Fig. 6.4(b), we have shown the schematic of the previously fabricated ML-BRW-SH3 as a reference for comparing the nonlinear performance of both designs. The epitaxial structure of ML-BRW-SH3 was earlier discussed in Section 5.2.

Linear properties of the two waveguides were extracted using the Fabry-Pèrot method around 1550 nm. Physical dimensions of devices including ridge width (W), etch depth (D) and sample length (L) along with propagation losses of TE and TM polarizations

($\alpha_\omega^{(\mathrm{TE})}$ and $\alpha_\omega^{(\mathrm{TM})}$), input coupling factor (C) and facet reflection (R) are given in Table 6.4.

Table 6.4: Physical dimensions and linear properties of ML-BRW-SH3 and ML-BRW-SH4 waveguides

Device	L [mm]	W [μm]	D [μm]	$\alpha_\omega^{(\mathrm{TE})}$ [cm^{-1}]	$\alpha_\omega^{(\mathrm{TM})}$ [cm^{-1}]	C	R
ML-BRW-SH3	2.2	4.8	3.6	2.2	2.0	0.49	0.28
ML-BRW-SH4	1.7	4.1	3.7	2.2	2.9	0.48	0.29

From the table, the linear loss of the pump for ML-BRW-SH4 waveguide were slightly larger in comparison to ML-BRW-SH3 for TM-polarizations. This larger loss value is believed to appear as a result of increased side-wall roughness and could be addressed by improving the etch quality during waveguide patterning.

Nonlinear characterizations were carried out for Type-II SHG where a hybrid pump with TE+TM polarization generated TE-polarized SH. Both waveguides were characterized using three different pulsed laser systems namely 1)- a Programmable Laser from Genia Photonics Inc. [111] generating pulses with 30-ps temporal width ($\Delta\tau$) at 15 MHz repetition rate, 2)- an OPO synchronously pumped by a mode-locked Ti:sapphire laser generating 2-ps pulses at 76 MHz repetition rate and 3)- the same OPO system converted to 250-fs pulses at 76 MHz repetition rate. Characterizing the waveguides using different pump pulse conditions was expected to reveal better insight into the nonlinear interaction in the two devices.

Table 6.5: Summary of SHG characterization of ML-BRW-SH3 and ML-BRW-SH4 waveguides obtained for the pump power of $\mathcal{P}_\omega = 2.2$ mW

Device	$\Delta\tau$ [ps]	$\Delta\lambda_\omega$ [nm]	$\mathcal{P}_{2\omega}$ [μW]	λ_{PM} [nm]	η_{norm} [%W^{-1}cm^{-2}]	$\Delta\lambda_{\text{SHG}}$ [nm]	$\Delta\lambda_{\text{SHG}}/\Delta\lambda_\omega$
ML-BRW-SH3	30	0.3	32.8	1554.0	1.4×10^4	0.6	2.0
ML-BRW-SH4	30	0.3	6.3	1559.6	4.6×10^3	1.0	3.3
ML-BRW-SH3	2	2.6	32.0	1553.4	1.4×10^4	1.9	0.7
ML-BRW-SH4	2	2.6	4.7	1558.9	3.2×10^3	2.7	1.0
ML-BRW-SH3	0.25	75	11.2	1556.2	4.8×10^3	36	0.5
ML-BRW-SH4	0.25	75	1.6	1560.1	1.1×10^3	38	0.5

Table 6.5 contains a summary of the examined SHG parameters, obtained for an internal pump power of $\mathcal{P}_\omega = 2.2$ mW, including phase-matching wavelength (λ_{PM}), full-width at half-maximum (FWHM) pump spectral bandwidth ($\Delta\lambda_\omega$), internal SH power ($\mathcal{P}_{2\omega}$), SHG wavelength acceptance bandwidth ($\Delta\lambda_{\text{SHG}}$) and normalized nonlinear conversion efficiency (η_{norm}).

The interpretation of the data in the Table 6.5 requires consideration of pump peak power, pump spectral usage as well as third-order nonlinear effects in each laser system. From the table, it can be seen that for either device, shifting the pulse width from 30-ps to 2-ps did not significantly affect the generated SH power and the efficiency. Although the temporal pulse widths of 30-ps and 2-ps laser systems were different by over an order of magnitude, the duty cycle of the pulses were within the same orders of magnitude. For the 30-ps laser, the duty cycle was 4.5×10^{-4} while it was 1.5×10^{-4} for 2-ps laser. Assuming nearly transform limited pulses, for a fixed pump power, it can be deduced that the peak powers in 30-ps and 2-ps systems were within the same order of magnitude, hence the generated SH should be comparable. On the other hand, pulses in 250-fs system had a duty cycle of 1.9×10^{-5} which was approximately an order of magnitude less than those in the other two laser systems. The smaller duty cycle of femtosecond pulses translated to roughly an order of magnitude enhancement in peak power. However, from Table 6.5, the generated SH power as well as the nonlinear conversion efficiency in 250-fs system was notably decreased in comparison to those estimated using longer pulse widths. The comparisons between the pump spectral bandwidth, $\Delta\lambda_\omega$ and the wavelength acceptance bandwidth $\Delta\lambda_{\mathrm{SHG}}$ measured in these laser systems can partly explain the lower efficiency of the femtosecond SHG. To show this, we consider the ratio $\Delta\lambda = \Delta\lambda_{\mathrm{SHG}}/\Delta\lambda_\omega$ as a figure of merit indicating the portion of the pump spectra utilized in the SHG. Using Table 6.5, for 30-ps pulses, the entire pump spectra is utilized in both devices. For ML-BRW-SH3, $\Delta\lambda = 2.0$ while for ML-BRW-SH4, $\Delta\lambda = 3.3$. On the other hand, for 250-fs pulses, only half of the pump spectra was utilized in both devices. There are other phenomena which additionally contribute to the inefficiency of nonlinear interaction in the devices using ultrashort pulses. In a recent study [112], we have theoretically investigated the influence of $\boldsymbol{\chi}^{(3)}$ effects including 2PA and SPM and the dispersion effects including group velocity mismatch and group velocity dispersion on femtosecond SHG. Using 190-fs pulses with 3.3 mW pump power in a 1.1 mm long ML-BRW-SH3 sample, we have shown that the dominating degrading factors in limiting the generated harmonic power and the efficiency of the nonlinear process are the group velocity mismatch between the pump and SH signal and the group velociy dispersions. The dispersion effects might seem mainly accountable for reduced harmonic generation in femtosecond regim, at high pump power third order nonlinearities are expected to have stronger contribution in down grading the performance of BRW devices. No studies has been carried out yet that addresses the effect of pump power on harmonic generation. The results discussed in the next section tries to do such study.

Figures. 6.5(a) and (b) illustrate the variations of $\mathcal{P}_{2\omega}$ as functions of pump power for both devices obtained using the three pulsed laser systems. From the figures, both devices show the maximum increase of second-harmonic power for 30-ps pulses. The

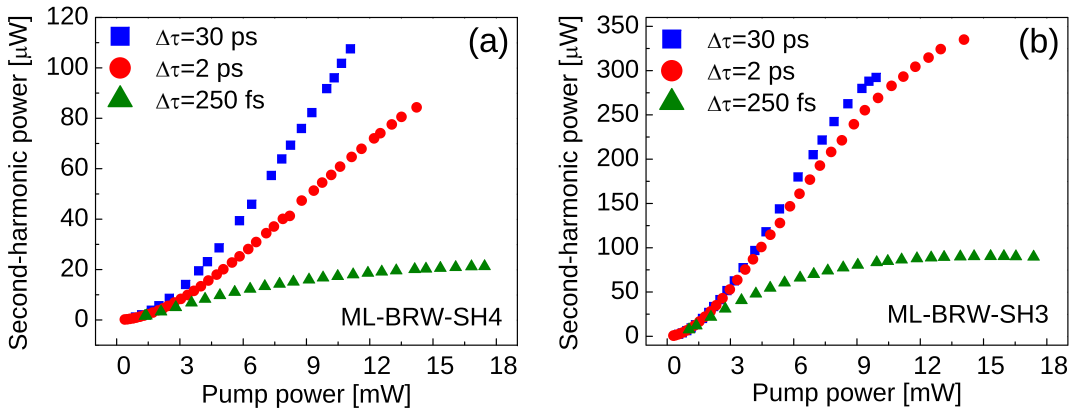

Figure 6.5: (a) Dependency of generated SH power on pump power in ML-BRW-SH4 waveguide ($L = 1.7$ mm) characterized using 30-ps, 2-ps and 250-fs pulse widths and (b) those obtained in ML-BRW-SH3 ($L = 2.2$ mm).

operation of the two devices at high pump powers is remarkable. From Fig. 6.5(b) it can be seen that, for 30-ps and 2-ps pulses, $\mathcal{P}_{2\omega}$ saturates around $\mathcal{P}_{\omega} = 9$ mW to a value of approximately $\mathcal{P}_{2\omega} = 300$ μW. For pump powers above ≈ 9 mW, $\mathcal{P}_{2\omega}$ fails to follow the quadratic relation with the pump power as a result of $\boldsymbol{\chi}^{(3)}$ nonlinearities. In contrast, from Fig. 6.5(a), for 30-ps pulses, the quadratic relation between $\mathcal{P}_{2\omega}$ and $\mathcal{P}_{\omega}$ is mainly maintained over the entire range of $\mathcal{P}_{\omega}$ as a result of less pronounced $\boldsymbol{\chi}^{(3)}$ nonlinearities. This suggested that the generated second-harmonic power could be further improved by increasing the length of the ML-BRW-SH4 sample.

To further distinguish the operation of both samples under high pump power condition, we normalized the generated SH power in each by the square of the sample length to obtain a figure of merit for harmonic generation independent of the device length. This normalization was supported by the fact that the measured losses in the two samples where comparable. Figure 6.6 illustrates the variation of $\mathcal{P}_{2\omega}/L^2$ as a function of pump powers. The solid-lines in both graphs are the theoretical quadratic fits of $\mathcal{P}_{2\omega} \propto \mathcal{P}_{\omega}^2$. From the figure, for ML-BRW-SH3, the measured data follows the quadratic power relation up to a pump power of ≈ 6 mW and saturates for pump powers above 9 mW. In contrast, for ML-BRW-SH4, a good match between the experimental data and the

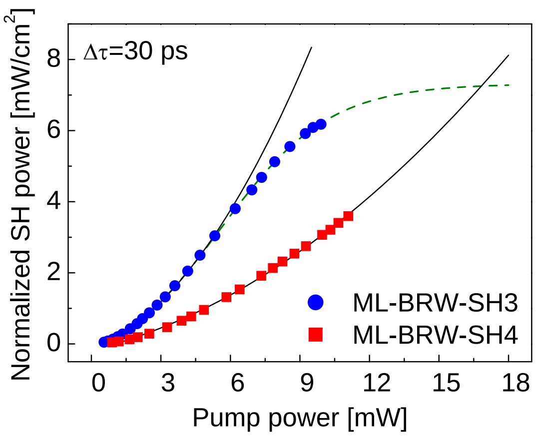

Figure 6.6: Second-harmonic power normalized to square of device length as a function of internal pump power measured for ML-BRW-SH3 (filled-circle) and ML-BRW-SH4 (filled-square) using 30-ps pulses. The solid-lines are theoretical quadratic fits showing the relation of $\mathcal{P}_{2\omega} \propto \mathcal{P}_{\omega}^2$. The dashed-line is a nonlinear fit as an extrapolation of the measured data in ML-BRW-SH3.

quadratic fit is observed over the entire range of employed pump power. Also in Fig. 6.6, the dashed-line shows a nonlinear fit which was used to fit the normalized SH power of ML-BRW-SH3 for extrapolating $\mathcal{P}_{2\omega}/L^2$ at high pump powers. From the figure, it can be expected that SHG favours ML-BRW-SH4 design for high pump powers levels.

Finally, Figs. 6.7(a) and (b) depict the normalized conversion efficiencies of the characterized devices. Both samples showed that their efficiencies decreased with the increase of pump power. For ML-BRW-SH3 the decay rate of η_{norm} was considerably greater in comparison to ML-BRW-SH4 owing to stronger $\boldsymbol{\chi}^{(3)}$ effects.

The generalized ML-BRW structure discussed in this chapter denoted the flexibility offered by this class of phase-matched BRWs in enhancing nonlinear interactions. Certain nonlinear interactions require high pump power operation. Although, using low aluminum fraction $Al_xGa_{1-x}As$ elements with small bandgap can enhance structure effective nonlinearity as well as nonlinear overlap factor in BRWs, in practice the operation of such devices is limited to low pump power levels because of third-order nonlinearities. Incorporation of materials with high aluminum concentration is thus essential for

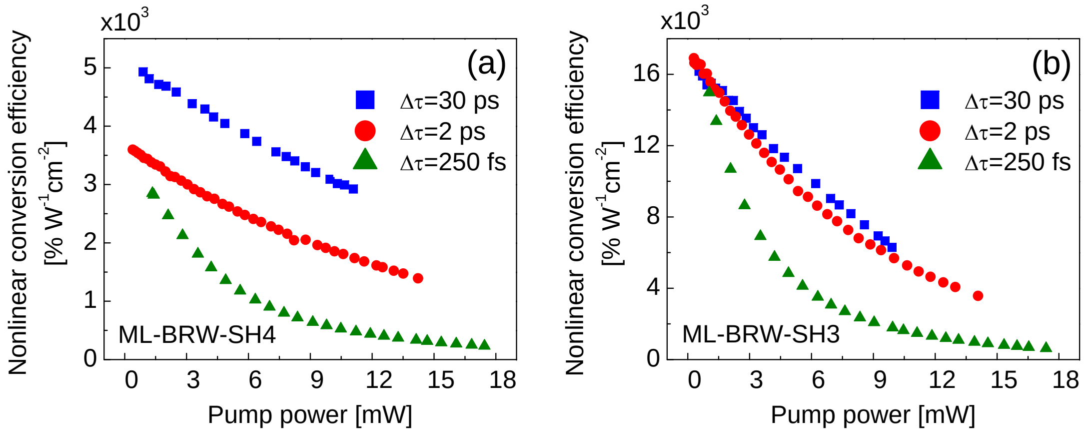

Figure 6.7: (a) Normalized SHG conversion efficiency in ML-BRW-SH4 waveguide characterized using 30-ps, 2-ps and 250-fs pulse widths and (b) those obtained in ML-BRW-SH3.

high power operation. However, a trade-off exists between using high aluminum content $Al_xGa_{1-x}As$ and the efficiency of nonlinear interaction. Using the proposed generalized ML-BRW design, it is possible to address this trade off to a large extent.

Chapter 7

BRWs as Integrated Sources of Photon-Pairs

An increasing number of quantum optics applications, ranging from quantum information processing and quantum communications [113] to quantum metrology and quantum imaging [114], show enhanced capabilities due to the use of biphotons with quantum frequency correlations. The successful technological implementation of these new applications depends primarily on efficient and robust sources which offer a high flux rate of photon-pairs while also allowing tailoring of the biphotons quantum state for a particular application.

Spontaneous-parametric down-conversion (SPDC) is the most popular method for the generation of photon-pairs with entangled properties [115]. The SPDC process is often implemented in bulk crystals using a solid-state laser source as a pump. The use of these crystals does not generally offer significant control over the spatio-temporal properties of the down-converted photons. Utilizing tilted pulses [116] or chirping the domains of the nonlinearity in quasi-phase-matched configurations [117] can provide control over the frequency-temporal characteristics. In addition to this, guided-wave alternatives can improve the spatial characteristics [118].

Photon-pair sources based on compound semiconductors, particularly $Al_xGa_{1-x}As$, define a new class of biphoton sources [119, 36]. Such devices are promising in an integrated platform where the frequency-down converting element is monolithically integrated with a diode laser pump. In this chapter, we discuss the potential of phase-matched BRWs as integrated sources of photon-pairs with frequency correlation properties. The required conditions for generating various forms of frequency correlation, including frequency anti-correlated, correlated and uncorrelated biphotons are theoretically discussed. As a case study, the CW source of anti-correlated photon-pairs, where strong waveguide dispersion in BRW is utilized for generating biphotons with ultra wideband and ultra narrowband spectral widths, is reviewed.

7.1 Frequency-correlated photon-pairs

The SPDC process involves the down-conversion of a pump photon (p) to a frequency correlated photon-pair of signal (s) and idler (i). The angular frequencies of the interacting waves are taken as

$$\omega_p = \omega_p^0 + \Omega_p \tag{7.1a}$$

$$\omega_s = \omega_s^0 + \Omega_s \tag{7.1b}$$

$$\omega_i = \omega_i^0 + \Omega_i \tag{7.1c}$$

where $\omega_j,\ j \in \{p, s, i\}$ is the central angular frequency with $\omega_s^0 = \omega_i^0 = \omega_p^0/2$ and Ω_j is the associated angular frequency deviation. The conservation of energy requires that

$$\omega_p^0 = \omega_s^0 + \omega_i^0 \tag{7.2a}$$

$$\Omega_p = \Omega_s + \Omega_i \tag{7.2b}$$

The quantum state of biphotons at the output facet of the nonlinear waveguide can be expressed as

$$|\psi\rangle = \iint_{-\infty}^{+\infty} d\Omega_s d\Omega_i \Phi(\Omega_s, \Omega_i)\Gamma(\Omega_s, \Omega_i)|\omega_s^0 + \Omega_s\rangle|\omega_i^0 - \Omega_s\rangle \tag{7.3}$$

where $\Gamma(\Omega_s, \Omega_i)$ is the spatial overlap of the interacting modes and

$$\Phi(\Omega_s, \Omega_i) = E_p(\Omega_s + \Omega_i)\mathrm{sinc}(\Delta\beta L/2)\exp(is_\beta L/2) \tag{7.4}$$

is the biphoton spectral amplitude. In Eq. (7.4), E_p is the pump spectral function, L is the length of the nonlinear waveguide, $\Delta\beta$ and s_β are the wavenumber mismatch and a phase term, respectively, which are defined as

$$\Delta\beta = \beta_p(\omega_p^0 + \Omega_p) - \beta_s(\omega_s^0 + \Omega_s) - \beta_i(\omega_i^0 + \Omega_i) \tag{7.5a}$$

$$s_\beta = \beta_p(\omega_p^0 + \Omega_p) + \beta_s(\omega_s^0 + \Omega_s) + \beta_i(\omega_i^0 + \Omega_i) \tag{7.5b}$$

Using Taylor's series, $\Delta\beta$ and s_β can be expanded around the central frequencies according to

$$\Delta\beta = N_p\Omega_p - N_s\Omega_s - N_i\Omega_i + \frac{1}{2}\left[D_p\Omega_p^2 - D_s\Omega_s^2 - D_i\Omega_i^2\right] + \dots \tag{7.6a}$$

$$s_\beta = N_p\Omega_p + N_s\Omega_s + N_i\Omega_i + \frac{1}{2}\left[D_p\Omega_p^2 + D_s\Omega_s^2 + D_i\Omega_i^2\right] + \dots \tag{7.6b}$$

where $N_j = \partial\beta_j/\partial\omega$ and $D_j = \partial^2\beta_j/\partial\omega^2$ are the first- and second-order dispersion values, respectively.

The wavenumber mismatch in Eq. (7.6a) plays an important role in determining the various forms of frequency correlation. In a nonlinear waveguide, $\Delta\beta$ is controlled by tailoring the modal dispersion of interacting frequencies by means of material dispersion, waveguide dispersion, or a combination of both. In the following, we address the effects of dispersion on the type of frequency correlation for different phase-matching schemes, including type-I and type-II interactions. Both continuous-wave and pulsed pump are considered.

7.1.1 Continuous-wave pump

For continuous-wave operation, the angular frequency of the pump is written as $\omega_p = \omega_p^0$ and $\Omega_p = 0$. Also, the pump spectral function is taken as $E_p(\omega_p) = 1$. Using the conservation of energy given in Eq. (7.2), we obtain $\Omega_s = -\Omega_i$. This implies that, by using a CW pump, the correlation of the generated photon-pair is always frequency anti-correlated. In this case, using Eq. (7.6), the wavenumber mismatch, $\Delta\beta$, and the phase factor, s_β, are expressed as

$$\Delta\beta = (N_i - N_s)\Omega_s - \frac{1}{2}\left[D_s + D_i\right]\Omega_s^2 + \dots \tag{7.7a}$$

$$s_\beta = s_0 + (N_i - N_s)\Omega_s + \frac{1}{2}\left[D_s + D_i\right]\Omega_s^2 + \dots \tag{7.7b}$$

where $s_0 = \beta_p(\omega_p^0) + \beta_s(\omega_s^0) + \beta_i(\omega_i^0)$ is a constant phase term. In a type-I SPDC in $Al_xGa_{1-x}As$ BRW, the polarization states of the photon-pair are identical (TE-polarized). As such, both signal and idler have identical dispersion properties. Ignoring dispersion terms with orders more than two, one can write that $N_s = N_i$ and $D_s = D_i$. Equations (7.7) can be simplified to

$$\Delta\beta \approx -D_s\Omega_s^2 \tag{7.8a}$$

$$s_\beta \approx s_0 + D_s\Omega_s^2 \tag{7.8b}$$

Taking the second-order derivative of Eq. (7.8b) with respect to Ω_s, we obtain a useful expression which relates the wavenumber mismatch to the second-order dispersion of signal/idler according to

$$D_s \approx -\frac{1}{2}\frac{\partial^2 \beta}{\partial \Omega_s^2} \tag{7.9}$$

In simulations, the numerical value of D_S is usually estimated using the finite-difference method. Equation (7.9) can be used as a tool for validating the numerical values of D_s obtained from finite-differences.

For the type-I SPDC, from Eqs. (7.7), the terms which involve an odd exponent of Ω_s cancel out from the Taylor series, since $N_i = N_s$. As a result, only the terms with an even exponent of Ω_s remain. It turns out that, in expanding $\Delta\beta$, generally including the group-velocity dispersion of the signal is sufficient. However, since the phase is more sensitive to the accuracy of the approximation, it might be necessary to include dispersion terms higher than two when expanding s_β in Taylor's series. This is particularly essential in studying the quantum-state of photon-pairs with wide spectral bandwidth.

For type-II anti-correlated bi-photons, the polarization states of the signal-idler pair are orthogonal, indicating that their modal dispersion is different. The general form of Eqs. (7.7) should then be employed. For a type-II process, the first order dispersion in Taylor's series usually dominates higher order terms. Approximation of $\Delta\beta$ and s_β can then be obtained as

$$\Delta\beta \approx (N_i - N_s)\Omega_s \tag{7.10a}$$

$$s_\beta \approx s_0 + (N_i - N_s)\Omega_s \tag{7.10b}$$

In a nonlinear waveguide, tailoring $N_i - N_s$ can be achieved, to a large extent, through waveguide dispersion. For example, $N_i - N_s$ can be made large in order to increase $\Delta\beta$; hence reducing the spectral bandwidth of biphotons. As a result, a type-II anti-correlated photon-pair is generally used for the generation of narrowband bi-photons with wide temporal correlation.

7.1.2 Pulsed pump

Using a pulsed pump offers more variety in generating bi-photons with frequency correlation properties, including frequency anti-correlated, correlated and uncorrelated photon-pairs. Here, the general assumption is that in the Taylor's series of Eqs. (7.6), the first order dispersion term is the dominating term. Using the conservation of energy of $\Omega_p = \Omega_s + \Omega_i$, we obtain

$$\Delta\beta \approx (N_p - N_s)\Omega_s + (N_p - N_i)\Omega_i \tag{7.11a}$$

$$s_\beta \approx s_0 + (N_p + N_s)\Omega_s + (N_p + N_i)\Omega_i \tag{7.11b}$$

For perfect phase-matching where $\Delta\beta = 0$, Eq. (7.11a) results in

$$\frac{N_p - N_s}{N_p - N_i} = -\frac{\Omega_i}{\Omega_s} \tag{7.12}$$

For anti-correlated bi-photons, $\Omega_s = -\Omega_i$. Using Eq. (7.12), the required conditions of having an integrated source of anti-correlated bi-photons with pulsed pump become

$$N_s = N_i \tag{7.13a}$$

$$s_\beta = s_0 \tag{7.13b}$$

which can be satisfied in a type-I SPCD process. For correlated photon-pairs, $\Omega_s = \Omega_i$. Using Eq. (7.11a), we obtain

$$N_p = \frac{1}{2}(N_s + N_i) \tag{7.14a}$$

$$s_\beta = s_0 + 2(N_s + N_i)\Omega_s \tag{7.14b}$$

For frequency uncorrelated photon-pairs, Ω_s and Ω_i can vary independently. Mathematically, this can be expressed as

$$\frac{N_p - N_s}{N_p - N_i} = 0, \infty \Rightarrow \begin{cases} N_p = N_s \\ N_p = N_i \end{cases} \tag{7.15}$$

Equations (7.13)–(7.15) determine the required modal dispersion conditions to be satisfied for generating bi-photons with various forms of frequency correlation in a pulsed regime.

In BRWs, generating anti-correlated photon-pairs by using either a CW pump or a pulsed pump is the most straightforward to implement. For correlated and uncorrelated photon-pairs, satisfying the dispersion conditions of Eq. (7.14) and (7.15) involves significant challenges. This is owing to the strong modal dispersion of the pump in BRWs, which propagates as a Bragg mode in comparison to the weaker dispersion of the signal-idler pair propagating as TIR modes. Searching for novel BRW designs which allow for the satisfying of the dispersion conditions in Eq. (7.14) and (7.15) is an open research topic currently under investigation. In the next section, we discuss the potential of BRWs as an integrated source of anti-correlated photon-pairs using a CW pump. As a case study, we examine the effect of waveguide dispersion in tailoring the spectral bandwidth of generated biphotons.

7.2 Bandwidth control of photon-pairs using BRWs

In an SPDC process, tailoring spectral bandwidth or temporal correlation of photo-pairs is essential for practical purposes. Certain applications demand short correlation times. Examples include the enhancement of the accuracy of protocols for quantum positioning and timing [120], or the enhancement of sensitivity due to quantum illumination [114]. Other applications also benefit from photon-pairs with large temporal correlation or nar-

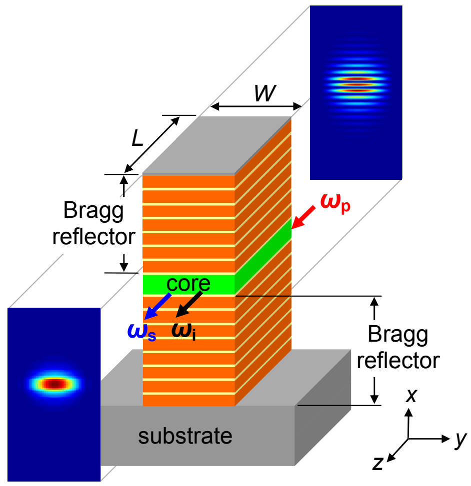

Figure 7.1: Schematic of a typical BRW with intensity profiles of TM-polarized pump (ω_p) and TE-polarized down-converted photons (ω_s, ω_i) in type-I phase-matching [13].

row spectral bandwidth, including efficient atom-photon interfaces or long haul transmission of entangled photons in optical fibers for quantum communication [121].

In an integrated source of down-converted biphotons, dispersion can be controlled using material dispersion, waveguide dispersion or a combination of both mechanisms [122, 123]. Most widely-used approaches for controlling material dispersion rely on temperature and electro-optic tuning which are, in general, weak effects in $Al_xGa_{1-x}As$ system. However, significant control over the waveguide dispersion is offered through proper choice of the ridge width, W, which could be simply implemented using standard lithographic techniques. The dependency of waveguide dispersion on ridge dimension is more pronounced in sub wavelength ridges where highly confined optical modes are established due to large index contrast between the waveguide's core and claddings [124, 125]. Here, we employ the strong dependency of waveguide dispersion on the ridge size as an effective mechanism for controlling the dispersion of biphotons in a monolithically integrated source.

A representative ridge BRW along with typical intensity profiles of the pump and down-converted photons is schematically illustrated in Fig. 7.1. The periodic claddings consisted of eight periods of $Al_{0.40}Ga_{0.60}As/Al_{0.90}Ga_{0.10}As$, with associated thicknesses of

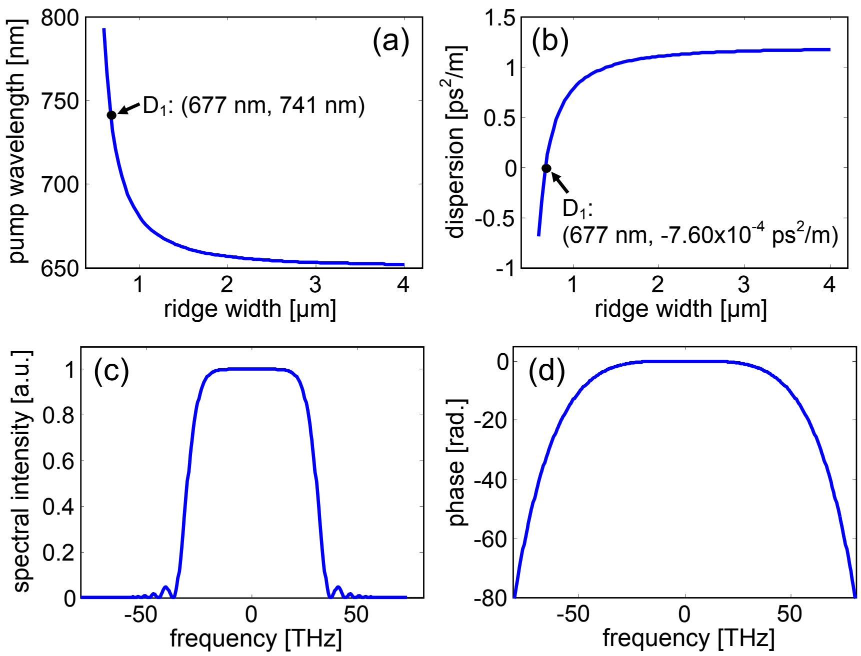

Figure 7.2: Pump wavelength (λ_p) vs. ridge width (W) (a) and the group velocity dispersion of signal (D_s) vs. ridge width (b) in type-I phase-matched BRW of Fig. 7.1. Normalized joint spectral intensity (c) and phase of spectral amplitude (d) of biphotons for the design D_1 with $\Delta\nu_{\mathrm{FWHM}} \approx 61$ THz (≈ 456 nm) [13].

89 nm/200 nm. The core was $Al_{0.65}Ga_{0.35}As$, with a thickness of 246 nm. The material indices of the $Al_xGa_{1-x}As$ elements were derived using the model proposed in [80] at a temperature of $T = 293$°K. The sample length was taken as $L = 3$ mm. Dispersion values of different orders were calculated using standard finite-difference techniques.

We are interested here in generating CW anti-correlated photon-pairs using a CW pump. As discussed earlier, broadband SPDC is attainable in type-I phase-matching where TM-polarized pump photons generate photon-pairs with TE polarization state. For type-I process, according to Eq. (7.8b), the SPDC phase mismatch is predominantly determined by the group velocity dispersion of the signal/idler. In order to maximize the spectral bandwidth, the waveguide design should satisfy the condition $D_s \approx 0$. We monitor the pump wavelength for exact PM, λ_p, and D_s as functions of the ridge width, W, over the range of $0.6 - 4.0$ μm. The results are illustrated in Figs. 7.2(a) and (b), where $D_s = -7.6 \times 10^{-4}$ ps^2/m was obtained for the design point D_1, which corresponds to $(W, \lambda_p) = (677 \text{ nm}, 741 \text{ nm})$. The joint spectral intensity $|\Phi(\Omega_s, \Omega_i)|^2$ of the source is illustrated in Fig. 7.2(c). The full-width at half-maximum bandwidth, $\Delta\nu_{\text{FWHM}}$, is estimated to be 61 THz ($\approx$ 456 nm). The phase of the spectral amplitude, $s_k L/2$, is shown in Fig. 7.2(d). It is instructive to note the flat phase-response around the degeneracy point, ($\Omega_s = 0$) which is expected as a result of the dependency of s_k on Ω_s. In this case, a fourth-order polynomial was used as the fit. This large bandwidth can be translated into an extremely narrow correlation time. The temporal shape of the correlation between signal and idler photons is given by the Fourier transform of $\Phi(\Omega_s, \Omega_i)$. This translates to $\approx$ 15 fs temporal delay between the arrival times of the down-converted photons for D_1. Comparable temporal correlations (23 fs) can be achieved by choosing an adequate nonlinear material and working at a specific wavelength and PM conditions [126]. In order to investigate the potential of the proposed design in reducing the SPDC bandwidth, we further examined the biphoton generation in a type-II phase-matching scheme where a TE-polarized pump generates a pair of signal and idler with TE and TM-polarization states. In this case, Δ_k is dominantly determined by the group velocity mismatch (GVM) between signal and idler according to Eq. (7.10). The minimization of the process bandwidth requires $N_i - N_s$ to be maximum. We thus monitored the variation of λ_p and $N_i - N_s$ as functions of the ridge width over the range of 0.375 μm $-$ 4.0 μm, as shown in Figs. 7.3(a) and (b), respectively. From the figures, the maximum value of $N_i - N_s = -2.92$ ns/m was obtained for the design D_2 with $(W, \lambda_p) = (375 \text{ nm}, 944 \text{ nm})$. The joint spectral intensity is illustrated in Fig. 7.3(c), where the SPDC bandwidth of $\Delta\nu_{\text{FWHM}} = 101$ GHz ($\approx$ 1.2 nm) was obtained. The phase of $\Phi(\Omega_s, \Omega_i)$ is illustrated in Fig. 7.3(d), where the linear phase response confirms the linearity of s_k on Ω_s in type-II PM. The obtained temporal correlation of the biphoton was 10 ps.

It would be interesting to verify the SPDC bandwidth offered by the design D_1 using type-II PM. From the simulation, $N_i - N_s$ was -0.34 ns/m for $\lambda_p = 706$ nm. The bandwidth obtained is $\Delta\nu_{\text{FWHM}} = 869$ GHz ($\approx$ 5.7 nm) with a temporal correlation of

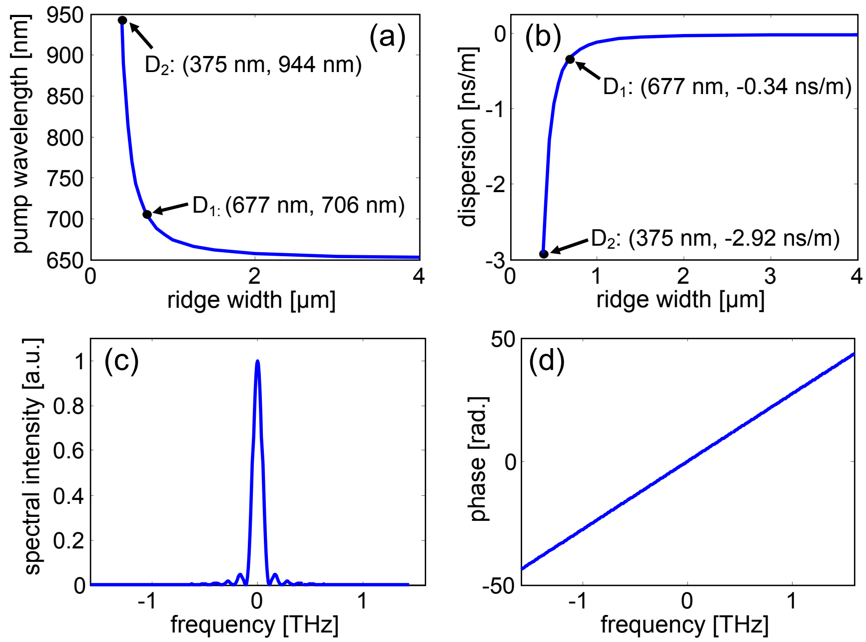

Figure 7.3: Pump wavelength (λ_p) vs. ridge width (W) (a) and the group velocity mismatch between signal and idler ($N_i - N_s$) vs. ridge width (b) in type-II phase-matched BRW of Fig. 7.1. Normalized joint spectral intensity (c) and phase of spectral amplitude (d) of biphotons for the design D_2 with $\Delta\nu_{\text{FWHM}} \approx 101$ GHz (≈ 1.2 nm) [13].

1.2 ps. Note that this can be achieved by simply rotating the polarization of the pump beam. The comparison between the estimated bandwidths in type-I and type-II PM indicates that the proposed structure is capable of controlling the SPDC bandwidth over 2.5 orders of magnitude. Such flexibility is attractive for the development of integrated photon-pair sources where large tunability over the temporal correlation of biphotons is desired.

It should be noted that no theoretical limitation exists on the attainable upper and lower limits of the spectral bandwidth. Instead, the limitations were posed by the practical considerations of designing the structures discussed. For example, for narrowband SPDC, larger $N_i - N_s$ and hence smaller bandwidth is possible by reducing the ridge width. However, this would degrade the conversion efficiency of the process, as the TE-polarized down-converted photons will experience greater losses inside the waveguide. A possible route for further broadening the SPDC spectrum in type-I PM is operating in a regime with lower material dispersion. For both narrowband and broadband sources, the fabrication step in fabricating the waveguides is expected to be challenging due to the small ridge sizes. The sharp slopes observed in Fig. 7.2(b) and Fig. 7.3(b) denote large design sensitivity to fabrication tolerances, which can be compensated, partly, through temperature tuning or electro-optically.

Chapter 8

Conclusions and Future Roadmap

This final chapter presents a summary of the major findings and contributions of this dissertation. In addition, we outline the future roadmap of this research topic to be undertaken for expanding and improving the discussed results. We have shown that phase-matched Bragg reflection waveguides offer a robust platform for the implementation of all-semiconductor integrated parametric devices. Such photonics elements favour many established and emerging applications, such as infrared spectroscopy, environmental monitoring, chemical sensing, medical diagnostics and treatments, generation of THz radiation and quantum communications.

8.1 Summary of Contributions

We started this dissertation by highlighting the challenges involved in phase-matching second-order optical nonlinearities in compound semiconductors, particularly $Al_xGa_{1-x}As$. Using Bragg reflection waveguides or one-dimensional photonic bandgap structures, we proposed an exact modal phase-matching technique which attains PM between a photonic bandgap mode (Bragg mode) and total-internal reflection modes. The proposed technique was demonstrated for the first time for frequency doubling of a 2 ps pump in a telecommunication spectral window in a fabricated slab BRW with a length of 2.4 mm.

For an internal pump power of 25 mW, we estimated a second-harmonic power of 729 nW with a conversion efficiency of 2.0 $\%\text{W}^{-1}\text{cm}^{-2}$. To further improve the nonlinear interaction in BRWs, ridge structures were fabricated and characterized. The two-dimensional optical confinement offered by ridge structures where shown to enhance the efficiency of nonlinear interaction by over an order of magnitude in comparison to slab devices.

Phase-matched BRWs are additionally demonstrated for continuous-wave second-harmonic generation. In a characterized sample with a length of 2.0 mm, we observed continuous-wave second-harmonic power of 23 nW using a pump around 1560 nm with an internal power of 98 mW. Using the beam propagation method for fitting the bandwidth of the experimental SHG tuning curve to the theoretical one, we estimated an upper limit of the SH loss around 780 nm to be 41 cm^{-1}.

Despite initial successes in demonstrating phase-matched BRW, generated harmonics in these structures were limited to power levels below 1 μW. In order to enhance the efficiency of BRWs, we proposed and demonstrated the modified design of matching-layer enhanced BRW (ML-BRW). Using fabricated devices of ML-BRWs, we reported highly efficient SHG in type-I and type-II phase-matching schemes. For type-II SHG in a 2.2 mm long sample, a second-harmonic power of 60 μW was estimated for a 2 ps pulsed pump with a power of 3.3 mW. For the first time, we theoretically investigated and demonstrated the existence of type-0 SHG, where both pump and SH signal has an identical TM polarization state. It was shown that the efficiency of the type-0 scheme was comparable to those of type-I and type-II processes with the phase-matching wavelengths of all three interactions lying within a spectral window as small as 17 nm.

We reported the observation of CW sum-frequency generation and pulsed difference-frequency generations in ML-BRWs. For SFG, conversion efficiency was estimated to be 298 $\%\text{W}^{-1}\text{cm}^{-2}$ with a wavelength acceptance bandwidth in excess of 60 nm. For DFG, the challenges involved in preferential coupling into Bragg hindered us to observed difference frequency power more than 0.95 nW for a respective pump and signal powers of 62.9 mW and 2.9 mW. The bandwidth of DFG was observed to be more than 40 nm.

For applications requiring high pump power, a generalized ML-BRW design was proposed and demonstrated. The proposed structure offsets the destructive effects of third-order nonlinearities on $\boldsymbol{\chi}^{(2)}$ processes when high power harmonics are involved. This is carried out through incorporation of larger bandgap materials by using high aluminum content $Al_xGa_{1-x}As$ layers without undermining the nonlinear conversion efficiency. To demonstrate the validity of generalized ML-BRW design, characterizations of different designs under different pulsed pump conditions including 30 ps, 2 ps and 250 fs were reported. For SHG, results indicated that nonlinear interaction was most efficient using 30 ps pulses, where third-order nonlinear effects were less pronounced and an excellent match between the spectra of employed pulses and wave acceptance bandwidth of the SHG process existed.

Finally, we theoretically investigated the potential of BRWs as integrated sources of frequency-correlated photon-pairs. Using versatile modal dispersion properties in BRWs, we showed that BRWs can serve an integrated source of telecommunication anti-correlated photon-pairs with bandwidth tunablity between 1 nm and 450 nm.

8.2 Future Roadmap

Several routes can be undertaken to further advance phase-matched BRW for integrated parametric devices. These are addressed here.

In discussed designs in this dissertation, we focused on symmetric BRWs. For the case of QtW-BRWs, the claddings consisted of quarter-wave-thick bi-layers. For matching-layer enhanced BRWs, we aimed at relaxing the perfect periodicity of the claddings while the bi-layers were designed to be quarter-wave thick. Investigation of non-symmetric BRWs with non-quarter-wave periodic claddings [55] is an open research which deserves additional attention. For example, one can study BRWs with chirped Bragg reflectors [73]. Chirping of the claddings can be applied to the thicknesses of the Bragg reflectors, to the material indices or to a combination of both thicknesses and material indices. In the linear regime, the effect of chirp can be examined on the first and second-order dispersion properties of different harmonics as well as on the propagation losses. In the nonlinear regime, the parameters of interest include, but are not limited to, wavelength acceptance bandwidth and nonlinear conversion efficiency.

In Section 6.1.2, we proposed phase-matched mono-stack Bragg reflection waveguides (MS-BRWs) without reporting the fabrication and characterization of any such devices. Some preliminary simulations have shown that these structures can provide phase-matched processes with efficiencies comparable to those demonstrated in symmetric ML-BRWs. The benefits of using mono-stack BRWs are anticipated to be manifold. First, as mentioned earlier, there exists a fabrication constraint regarding the overall thickness of BRWs. Using the MOCVD growth technique, as used for the fabrication of devices in this project, the overall design thickness is limited to ≈ 9 μm to avoid large surface morphology. The asymmetric design of MS-BRWs allows replacing the top periodic cladding by a single-layer cladding. This helps in reducing the overall thickness of the top cladding and hence facilitates the growth of the wafer and reduces the required etch depth in patterning the ridge structures. Second, in cases where the growth of wafers with thicknesses above 9 μm can be tolerated, the MS-BRWs favours the allocation of more periodicity to the bottom cladding, which helps at reducing leakage losses of propagating harmonics. Third, the ML-BRW with single-layer top cladding allows patterning longitudinal gratings along the propagation direction. The existence of a properly designed grating on top of the sample can be used for excitation of Bragg mode in BRWs through grating coupling [127, 128, 129, 130, 131]. This is essential for the DFG and SPDC processes where the pump propagates as a Bragg mode.

Investigation of more complicated passive elements in phase-matched BRWs is an open research topic. Examples include longitudinal and transverse directional couplers [132] and ring resonators. BRW ring resonator are particularly attractive for the implantation of novel integrated elements such as spectral filters, phase-shifters and delay lines.

Potential applications for harmonic generations in short wavelength regimes certainly exist [133, 134]. For example for high resolution spectroscopy, the availability of compact, low-cost and tunable sources in green, blue and UV spectral range is highly desired. Due to large absorption of $Al_xGa_{1-x}As$ in such short wavelengths, compound semiconductors with larger bandgaps, such as GaP/AlP with respective bandgaps of 549/506 nm and GaN/AlN with respective bandgaps of 364/197 nm, should be considered for optical frequency mixing using BRWs.

In the quantum optics domain, BRWs deserve additional attention as the source of photon-pairs with frequency correlation properties. We theoretically investigated the generation of frequency anti-correlation biphotons; however, no experimental results were reported on this subject. Also attractive is the generation of frequency correlated biphotons as well as frequency uncorrelated biphotons using a pulsed pump, though these forms of frequency correlations are challenging to attain owing to strong dispersions in $Al_xGa_{1-x}As$ waveguides. Regardless of the type of frequency correlation of photon-pairs generated using BRWs, the biggest challenge in realizing these devices involves with the excitation of the pump mode which propagates as a photonic bandgap mode. Several routes are currently under investigation. The first technique involves manipulating the profile of a Gaussian beam in free-space to form a Bragg mode using programmable spatial light modulators (SLMs). This technique requires simultaneous tailoring of both the amplitude and the phase of the wavefront of an incident Gaussian beam. Another approach is the preferential coupling into the pump Bragg mode using a grating coupler patterned on the top of the device using standard lithographical techniques [127, 128, 129, 130, 131]. Grating couplers have successfully been implemented in Silicon waveguides with high coupling efficiencies between the waveguides and optical fibers. Similar grating couplers are potential for addressing the excitation of the pump Bragg mode in BRWs for generation of photon-pairs. Also, using a grating coupler can assist with the generation of counter-propagating bi-photons where the signal-idler pair propagate in opposite directions along the waveguide length.

Novel cascaded devices including SHG+DFG and SFG+DFG have already been demonstrated in bulk crystals such as $LiNbO_3$ [135, 136, 137, 138]. However, no research has been carried out so far to investigate BRWs for cascaded optical frequency conversion in $Al_xGa_{1-x}As$. This topic deserve attention for future works in nonlinear Bragg reflection waveguides.

Two-dimensional photonic crystals have received some attentions in recent years as candidates for on-chip optical frequency mixing [139, 140, 141]. Extension of the slab and ridge BRWs to two-dimensional photonic bandgap structures is believed to offer additional flexibilities in controlling modal properties of optical signals. Considering significant advances in the fabrication of photonic crystals with nanoscale feature sizes, detailed research in this area is demanded.

Finally, integrated optical parametric oscillators (OPOs) [142] and optical parametric amplifiers (OPAs) [143] cannot be realized without the availability of an on-chip high power pump laser. Significant research is demanded for integration of the devices discussed in this project with recently demonstrated Bragg lasers [106]. The successful integration of these two elements will pave the way for the realization of electrically pumped all-semiconductor integrated OPOs/OPAs.

Appendix A

List of Journal Publications and Conference Presentations

A1. Journal Publications

1. J. Han, D. P. Kang, **P. Abolghasem**, B. J. Bijlani and A. S. Helmy, "Pulsed- and continuous-wave difference-frequency generation in AlGaAs Bragg reflection waveguides," *J. Opt. Soc. Am. B.*, 27(12): 2488–2494, 2010.

2. A. S. Helmy, **P. Abolghasem**, J. S. Aitchison, B. J. Bijlani, J. Han, B. M. Holmes, D. C. Hutchings, U. Younis and S. J. Wagner, "Recent advances in phase-matching second order nonlinearities in monolithic semiconductor waveguides," *Laser & Photonics Rev.*, 2010 .

3. J. B. Han, **P. Abolghasem**, D. Kang, B. J. Bijlani and A. S. Helmy. Difference-frequency generation in AlGaAs Bragg reflection waveguides, *Opt. Lett.*, 35(14): 2334–2336, 2010.

4. J. B. Han, **P. Abolghasem**, B. J. Bijlani, A. Arjmand, S. C. Kumar, A. Esteban-Martin, M. Ebrahim-Zadeh and A. S. Helmy. Femtosecond second-harmonic generation in AlGaAs Bragg reflection waveguides: theory and experiment, *J. Opt. Soc. Am. B.*, 27(6): 1291–1298, 2010.

5. **P. Abolghasem**, J. B. Han, B. J. Bijlani and A. S. Helmy. Type-0 second order nonlinear interaction in monolithic waveguides of isotropic semiconductors, *Opt. Express*, 18(12): 12681–12689, 2010.

6. J. B. Han, **P. Abolghasem**, B. J. Bijlani and A. S. Helmy. Continuous-wave sum-frequency generation in AlGaAs Bragg reflection, *Opt. Lett.*, 34(23): 3656–3658, 2009.

7. **P. Abolghasem**, J. B. Han, A. Arjmand, B. J. Bijlani and A. S. Helmy. Highly efficient second-harmonic generation in monolithic matching-layer enhanced AlGaAs Bragg reflection waveguides, *IEEE Photon. Tech. Lett.*, 21(19): 1462–1464, 2009.

8. **P. Abolghasem**, M. Hedrych, X. J. Shi, J. P. Torres and A. S. Helmy. Bandwidth control of paired photons generated in monolithic Bragg reflection waveguides, *Opt. Lett.*, 14(13): 2000–2002, 2009.

9. **P. Abolghasem**, J. B. Han, B. J. Bijlani, A. Arjmand and A. S. Helmy. Continuous-wave second harmonic generation in Bragg reflection waveguides, *Opt. Express*, 17(11): 9460–9467, 2009.

10. **P. Abolghasem** and A. S. Helmy. Matching layers in Bragg Reflection Waveguides for enhance nonlinear interaction, *IEEE J. Quantum Electron.*, 45(5–6): 646–653, 2009.

11. B. Bijlani, **P. Abolghasem** and A. S. Helmy. Second Harmonic Generation in Ridge Bragg Reflection Waveguides, *Appl. Phys. Lett.*, 92(10): 101124, 2008.

12. A. S. Helmy, B. Bijlani and **P. Abolghasem**. Phase matching in monolithic Bragg reflection waveguides, *Opt. Lett.*, 32(16): 2399–2401, 2007.

A2. Conference Presentations

1. **P. Abolghasem**, B. J. Bijlani, B. Burgoyne, P. Martek, V. Mercier, A. Villeneuve and A. S. Helmy. Bandgap effects in $Al_xGa_{1-x}As$ Bragg reflection waveguides

for enhanced $\chi^{(2)}$ interactions, The 23rd Annual Meeting of the IEEE Photonics Society (PHO), Denver, CO (Nov. 2010) (submitted).

2. J. B. Han, **P. Abolghasem**, D. Kang, B. J. Bijlani and A. S. Helmy. Broadband Difference-Frequency Generation in AlGaAs Bragg Reflection Waveguides, The 23rd Annual Meeting of the IEEE Photonics Society (PHO), Denver, CO (Nov. 2010) (submitted).

3. **P. Abolghasem**, J. Han, B. J. Bijlani and A. S. Helmy. Broadband sum-frequency generation in AlGaAs Bragg reflection waveguides, Conference on Lasers and Electro-Optics (CLEO), San Jose, CA (May 2010).

4. J. Han, **P. Abolghasem**, B. J. Bijlani and A. S. Helmy. New modality of type-I second order nonlinear interaction in bulk AlGaAs Bragg reflection waveguide, Conference on Lasers and Electro-Optics (CLEO), San Jose, CA (May 2010).

5. M. Hendrych, R. L. Montiel, R. Machulka, J. P. Torres, **P. Abolghasem** and A. S. Helmy. Generation of entangled photons in Bragg reflection waveguides, 15th European Conference on Integrated Optics (ECIO), Cambridge, UK (April 2010).

6. **P. Abolghasem**, J. Han, B. J. Bijlani, A. Arjmand and A. S. Helmy. Record secondharmonic conversion efficiency in improved $Al_xGa_{1-x}As$ Bragg reflection waveguides, Frontiers in Optics (FiO), San Jose, CA (Nov. 2009).

7. **P. Abolghasem**, J. Han, B. J. Bijlani, A. Arjmand and A. S. Helmy. Second-harmonic generation using matching-layer enhanced Bragg reflection waveguide, The 22st Annual Meeting of IEEE Lasers and Electro-Optics Society (LEOS), Belek-Antalya, Turkey (Oct. 2009).

8. **P. Abolghasem**, B. J. Bijlani and A. S. Helmy. Continuous-wave frequency doubling of near-infrared Light using $Al_xGa_{1-x}As$ Bragg reflection waveguides, Conference on Lasers and Electro-Optics (CLEO), Baltimore, ML (May 2009).

9. **P. Abolghasem**, S. Xhi, M. Hendrych, J. P. Torres and A. S. Helmy. Bragg reflection waveguides: a novel platform for enhanced control of photon-pair generation

via spontaneous parametric down conversion, Conference on Quantum Electronics and Laser Science (IQEC), Baltimore, MD (May 2009).

10. A. S. Helmy, **P. Abolghasem** and B. J. Bijlani. Harnessing second order optical nonlinearities in compound semiconductors, SPIE Photonics West, San Jose, CA (Jan. 2009) (invited).

11. **P. Abolghasem**, B. J. Bijlani and A. S. Helmy. Enhanced second harmonic generation in ridge Bragg Reflection Waveguides, The 21st Annual Meeting of IEEE Lasers and Electro-Optics Society (LEOS), Newport Beach, CA (Nov. 2008).

12. **P. Abolghasem**, B. J. Bijlani and A. S. Helmy. Phase matching second order optical nonlinearities in compound semiconductors, The 214th Electrochemical Society's International Symposium on Integrated Optoelectronics (ECS), Honolulu, HI (Oct. 2008) (Invited).

13. B. J. Bijlani, **P. Abolghasem** and A. S. Helmy. Phase matching using ridge Bragg reflection waveguides, Conference on Lasers and Electro-Optics (CLEO), San Jose, CA (May 2008).

14. B. Bijlani, **P. Abolghasem**, A. S. Helmy. Phase matching in monolithic Bragg reflection waveguides, The 20th Annual Meeting of the IEEE Lasers and Electro-Optics Society (LEOS), Lake Buena Vista, FL (Oct. 2007).

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